

Near-Lossless Knowledge Distillation: Definitions, Error Decompositions, Capacity Lower Bounds and Objective Design

A theoretical treatment

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Abstract

We give a mathematically explicit account of distilling a teacher $p_T(y | x)$ into a smaller autoregressive student $q_\theta(y | x)$. We (i) formalise “lossless” through divergences between sequence-level laws under a deployment input distribution and prove the relations among forward KL, reverse KL, total variation, JSD_β , Hellinger and task-level excess risk, including exact chain-rule identities that pair each KL direction with a specific prefix distribution; (ii) decompose the student’s gap into approximation, estimation, optimisation, prefix-shift (exposure bias) and covariate-shift terms and bound each, recovering the $O(\varepsilon T)$ versus $O(\varepsilon T^2)$ compounding dichotomy and showing exactly when on-policy training removes it; (iii) prove information-theoretic (rate–distortion) lower bounds on the achievable divergence as a function of the student’s description length, together with a softmax-bottleneck rank bound, and derive the power-law forms they predict; (iv) analyse the logit gradients of the standard objectives, the role of temperature, and the Fisher-information content of soft labels, and propose a sequence-level, prefix-adaptive JSD_β objective with proven exactness, boundedness, agnostic total-variation competitiveness, consistency, and an unbiased Rao–Blackwellised gradient estimator; (v) analyse intermediate-representation matching, proving when it is redundant, when it helps (noisy supervision, conditioning, compositional hardness) and when it over-constrains, and propose a Fisher-aligned variant with no approximation cost in a linear-Gaussian model; (vi) assemble an end-to-end bound, separate what is proved from what is conjectured, and specify experiments and open problems; (vii) prove a corrected distillation scaling law in a solvable model with a softmax bottleneck and depth: an exact finite-sample learning curve, soft- and hard-label exponents, the capacity gap as aliasing, depth as implicit capacity selection, and the finding that the exponential phase of the capacity floor does not survive spectra with infinitely many nonzero eigenvalues. Every inequality marked *numerically checked* is verified by the scripts in `verification/`.

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Setting, notation and standing assumptions

Context. Distillation in its modern form goes back to [Buciluă et al. \(2006\)](#), [Ba and Caruana \(2014\)](#) and [Hinton et al. \(2015\)](#). The sequence-level variant for autoregressive models is due to [Kim and Rush \(2016\)](#). [Xu et al. \(2024\)](#) survey methods for large language models. Here we care about the theory: what “lossless” should mean, what limits it, and which objectives provably approach it.

Objects. $\mathcal{X}$ is a countable input space and $\mathcal{V}$ a finite vocabulary, $|\mathcal{V}| = V$. Outputs are sequences $y = (y_1, \dots, y_T) \in \mathcal{Y} := \mathcal{V}^T$. Variable-length outputs are handled by an absorbing end-of-sequence token: after it, every model emits it again with probability one. This does not change any divergence between sequence laws, so a fixed horizon T loses no generality. To avoid a clash with the horizon T , we write the *teacher* as

$$p(y | x) := p_T(y | x) = \prod_{t=1}^T p(y_t | x, y_{<t}), \quad q_\theta(y | x) = \prod_{t=1}^T q_\theta(y_t | x, y_{<t}), \quad \theta \in \Theta_S.$$

The *state* (context) at step t is $s_t = (x, y_{<t}) \in \mathcal{S}_t$. We abbreviate the next-token conditionals as $p_t(\cdot | s) := p(\cdot | s)$ and $q_t(\cdot | s) := q_\theta(\cdot | s)$. The sequence laws are $P^x := p(\cdot | x)$ and $Q_\theta^x := q_\theta(\cdot | x)$ on $\mathcal{Y}$. Inputs follow a *training* distribution $\mathcal{D}$. Behaviour is evaluated under a *deployment* distribution μ , which may differ from $\mathcal{D}$.

Prefix (state) distributions. For a policy $\pi \in \{p, q_\theta\}$, d_t^π is the law of $s_t = (x, y_{<t})$ when $x \sim \mathcal{D}$ and $y \sim \pi(\cdot | x)$. We write d_t^p for the teacher and d_t^θ for the student. Distillation on teacher-generated prefixes (“teacher forcing”) trains on d_t^p . *On-policy* distillation trains on d_t^θ .

Per-token errors. For a divergence D between next-token distributions, define the off- and on-policy average per-token errors

$$\varepsilon_{\text{off}}^D := \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{s \sim d_t^p} D(p_t(\cdot | s) \| q_t(\cdot | s)), \quad \varepsilon_{\text{on}}^D := \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{s \sim d_t^\theta} D(p_t(\cdot | s) \| q_t(\cdot | s)). \quad (1)$$

For reverse KL, “ $D(p \| q)$ ” in (1) means $\text{KL}(q \| p)$. We write e_t^{off} and e_t^{on} for the individual summands, so that $\varepsilon_{\text{off}} = \frac{1}{T} \sum_t e_t^{\text{off}}$.

Logits. The student’s next-token distribution is $q_t(\cdot | s) = \text{softmax}(z_\theta(s))$ with logits $z_\theta(s) \in \mathbb{R}^V$. The teacher’s logits are $v(s)$. $C := I - \frac{1}{V} \mathbf{1}\mathbf{1}^\top$ is the centring projector and $\tilde{z} := Cz$ denotes centred logits.

Conventions. $\log$ is natural. Rates R are in bits unless stated otherwise, so a class of 2^R students has $\log |\Theta_S| = R \ln 2$ nats. $a \lesssim b$ hides absolute constants. $h(\beta) = -\beta \ln \beta - (1 - \beta) \ln(1 - \beta)$ is the binary entropy in nats.

Standing assumptions. Each result states exactly which of the following it uses.

Assumption A1 (Bounded student log-probabilities). There is $B < \infty$ with $\log q_\theta(a | s) \geq -B$ for all θ, s and all a with $p(a | s) > 0$. This can be enforced, for example, by mixing with the uniform distribution at every state before end-of-sequence: $q = (1 - \eta)q' + \eta/V$ gives $B \leq \log(V/\eta)$. Restricting to the teacher’s support keeps the absorbing end-of-sequence convention intact, and it is all that Theorem 2.6 uses.

Assumption A2 (Bounded covariate shift). $\mu \ll \mathcal{D}$ and $\rho := \|\text{d}\mu/\text{d}\mathcal{D}\|_\infty < \infty$.

Assumption A3 (Realisability). There is $\theta^\circ \in \Theta_S$ with $q_{\theta^\circ}(\cdot | x) = p(\cdot | x)$ for $\mathcal{D}$ -a.e. x .

Assumption A4 (Finite description length). $|\Theta_S| \leq 2^R$, for example P parameters stored with b bits each, so that $R = Pb$. Alternatively Θ_S has ε -covering number $\mathcal{N}(\varepsilon)$ in the metric $\|\log q_\theta - \log q_{\theta'}\|_\infty$.

Assumption A5 (Linear read-out). $z_\theta(s) = Wh_\theta(s) + b$ with $W \in \mathbb{R}^{V \times d_S}$, $h_\theta(s) \in \mathbb{R}^{d_S}$. The same form holds for the teacher, with width d_T .

Assumption A6 (Stochastic optimisation). The training objective F is L -smooth and bounded below. The optimiser sees unbiased stochastic gradients with variance at most σ^2 .

Status labels. Each result carries one of these tags: [proved] means proved here with no assumption beyond the definitions; [proved under ...] means proved under the listed assumptions; [known result; proof sketch] means a result from the literature with a pointer to its key lemma; [proved within the stated model] means an exact calculation inside a stylised model, whose relevance to real networks is itself an assumption. Conjectures and heuristics are labelled as such. Several inequalities are additionally [numerically checked] by the scripts in `verification/`.

1 Formalising “lossless”

1.1 Divergences, distillation risk and task-level excess risk

Definition 1.1 (f -divergences used throughout). For convex $f : (0, \infty) \rightarrow \mathbb{R}$ with $f(1) = 0$, set $D_f(P\|Q) = \sum_y Q(y) f(P(y)/Q(y))$. We use the conventions $0f(0/0) = 0$ and $Qf(P/0) = Pf^*(0)$, where $f^*(0) := \lim_{u \rightarrow \infty} f(u)/u$. The instances we use are:

divergence	$f(u)$	$f''(1)$	bounded?
forward KL, $\text{KL}(P\ Q)$	$u \log u$	1	no
reverse KL, $\text{KL}(Q\ P)$	$-\log u$	1	no
total variation $\text{TV} = \frac{1}{2}\ P - Q\ _1$	$\frac{1}{2} u - 1 $	—	≤ 1
Hellinger ² , $H^2 = \frac{1}{2} \sum (\sqrt{P} - \sqrt{Q})^2$	$\frac{1}{2}(\sqrt{u} - 1)^2$	$\frac{1}{4}$	≤ 1
$\chi^2(P\ Q)$	$(u - 1)^2$	2	no
$\text{JSD}_\beta(P\ Q)$	$\beta u \log u - m \log m$, $m = \beta u + 1 - \beta$	$\beta(1 - \beta)$	$\leq h(\beta)$

Here $\text{JSD}_\beta(P\|Q) := \beta \text{KL}(P\|M_\beta) + (1 - \beta) \text{KL}(Q\|M_\beta)$ with $M_\beta = \beta P + (1 - \beta)Q$. This follows the convention of [Agarwal et al. \(2024\)](#); $\beta = \frac{1}{2}$ is the classical Jensen–Shannon divergence ([Lin, 1991](#)). Conventions differ between papers (some swap β and $1 - \beta$), so we fix this one. A direct computation gives $f''_\beta(u) = \beta(1 - \beta)/(um) > 0$. The Rényi divergence of order $\alpha > 1$ is $D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \sum_y P(y)^\alpha Q(y)^{1 - \alpha}$ ([van Erven and Harremoës, 2014](#)).

Definition 1.2 (Distillation risk; (ε, D, μ) -lossless). For a divergence D on $\Delta(\mathcal{Y})$ and an input law μ , the *sequence-level distillation risk* is $\mathcal{L}_D(\theta; \mu) := \mathbb{E}_{x \sim \mu} D(P^x \| Q_\theta^x)$. The student is (ε, D, μ) -*lossless* if $\mathcal{L}_D(\theta; \mu) \leq \varepsilon$. For a family of prefix laws $\nu = (\nu_t)_{t \leq T}$, the *token-level risk* is $\ell_D(\theta; \nu) := \sum_{t=1}^T \mathbb{E}_{s \sim \nu_t} D(p_t(\cdot | s) \| q_t(\cdot | s))$.

Definition 1.3 (Task-level excess risk). A task is a measurable reward $r : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$; a loss ℓ is handled as $r = 1 - \ell$. Its excess risk under sampling is $\Delta_r(\theta; \mu) := \mathbb{E}_{x \sim \mu} [\mathbb{E}_{y \sim P^x} r(x, y) - \mathbb{E}_{y \sim Q_\theta^x} r(x, y)]$. For a class $\mathcal{R}$ of tasks, $\Delta_{\mathcal{R}}(\theta; \mu) := \sup_{r \in \mathcal{R}} |\Delta_r(\theta; \mu)|$.

1.2 Relations between the definitions

Proposition 1.4 (TV is the worst-case task regret). *[proved]*

(a) $\sup_{r: \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]} \Delta_r(\theta; \mu) = \sup_r |\Delta_r(\theta; \mu)| = \mathbb{E}_{x \sim \mu} \text{TV}(P^x, Q_\theta^x)$.

- (b) If the task sees y only through a statistic $\phi(x, y)$, for example the final answer, so that $r = g(x, \phi(x, y))$, then $\sup_g |\Delta_r| = \mathbb{E}_{x \sim \mu} \text{TV}(\phi_{\#} P^x, \phi_{\#} Q_{\theta}^x) \leq \mathbb{E}_{x \sim \mu} \text{TV}(P^x, Q_{\theta}^x)$.
- (c) For each x , the minimum error probability of any test that tells, from one draw, whether the draw came from P^x or from Q_{θ}^x (equal priors) is $\frac{1}{2}(1 - \text{TV}(P^x, Q_{\theta}^x))$.

Proof. (a) For fixed x , $\sum_y r(x, y)(P^x(y) - Q_{\theta}^x(y)) \leq \sum_y (P^x(y) - Q_{\theta}^x(y))_+ = \text{TV}$, with equality at $r = \mathbf{1}\{P^x > Q_{\theta}^x\}$. This pointwise maximiser is measurable because $\mathcal{X} \times \mathcal{Y}$ is countable. Taking $1 - r$ covers the absolute value. Integrate over $x \sim \mu$. (b) Apply (a) to the push-forward laws, then use data processing for TV. (c) This is Le Cam's two-point identity: the Bayes test is $\mathbf{1}\{P^x > Q_{\theta}^x\}$. $\square$

Proposition 1.5 (Pinsker, Bretagnolle–Huber, Hellinger). *[known result; proof sketch] For any P, Q and $K \in \{\text{KL}(P\|Q), \text{KL}(Q\|P)\}$: (i) $\text{TV} \leq \sqrt{K/2}$ (Pinsker, 1964; Tsybakov, 2009); (ii) $\text{TV} \leq \sqrt{1 - e^{-K}}$ (Bretagnolle and Huber, 1979), which is the informative bound when $K > 2$; (iii) $H^2 \leq \text{TV} \leq H\sqrt{2 - H^2}$; (iv) $K \geq -2\log(1 - H^2) \geq 2H^2$.*

Proof of (iv). By Jensen, $\text{KL}(P\|Q) = -2\mathbb{E}_P \log \sqrt{Q/P} \geq -2\log \mathbb{E}_P \sqrt{Q/P} = -2\log(1 - H^2)$. The same argument applies with P and Q exchanged. The other parts are textbook; see Tsybakov (2009, Lemmas 2.3–2.6). $\square$

Lemma 1.6 (f -divergences are at most linear in TV). *[proved] For convex f with $f(1) = 0$, $D_f(P\|Q) \leq (f(0) + f^*(0)) \text{TV}(P, Q)$.*

Proof. Write $u = P/Q$. On $[0, 1]$, convexity gives $f(u) \leq (1 - u)f(0) + uf(1) = (1 - u)f(0)$. For $u > 1$ the slope $(f(u) - f(1))/(u - 1)$ is non-decreasing in u and tends to $f^*(0)$, so $f(u) \leq (u - 1)f^*(0)$. Hence $D_f \leq f(0)\sum_y (Q - P)_+ + f^*(0)\sum_y (P - Q)_+ = (f(0) + f^*(0))\text{TV}$. This is classical; see Sason and Verdú (2016). $\square$

Proposition 1.7 (Properties of JSD_{β}). *[proved] [numerically checked] For all P, Q and $\beta \in (0, 1)$: (i) $2\beta(1 - \beta) \text{TV}^2 \leq \text{JSD}_{\beta}(P\|Q) \leq h(\beta) \text{TV} \leq h(\beta)$; (ii) $\text{JSD}_{\beta}(P\|Q) \leq \beta(1 - \beta)[\text{KL}(P\|Q) + \text{KL}(Q\|P)]$; (iii) if $\text{supp } P = \text{supp } Q$ is finite, then $\text{JSD}_{\beta}/\beta \rightarrow \text{KL}(P\|Q)$ as $\beta \rightarrow 0$ and $\text{JSD}_{\beta}/(1 - \beta) \rightarrow \text{KL}(Q\|P)$ as $\beta \rightarrow 1$.*

Proof. (i) Since $\text{TV}(P, M_{\beta}) = (1 - \beta)\text{TV}(P, Q)$ and $\text{TV}(Q, M_{\beta}) = \beta \text{TV}(P, Q)$, Pinsker applied to each term gives $\text{JSD}_{\beta} \geq 2\beta(1 - \beta)^2 \text{TV}^2 + 2(1 - \beta)\beta^2 \text{TV}^2$. For the upper bound, apply Lemma 1.6 with $f_{\beta}(0) = (1 - \beta) \log \frac{1}{1 - \beta}$ and $f_{\beta}^*(0) = \lim_u [\beta \log u - \beta \log(\beta u)] = \beta \log \frac{1}{\beta}$, whose sum is $h(\beta)$. (ii) KL is convex in its second argument, so $\text{KL}(P\|\beta P + (1 - \beta)Q) \leq (1 - \beta)\text{KL}(P\|Q)$ and $\text{KL}(Q\|M_{\beta}) \leq \beta \text{KL}(Q\|P)$. (iii) As $\beta \rightarrow 0$, $M_{\beta} \rightarrow Q$, so $\text{KL}(P\|M_{\beta}) \rightarrow \text{KL}(P\|Q)$. Also $\text{KL}(Q\|Q + \beta(P - Q)) = \frac{\beta^2}{2} \chi^2(P\|Q) + O(\beta^3)$ on a finite common support. Hence $\text{JSD}_{\beta} = \beta \text{KL}(P\|Q) + O(\beta^2)$. The case $\beta \rightarrow 1$ is symmetric. $\square$

Proposition 1.8 (No global comparison; local equivalence). *[proved] [numerically checked]*

- (i) Forward and reverse KL are not comparable in general. For $P = \text{Bern}(\delta)$ and $Q = \text{Bern}(0)$, $\text{KL}(P\|Q) = \infty$ while $\text{KL}(Q\|P) = \log \frac{1}{1 - \delta} \rightarrow 0$. Exchanging the roles gives the opposite.
- (ii) Suppose $a \leq P(y)/Q(y) \leq b$ for all y (necessarily $a \leq 1 \leq b$). For every twice differentiable f with $f(1) = 0$,

$$\frac{1}{2} \inf_{[a, b]} f'' \cdot \chi^2(P\|Q) \leq D_f(P\|Q) \leq \frac{1}{2} \sup_{[a, b]} f'' \cdot \chi^2(P\|Q).$$

In particular $\frac{\chi^2}{2b} \leq \text{KL}(P\|Q) \leq \frac{\chi^2}{2a}$ and $\frac{\chi^2}{2b^2} \leq \text{KL}(Q\|P) \leq \frac{\chi^2}{2a^2}$. Hence $\text{KL}(P\|Q) \leq \frac{b^2}{a} \text{KL}(Q\|P)$ and $\text{KL}(Q\|P) \leq \frac{b}{a^2} \text{KL}(P\|Q)$. Also $\text{JSD}_{\beta} = \frac{\beta(1 - \beta)}{2} \chi^2(1 + O(b - a))$.

Proof. (ii) Replacing f by $\tilde{f}(u) = f(u) - f'(1)(u-1)$ leaves D_f unchanged, because $\sum_y Q(u-1) = 0$. Taylor's theorem gives $\tilde{f}(u) = \frac{1}{2}f''(\xi_u)(u-1)^2$ with $\xi_u \in [a, b]$. Multiply by Q and sum. For forward KL, $f''(u) = 1/u$; for reverse KL, written as $D_{-\log}(P\|Q)$, $f''(u) = 1/u^2$. $\square$

Remark 1.9 (What Proposition 1.8 means for “near-lossless”). In the near-lossless regime with bounded likelihood ratios, every smooth f -divergence equals $\frac{1}{2}f''(1)\chi^2$ to leading order. So forward KL, reverse KL and JSD_β (rescaled by $\beta(1-\beta)$) agree when the student is nearly lossless. The choice of divergence matters only (a) in the lossy, capacity-limited regime, (b) through tails and support mismatch, where likelihood ratios are unbounded, and (c) through *which prefix law* the per-token divergence is averaged over (Section 1.3).

Proposition 1.10 (Coverage versus precision: event bounds). *[proved] [numerically checked]* For every event $A \subseteq \mathcal{Y}$:

- (i) (coverage) $P(A) \leq \frac{\text{KL}(P\|Q) + \log 2}{\log(1/Q(A))}$;
- (ii) (precision) $Q(A) \leq \frac{\text{KL}(Q\|P) + \log 2}{\log(1/P(A))}$;
- (iii) (Rényi, $\alpha > 1$) $Q(A) \leq (e^{D_\alpha(Q\|P)} P(A))^{(\alpha-1)/\alpha}$;
- (iv) (additive) $|P(A) - Q(A)| \leq \text{TV}(P, Q)$.

Proof. (ii) By data processing onto $\{A, A^c\}$, $\text{KL}(Q\|P) \geq \text{kl}(Q(A)\|P(A))$, and $\text{kl}(a\|b) \geq a \log \frac{1}{b} - h(a) \geq a \log \frac{1}{b} - \log 2$. Part (i) is (ii) with the roles exchanged. (iii) By Hölder with exponents α and $\alpha/(\alpha-1)$, $Q(A) = \sum_A P(Q/P) \leq P(A)^{(\alpha-1)/\alpha} (\sum_y P(Q/P)^\alpha)^{1/\alpha}$, and $\sum_y P(Q/P)^\alpha = e^{(\alpha-1)D_\alpha(Q\|P)}$ (Mironov, 2017, Prop. 10). (iv) is the definition of TV. $\square$

Forward KL guarantees *coverage*: the student cannot give negligible probability to events the teacher finds likely. Reverse KL, and Rényi divergences of Q from P , guarantee *precision*: the student cannot put appreciable mass on events the teacher considers essentially impossible (for example hallucinated or unsafe continuations). TV controls neither multiplicatively. A student at $\text{TV} = 10^{-2}$ may emit a teacher-impossible output with probability 10^{-2} , which is invisible to TV but catastrophic at deployment scale.

Proposition 1.11 (Greedy decoding needs margins). *[proved]* Let $a^*(s) = \arg \max_a p(a | s)$ have margin $\gamma(s) = p(a^* | s) - \max_{a \neq a^*} p(a | s)$. If $\text{TV}(p_t(\cdot | s), q_t(\cdot | s)) < \gamma(s)/2$ at every state s on the teacher's greedy path from x , then the student's greedy decoding of x equals the teacher's. The constant is tight: moving mass $\gamma/2$ from a^* to the runner-up creates a tie.

Proof. $|p(a | s) - q(a | s)| \leq \text{TV}$ for every a , so $q(a^* | s) - q(a | s) \geq \gamma(s) - 2\text{TV} > 0$. Induct along the path. $\square$

1.3 Sequence level versus token level

Theorem 1.12 (Chain rules: each KL direction has its own prefix law). *[proved] [numerically checked]* For every x ,

$$\text{KL}(P^x\|Q_\theta^x) = \sum_{t=1}^T \mathbb{E}_{s_t \sim P^x} \text{KL}(p_t(\cdot | s_t) \| q_t(\cdot | s_t)), \quad (2)$$

$$\text{KL}(Q_\theta^x\|P^x) = \sum_{t=1}^T \mathbb{E}_{s_t \sim Q_\theta^x} \text{KL}(q_t(\cdot | s_t) \| p_t(\cdot | s_t)). \quad (3)$$

After averaging over $x \sim \mathcal{D}$: $\mathbb{E}_{\mathcal{D}} \text{KL}(P\|Q_\theta) = T \varepsilon_{\text{off}}^{\text{KL}}$ (teacher prefixes) and $\mathbb{E}_{\mathcal{D}} \text{KL}(Q_\theta\|P) = T \varepsilon_{\text{on}}^{\text{KL}}$ (student prefixes).

Proof. $\log \frac{P^x(y)}{Q_\theta^x(y)} = \sum_t \log \frac{p_t(y_t|s_t)}{q_t(y_t|s_t)}$. Take $\mathbb{E}_{y \sim P^x}$ and condition each summand on s_t (tower property). The reverse identity is the same computation with $y \sim Q_\theta^x$. $\square$

Remark 1.13. Supervised distillation on teacher-generated sequences with soft labels at every position (the token-level analogue of [Kim and Rush \(2016\)](#)) minimises (2) *exactly*. On-policy reverse-KL distillation, as in the objectives of [Gu et al. \(2024\)](#) and [Agarwal et al. \(2024\)](#) with $\lambda = 1$ and reverse KL, minimises (3) exactly, provided the gradient is taken through the sampling distribution (Proposition 4.9). The two remaining pairings are forward KL on student prefixes (for example GKD with forward KL on on-policy data) and reverse KL on teacher prefixes. These are genuine discrepancies (Proposition 1.15) but do not equal any sequence-level f -divergence.

Proposition 1.14 (Hybrid argument for TV, and its converse). *[proved] [numerically checked]*
For every x ,

$$\text{TV}(P^x, Q_\theta^x) \leq \sum_{t=1}^T \mathbb{E}_{s_t \sim P^x} \text{TV}_t(s_t) \quad \text{and} \quad \text{TV}(P^x, Q_\theta^x) \leq \sum_{t=1}^T \mathbb{E}_{s_t \sim Q_\theta^x} \text{TV}_t(s_t),$$

where $\text{TV}_t(s) = \text{TV}(p_t(\cdot | s), q_t(\cdot | s))$. Conversely, for each t , $\mathbb{E}_{s_t \sim P^x} \text{TV}_t(s_t) \leq 2 \text{TV}(P^x, Q_\theta^x)$, and the same holds with Q_θ^x .

Proof. Let H_k draw $y_{1:k}$ from p and $y_{k+1:T}$ from q_θ , so that $H_T = P^x$ and $H_0 = Q_\theta^x$. H_k and H_{k-1} share the law of s_k and the continuation kernel after step k . By data processing, $\text{TV}(H_k, H_{k-1}) \leq \mathbb{E}_{s_k \sim P^x} \text{TV}_k(s_k)$. Telescope. For the second bound, exchange the roles of p and q_θ . For the converse, let Π_t^P and Π_t^Q be the laws of (s_t, y_t) and $\Pi^H := d_t^p \otimes q_t$. Then $\mathbb{E}_{d_t^p} \text{TV}_t = \text{TV}(\Pi_t^P, \Pi^H) \leq \text{TV}(\Pi_t^P, \Pi_t^Q) + \text{TV}(d_t^p, d_t^q) \leq 2 \text{TV}(P^x, Q_\theta^x)$, because marginalisation cannot increase TV. $\square$

Proposition 1.15 (The mismatched pairings are divergences with a $\sqrt{T}$ -lossy Pinsker bound). *[proved]* Let $H_{\text{on}}^{\text{KL}} := \sum_t \mathbb{E}_{s \sim Q_\theta^x} \text{KL}(p_t \| q_t)$ and $H_{\text{off}}^{\text{rKL}} := \sum_t \mathbb{E}_{s \sim P^x} \text{KL}(q_t \| p_t)$. Each is ≥ 0 , and each vanishes iff $P^x = Q_\theta^x$. Moreover $\text{TV}(P^x, Q_\theta^x) \leq \sqrt{TH/2}$ for either one. By contrast, the exact pairings give $\text{TV} \leq \sqrt{\text{KL}/2}$ with KL the corresponding sequence-level KL. The factor T is an artefact of the proof, not a proven loss. A sequential Hellinger subadditivity inequality ([Foster et al., 2021](#)) gives $\text{TV} \leq C\sqrt{H \log T}$ instead. Multi-start local search over binary autoregressive pairs finds worst ratios $\text{TV}^2/H \approx 0.50, 0.60, 0.69, 0.74$ for $T = 1, \dots, 4$, far below $T/2$. These are lower estimates of the true worst case.

Proof. If $H_{\text{on}}^{\text{KL}} = 0$, then $p_1 = q_1$ at x . So $d_2^p = d_2^q$, hence $p_2 = q_2$ on the common support, and so on by induction. The bound follows from Proposition 1.14, Pinsker per token, and Cauchy–Schwarz: $\sum_t \mathbb{E} \text{TV}_t \leq \sum_t \sqrt{e_t/2} \leq \sqrt{T \sum_t e_t/2}$. $\square$

Corollary 1.16 (Sequence-level TV from per-token errors). *[proved]*

$$\mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_\theta) \leq \min \left\{ \sqrt{T \varepsilon_{\text{off}}^{\text{KL}}/2}, \sqrt{1 - e^{-T \varepsilon_{\text{off}}^{\text{KL}}}}, \sqrt{T \varepsilon_{\text{on}}^{\text{rKL}}/2}, T \varepsilon_{\text{off}}^{\text{TV}}, T \varepsilon_{\text{on}}^{\text{TV}} \right\}.$$

Proof. Combine Theorem 1.12 and Propositions 1.5 and 1.14 with Jensen. The maps $k \mapsto \sqrt{k}$ and $k \mapsto \sqrt{1 - e^{-k}}$ are concave. $\square$

Example 1.17 (Neither route dominates). *[proved]* (a) *Diffuse errors.* Let $p_t = \text{Bern}(\frac{1}{2})$ and $q_t = \text{Bern}(\frac{1}{2} + \delta)$ be i.i.d. Then $T \varepsilon_{\text{off}}^{\text{TV}} = T\delta$, whereas $\text{KL}(P \| Q) = -\frac{T}{2} \log(1 - 4\delta^2)$, so the Pinsker route gives $\text{TV} \leq \sqrt{T} \delta (1 + o(1))$. This is the correct order, by the central limit theorem for the log-likelihood ratio, and it improves on the hybrid route by $\sqrt{T}$. (b) *Concentrated errors (deterministic teacher).* Let $p_t = \delta_a$ and $q_t(a) = 1 - \delta$. Then $\text{TV} = 1 - (1 - \delta)^T \leq T\delta$ and

$KL = -T \log(1 - \delta)$, so $\sqrt{KL/2} \approx \sqrt{T\delta/2} \gg T\delta$ when $T\delta \ll 1$. Here the hybrid route is tight. In general, if P is a point mass then $TV(P, Q) = 1 - e^{-KL(P\|Q)} \leq KL(P\|Q)$: control is linear rather than square-root.

Proposition 1.18 (Covariate shift). *[proved] For any measurable $g \geq 0$ (for example a per-input divergence), under Assumption A2, $\mathbb{E}_\mu g \leq \rho \mathbb{E}_{\mathcal{D}} g$. If $0 \leq g \leq G$, then without any density-ratio assumption, $|\mathbb{E}_\mu g - \mathbb{E}_{\mathcal{D}} g| \leq G TV(\mu, \mathcal{D})$. Bounded criteria (TV , H^2 , JSD_β) therefore transfer across input shift additively. Unbounded ones (KL) need Assumption A2.*

1.4 Which definition is the right one?

Definition 1.19 (Hierarchy of near-losslessness). At tolerance ε and deployment law μ , the student is:

behaviourally lossless if $\mathbb{E}_\mu TV(P^x, Q_\theta^x) \leq \varepsilon$;

likelihood lossless if $\mathbb{E}_\mu KL(P^x \| Q_\theta^x) \leq \varepsilon$;

two-sided lossless if $\mathbb{E}_\mu [KL(P^x \| Q_\theta^x) + KL(Q_\theta^x \| P^x)] \leq 2\varepsilon$;

decoding lossless if it is behaviourally lossless and the margin condition of Proposition 1.11 holds on the teacher’s greedy paths.

Position taken in this document.

1. *The target is sequence-level, on the deployment law.* Behavioural losslessness, $\mathbb{E}_\mu TV(P^x, Q_\theta^x) \leq \varepsilon$, is *exactly* the statement that no bounded downstream task evaluated by sampling changes by more than ε (Proposition 1.4). It is also exactly the statistical indistinguishability of the two models from one sample. It is symmetric, bounded, and transfers across covariate shift additively (Proposition 1.18). Token-level criteria are *surrogates*; they relate to the target only through Theorem 1.12 and Proposition 1.14.
2. *TV alone is not enough when probabilities are consumed or events are rare.* If the student’s likelihoods are used (scoring, calibration, uncertainty, perplexity), the relevant quantity is the log-loss regret, which is *exactly* $\mathbb{E}_\mu KL(P^x \| Q_\theta^x)$. If the student is sampled at scale and rare events matter (hallucinations, unsafe outputs), the precision bound of Proposition 1.10(ii–iii) requires the *reverse* direction. Hence two-sided losslessness. If decoding is greedy or beam search, margins matter too (Proposition 1.11).
3. *Among trainable surrogates, sequence-level forward KL is canonical.* It dominates behavioural loss via Pinsker and gives log-loss exactly. By Theorem 1.12 its per-token decomposition uses *teacher prefixes only*, so it is estimable without student roll-outs and has no exposure-bias term (Section 2.5). It is convex in the logits (Proposition 2.11). More generally, KL is the only divergence on the simplex that is both an f -divergence and a Bregman divergence (Amari, 2009). That characterisation singles out the KL *family*, not a direction: both argument orders qualify.
4. *But when exactness is impossible, the surrogate must be chosen for the target.* Under misspecification, minimisers of either KL direction can be arbitrarily bad in TV, whereas minimisers of a bounded divergence such as JSD_β are TV-competitive (Proposition 4.13). This is why the objective proposed in Section 4 is a *sequence-level* JSD_β with an optional forward-KL anchor.
5. *In the truly near-lossless regime the question dissolves.* By Proposition 1.8, all smooth divergences agree to second order when likelihood ratios are bounded. The practically important choice is then the *prefix law* and the *estimator*, not the divergence.

In one sentence: a student should be called ε -lossless if it is two-sided lossless at level ε under the deployment input law (behavioural losslessness at level $\sqrt{\varepsilon/2}$ then follows). It should be trained with a sequence-level divergence whose minimiser remains TV-competitive when ε -losslessness is out of reach.

2 Decomposing the student’s gap

2.1 Pipeline and master decomposition

Pipeline. We draw n prompts $x_i \sim \mathcal{D}$ i.i.d. For each prompt the teacher generates $y^{(i)} \sim P^{x_i}$ and supplies either soft labels $p_t(\cdot \mid s_t^{(i)})$ at every position, or only the sampled tokens (hard labels). On-policy variants additionally sample $y \sim Q_\theta^{x_i}$. Training uses a prefix law $\nu(\theta) = (\nu_t(\theta))_t$: $\nu_t = d_t^p$ (off-policy), $\nu_t = d_t^\theta$ (on-policy), or a mixture $\nu_t = \lambda d_t^\theta + (1 - \lambda)d_t^p$ (Agarwal et al., 2024). The per-state loss is $\ell_\theta(s) = D(p_t(\cdot \mid s) \parallel q_t(\cdot \mid s))$. The population and empirical training objectives are

$$F(\theta) := \mathbb{E}_{x \sim \mathcal{D}} \sum_{t=1}^T \mathbb{E}_{s \sim \nu_t(\theta) \mid x} \ell_\theta(s), \quad \hat{F}_n(\theta) := \frac{1}{n} \sum_{i=1}^n \sum_{t=1}^T \ell_\theta(s_t^{(i)}).$$

For on-policy terms, $\hat{F}_n$ is defined with fixed sampling noise (for example Gumbel-max reparametrisation), so that it is a fixed random function of θ . Let $\theta^* \in \arg \min_{\Theta_S} F$, $\hat{\theta} \in \arg \min \hat{F}_n$, and let $\tilde{\theta}$ be the output of the optimiser. The *evaluation functional* $\mathcal{E}_\mu(\theta)$ is a sequence-level criterion under the deployment law, for example $\mathbb{E}_\mu \text{KL}(P^x \parallel Q_\theta^x)$ or a task regret. $\mathcal{E}_\mathcal{D}$ is the same functional under $\mathcal{D}$.

Theorem 2.1 (Master decomposition). *[proved]*

$$\mathcal{E}_\mu(\tilde{\theta}) = \underbrace{F(\theta^*)}_A \text{ approximation} + \underbrace{F(\tilde{\theta}) - F(\theta^*)}_{\text{estimation+optimisation}} + \underbrace{\mathcal{E}_\mathcal{D}(\tilde{\theta}) - F(\tilde{\theta})}_{\mathcal{E}_{\text{shift}} \text{ prefix shift}} + \underbrace{\mathcal{E}_\mu(\tilde{\theta}) - \mathcal{E}_\mathcal{D}(\tilde{\theta})}_{\mathcal{E}_{\text{cov}} \text{ covariate shift}}, \quad (4)$$

$$F(\tilde{\theta}) - F(\theta^*) \leq \underbrace{2 \sup_{\theta} |\hat{F}_n(\theta) - F(\theta)|}_{\mathcal{E}_{\text{est}}} + \underbrace{\hat{F}_n(\tilde{\theta}) - \hat{F}_n(\hat{\theta})}_{\mathcal{E}_{\text{opt}}}. \quad (5)$$

Proof. (4) telescopes. For (5), write $F(\tilde{\theta}) - F(\theta^*) = [F - \hat{F}_n](\tilde{\theta}) + [\hat{F}_n(\tilde{\theta}) - \hat{F}_n(\hat{\theta})] + [\hat{F}_n(\hat{\theta}) - \hat{F}_n(\theta^*)] + [\hat{F}_n - F](\theta^*)$, and note that the third bracket is ≤ 0 . This is the approximation–estimation–optimisation split of Bottou and Bousquet (2008) with two shift terms added. $\square$

When $\mathcal{E}$ is a KL, the behavioural criterion follows from Pinsker: $\mathbb{E}_\mu \text{TV}(P^x, Q_\theta^x) \leq \sqrt{\mathcal{E}_\mu(\tilde{\theta})/2}$. The key structural fact is that $\mathcal{E}_{\text{shift}} \equiv 0$ for exactly two pairings (Theorem 1.12). The table lists what each pairing’s training objective F equals and the resulting bound on sequence-level behaviour:

per-token loss	prefix law	F equals	$\mathbb{E}_\mathcal{D} \text{TV}(P, Q_\theta) \leq$
forward KL	teacher	$\mathbb{E}_\mathcal{D} \text{KL}(P \parallel Q_\theta)$; $\mathcal{E}_{\text{shift}} = 0$	$\sqrt{F/2}$
reverse KL	student	$\mathbb{E}_\mathcal{D} \text{KL}(Q_\theta \parallel P)$; $\mathcal{E}_{\text{shift}} = 0$	$\sqrt{F/2}$
forward KL	student	hybrid (Proposition 1.15)	$\sqrt{TF/2}$
reverse KL	teacher	hybrid (Proposition 1.15)	$\sqrt{TF/2}$
TV	either	hybrid (Proposition 1.14)	F
JSD_β	either	hybrid	$\sqrt{TF/(2\beta(1-\beta))}$

For the hybrid rows, the factor T inside the square root comes from our Cauchy–Schwarz argument. A sequential subadditivity inequality for squared Hellinger distance (Foster et al., 2021) improves it to $O(\log T)$; we have not re-derived that lemma’s constant. For hybrid rows, the on-policy per-token loss can also exceed the training loss. For bounded losses the excess is at most $G \sum_t \sum_{k < t} e_k^{\text{off,TV}}$, which is $O(T^2)$ in the per-token TV error (Proposition 2.19). For reverse KL on teacher prefixes it is unbounded relative to the training loss.

2.2 Approximation error

Lemma 2.2 (Log-ratio variance). *[proved] [numerically checked] Let $\delta = \log(p/q)$ on $\text{supp } p$ and $M := \max\{0, \max_y \delta(y)\}$. Then $\text{KL}(p\|q) \geq \frac{e^{-M}}{2} \mathbb{E}_p[\delta^2] \geq \frac{e^{-M}}{2} \text{Var}_p(\delta)$.*

Proof. $\mathbb{E}_p[e^{-\delta}] = q(\text{supp } p) \leq 1$. Taylor expansion gives $e^{-\delta} = 1 - \delta + \frac{\delta^2}{2}e^{-\xi}$ with ξ between 0 and δ , so $e^{-\xi} \geq e^{-M}$. Taking $\mathbb{E}_p$: $1 \geq 1 - \text{KL}(p\|q) + \frac{e^{-M}}{2} \mathbb{E}_p[\delta^2]$. $\square$

Theorem 2.3 (Softmax-bottleneck approximation floor). *[proved under Assumption A5] [numerically checked] Let $s_1, \dots, s_N$ be contexts at which the teacher has full support. Let $A \in \mathbb{R}^{N \times V}$ have rows $\log p(\cdot | s_i)$, let $\pi_i := \min_y p(y | s_i)$, and let $\Pi = \text{diag}(\pi_i)$. For every student of width d_S with $\max_{i,y} \log \frac{p(y|s_i)}{q_\theta(y|s_i)} \leq M$,*

$$\frac{1}{N} \sum_{i=1}^N \text{KL}(p(\cdot | s_i) \| q_\theta(\cdot | s_i)) \geq \frac{e^{-M}}{2N} \sum_{j > d_S + 1} \sigma_j^2(\Pi^{1/2} AC).$$

Proof. By Lemma 2.2, $\text{KL}_i \geq \frac{e^{-M}}{2} \text{Var}_{p_i}(\delta_i)$. Also $\text{Var}_{p_i}(\delta_i) = \min_c \mathbb{E}_{p_i}(\delta_i - c)^2 \geq \pi_i \min_c \|\delta_i - c\mathbf{1}\|^2 = \pi_i \|C\delta_i\|^2$. The student’s log-probability rows are $Wh_i + b - \text{lse}_i \mathbf{1}$, so the centred matrix ZC has rows $C(Wh_i + b)$ and rank at most $d_S + 1$. Left-multiplication by the diagonal $\Pi^{1/2}$ does not increase rank. Hence $\sum_i \pi_i \|C\delta_i\|^2 = \|\Pi^{1/2}(A - Z)C\|_F^2 \geq \min_{\text{rank}(Y) \leq d_S + 1} \|\Pi^{1/2}AC - Y\|_F^2$, which equals the tail sum by Eckart–Young–Mirsky (Eckart and Young, 1936). $\square$

Remark 2.4. (i) If the teacher also has a linear read-out of width d_T , then $\text{rank}(AC) \leq d_T + 1$. The floor is then a sum over the teacher’s singular values with indices in $(d_S + 1, d_T + 1]$, and it vanishes if $d_S \geq d_T$. This quantifies the “softmax bottleneck” (Yang et al., 2018). Its relevance to small language models was studied empirically by Godey et al. (2024); we are fairly, but not fully, confident of that reference’s exact claims. (ii) The factor π_i is crude. Replacing $\Pi^{1/2}C$ by any common lower bound $\Gamma^{1/2}$ on the Fisher matrices $\text{diag}(p_i) - p_i p_i^\top$ gives the tail spectrum of $A\Gamma^{1/2}$. In the check of `verification/check_capacity.py`, the row-weighted floor is within a factor 15 of a fitted student’s loss. (iii) The floor is a statement about the *read-out*. When $d_S \geq d_T$ it is silent; information-theoretic floors (Section 3) take over.

Corollary 2.5 (Approximation grows linearly with the horizon). *[proved] If the per-token approximation floor on teacher prefixes is $\geq \varepsilon_0$ at every t , then $\min_\theta \mathbb{E}_D \text{KL}(P\|Q_\theta) \geq T\varepsilon_0$. Likelihood losslessness at level ε therefore requires $\varepsilon_0 \leq \varepsilon/T$. For TV, Proposition 1.14 gives only $\mathbb{E} \text{TV}(P, Q_\theta) \geq \frac{1}{2} \max_t \mathbb{E}_{d_t^p} \text{TV}_t$. Whether TV grows with T depends on how the errors are structured (Example 1.17).*

2.3 Estimation error

Theorem 2.6 (Uniform deviation, soft-label off-policy KD). *[proved under Assumption A1] With soft labels, $\ell(x, y; \theta) = \sum_t \text{KL}(p_t(s_t) \| q_t(s_t)) \in [0, TB]$, since $\text{KL}(p\|q) \leq \mathbb{E}_p[-\log q] \leq B$. With probability $\geq 1 - \delta$,*

$$\sup_\theta |\hat{F}_n(\theta) - F(\theta)| \leq 2\mathfrak{R}_n(\ell \circ \Theta_S) + TB \sqrt{\frac{\log(2/\delta)}{2n}}.$$

For a finite class (Assumption A4), $\mathfrak{R}_n \leq TB \sqrt{2R \ln 2/n}$ by Massart’s lemma.

Proof. Symmetrisation and McDiarmid (Bartlett and Mendelson, 2002). $\square$

This bound is pessimistic: a slow rate, and linear in T . For the realisable case, a horizon-free fast rate holds even with *hard* labels. The logic is due to Zhang (2006) and Wong and Shen (1995); its horizon-independence in imitation learning was emphasised by Foster et al. (2024).

Theorem 2.7 (Horizon-free rate for sequence-level hard-label KD). *[proved under Assumptions A3 and A4] Let $\hat{\theta}$ maximise $\sum_{i=1}^n \log Q_{\theta}(y^{(i)} | x_i)$ over Θ_S , with $y^{(i)} \sim P^{x_i}$. With probability $\geq 1 - \delta$,*

$$\mathbb{E}_{x \sim \mathcal{D}} H^2(P^x, Q_{\hat{\theta}}^x) \leq \frac{\log(|\Theta_S|/\delta)}{n} \leq \frac{R \ln 2 + \log(1/\delta)}{n},$$

$$\mathbb{E}_{x \sim \mathcal{D}} \text{TV}(P^x, Q_{\hat{\theta}}^x) \leq \sqrt{\frac{2(R \ln 2 + \log(1/\delta))}{n}}.$$

Neither bound depends on T .

Proof. See Appendix A.1. The key lemma is $\mathbb{E}_{y \sim P} \sqrt{Q(y)/P(y)} = 1 - H^2(P, Q) \leq e^{-H^2}$, combined with a Chernoff bound and a union bound over Θ_S .

Proposition 2.8 (Soft labels Rao–Blackwellise hard labels). *[proved] Fix a state law ν with states drawn independently of labels. Let $\hat{h} = -\log q_{\theta}(y | s)$ and $\hat{s} = \text{CE}(p(\cdot | s), q_{\theta}(\cdot | s))$, where $s \sim \nu$ and $y \sim p(\cdot | s)$. Then $\mathbb{E}\hat{h} = \mathbb{E}\hat{s}$ and*

$$\text{Var } \hat{h} - \text{Var } \hat{s} = \mathbb{E}_{s \sim \nu} \text{Var}_{y \sim p(\cdot | s)} \log q_{\theta}(y | s) \geq 0,$$

$$\mathbb{E} \|\nabla \hat{h} - \nabla \mathbb{E}\hat{h}\|^2 - \mathbb{E} \|\nabla \hat{s} - \nabla \mathbb{E}\hat{s}\|^2 = \mathbb{E}_{\nu} \text{tr Cov}_{y \sim p(\cdot | s)} \nabla_{\theta} \log q_{\theta}(y | s).$$

At $\theta = \theta^{\circ}$ (Assumption A3), the gradient gap equals $\mathbb{E}_{\nu} \text{tr } \mathcal{I}(s)$. Here $\mathcal{I}(s) = J(s)^{\top} (\text{diag } p - pp^{\top}) J(s)$ is the per-state Fisher information and $J = \partial z / \partial \theta$. The gap vanishes wherever $p(\cdot | s)$ is a point mass.

Proof. $\hat{s} = \mathbb{E}[\hat{h} | s]$, so the law of total variance applies; likewise for the gradient, since $\nabla \hat{s} = \mathbb{E}[\nabla \hat{h} | s]$. At θ° , $\text{Cov}_{y \sim p} \nabla \log q = J^{\top} \text{Cov}_{y \sim p}(e_y) J = J^{\top} (\text{diag } p - pp^{\top}) J$. $\square$

Remark 2.9. For whole teacher-sampled sequences, later states depend on earlier labels. Position-wise Rao–Blackwellisation still holds, but cross-time covariances can in principle make the variance of the *sum* larger. We therefore claim no sequence-level variance ordering. The variance-reduction view of distillation is developed by Menon et al. (2021). Related analyses include Lopez-Paz et al. (2016), Phuong and Lampert (2019), Dao et al. (2021) and Harutyunyan et al. (2023).

Proposition 2.10 (Soft labels: exact recovery versus $k/2n$). *Let $q_{\theta}(\cdot | s) = \text{softmax}(\Theta \phi(s))$ with $\Theta \in \mathbb{R}^{V \times d}$, and let $p = q_{\Theta^{\circ}}$.*

- (a) *[proved] If $\phi(s_1), \dots, \phi(s_n)$ span $\mathbb{R}^d$ (so $n \geq d$), every minimiser of $\sum_i \text{KL}(p(\cdot | s_i) \| q_{\theta}(\cdot | s_i))$ satisfies $q_{\theta} = p$ at every s . Estimation error is zero.*
- (b) *[known result; proof sketch] If $\mathbb{E}[\phi \phi^{\top}] \succ 0$, the hard-label MLE from n i.i.d. pairs (s_i, y_i) satisfies $n \text{KL}(p \| q_{\hat{\theta}}) \Rightarrow \frac{1}{2} \chi_{(V-1)d}^2$ in distribution. The limiting mean is $\frac{(V-1)d}{2}$. This is the classical $k/2n$ law for a regular, well-specified model with k free parameters, the computation behind AIC (Akaike, 1974); see White (1982) for the misspecified case. The expectation $\mathbb{E} \text{KL}$ is infinite for every n , because with positive probability the MLE does not exist or puts zero mass on an observed class. The $k/2n$ law is therefore a statement about the limit law, not about $\mathbb{E} \text{KL}$.*

Proof of (a). The objective is 0 at Θ° . It vanishes at Θ iff $(\Theta - \Theta^{\circ})\phi(s_i) \in \text{span}(\mathbf{1})$ for all i , which means $C(\Theta - \Theta^{\circ})\phi(s_i) = 0$. By spanning, $C(\Theta - \Theta^{\circ}) = 0$. So the logits differ by a constant vector at every s , and the softmax is unchanged. $\square$

In the realisable case, a soft label is therefore worth “infinitely many” hard labels at its state. Once the teacher’s labels are exact, what remains is extrapolation, which is controlled by coverage of the feature space and by approximation, not by label noise.

2.4 Optimisation error

Proposition 2.11 (Curvature in logit space). *[proved] [numerically checked]*

- (i) $z \mapsto \text{KL}(p \parallel \text{softmax } z)$ is convex. Its Hessian satisfies $0 \preceq \text{diag}(q) - qq^\top \preceq \frac{1}{2}C$ (Böhning, 1992), so it is $\frac{1}{2}$ -smooth.
- (ii) $z \mapsto \text{KL}(\text{softmax } z \parallel p)$ is not convex, for any $V \geq 2$ and any p of full support. With $V = 2$, $f(z) = \text{KL}(\sigma(z) \parallel \text{Bern}(p))$ has $f''(z) = \sigma'(z) [1 + (1 - 2\sigma(z))(z - z_p)]$ with $z_p = \text{logit } p$, which is negative whenever $(2\sigma(z) - 1)(z - z_p) > 1$; for example $p = 0.3$, $z = z_p + 3$. For general V , along the ray $z = (t, 0, \dots, 0)$ the second derivative is $-(V - 1)(t - 2)e^{-t} + O(e^{-t}) < 0$ for large t .
- (iii) If p has full support, then for β close to 1, $z \mapsto \text{JSD}_\beta(p \parallel \text{softmax } z)$ is not convex: $\text{JSD}_\beta/(1 - \beta) \rightarrow \text{KL}(q \parallel p)$ together with its second derivatives, uniformly on compacts.
- (iv) Even with logits linear in a scalar θ , reverse KL can have spurious local minima. Take two contexts, $V = 2$, logits $z_1 = 5.2 - \theta/2$ and $z_2 = -1.5 - \theta/3$, and teacher probabilities 0.79 and 0.26 for the first token. The summed reverse KL has strict local minima at $\theta \approx -1.00$ (value 0.2191) and $\theta \approx 7.23$ (value 0.2300), separated by a barrier of height 0.2678 at $\theta \approx 3.61$. The forward KL has a single minimum.

Consequently, for any student whose logits are linear in θ (for example, training only the read-out of Assumption A5), off-policy forward-KL distillation is a convex problem. Reverse-KL objectives can have spurious local minima even then (iv).

Proof. (i) The Hessian of lse is $\text{diag}(q) - qq^\top$; Böhning’s bound gives the smoothness constant. (ii) $f'(z) = \sigma(z)(1 - \sigma(z))(z - z_p)$, because $\log \frac{q}{1-q} = z$; differentiate once more. On the ray, $q_j = e^{-t}(1 + O(e^{-t}))$ for $j > 1$, and the reverse KL equals $\log \frac{1}{p_1} - (V - 1)te^{-t} + O(e^{-t})$; differentiate twice. (iii) Write $\text{JSD}_\beta/(1 - \beta) = \text{KL}(q \parallel M_\beta) + \frac{\beta}{1-\beta} \text{KL}(p \parallel M_\beta)$. Since $p > 0$, $M_\beta \geq \beta p$ is bounded away from 0, so $\text{KL}(q \parallel M_\beta)$ is jointly smooth in (β, z) up to $\beta = 1$, where it equals $\text{KL}(q \parallel p)$. Let $\epsilon = 1 - \beta$ and $g(s) := \text{KL}(p \parallel p + s(q - p))$. Then $g(0) = g'(0) = 0$, so $g(\epsilon)/\epsilon = \epsilon \int_0^1 (1-t) g''(\epsilon t) dt$. This is jointly smooth in (ϵ, z) and vanishes at $\epsilon = 0$ together with its z -derivatives. Hence second z -derivatives converge uniformly on compacts, and the strict negative curvature of (ii) persists for β near 1. (iv) is a direct computation (`check_divergences.py`). $\square$

Theorem 2.12 (Stochastic optimisation). *[known result; proof sketch]* Under Assumption A6, SGD with a tuned constant step size satisfies $\min_{k < K} \mathbb{E} \|\nabla F(\theta_k)\|^2 \lesssim \frac{L\Delta}{K} + \sigma \sqrt{\frac{L\Delta}{K}}$ with $\Delta = F(\theta_0) - \inf F$ (Ghadimi and Lan, 2013). If F is convex (Proposition 2.11), then $\mathbb{E} F(\bar{\theta}_K) - \min F \lesssim \frac{L\|\theta_0 - \theta^*\|^2}{K} + \frac{\sigma\|\theta_0 - \theta^*\|}{\sqrt{K}}$. In both cases σ^2 enters, so the Rao–Blackwell gap of Proposition 2.8 translates directly into lower optimisation error at fixed compute.

Remark 2.13. No non-convex guarantee beyond stationarity is available for deep students. Empirically, optimisation is often the binding constraint on fidelity. Stanton et al. (2021) find that students frequently fail to match their teachers even when they have the capacity to do so. Beyer et al. (2022) find that long, “patient and consistent” schedules are needed. We treat $\mathcal{E}_{\text{opt}}$ as measurable (Section 6.3), not as provably small.

2.5 Distribution shift and exposure bias

Let $c_t : \mathcal{S}_t \times \mathcal{Y} \rightarrow [0, 1]$ be per-step costs and $J_c(\pi) := \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi} \sum_{t=1}^T c_t(s_t, y_t)$. Let $r : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$ be a terminal reward and $J_r(\pi) := \mathbb{E} r(x, y)$. The teacher’s cost-to-go is

$Q_t^p(s, a) := c_t(s, a) + \mathbb{E}_p[\sum_{k>t} c_k \mid s_t = s, y_t = a]$, and its *recoverability constant* is

$$u := \max_t \sup_s \left[\max_a Q_t^p(s, a) - \min_a Q_t^p(s, a) \right] \in [0, T].$$

Theorem 2.14 (Terminal rewards: linear in T). *[proved]* $|J_r(p) - J_r(q_\theta)| \leq \mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_\theta) \leq T \min\{\varepsilon_{\text{off}}^{\text{TV}}, \varepsilon_{\text{on}}^{\text{TV}}\}$. Also $|J_r(p) - J_r(q_\theta)| \leq \sqrt{T \varepsilon_{\text{off}}^{\text{KL}}/2}$.

Proof. Propositions 1.4 and 1.14 and Corollary 1.16. $\square$

Theorem 2.15 (Cumulative costs, off-policy: quadratic, and tight). *[proved] [numerically checked]*

$$|J_c(p) - J_c(q_\theta)| \leq \sum_{t=1}^T (T - t + 1) e_t^{\text{off,TV}} \leq T^2 \varepsilon_{\text{off}}^{\text{TV}}.$$

The order is attained. Take a deterministic teacher emitting a^ in every “on-track” state. Let every deviation lead to absorbing off-track states that cost 1 per step. Let the student deviate with probability ε on track. Then $\varepsilon_{\text{off}}^{\text{TV}} = \varepsilon$ and $J_c(q_\theta) - J_c(p) = \sum_{t=1}^T [1 - (1 - \varepsilon)^t] = \varepsilon \frac{T(T+1)}{2} (1 + O(\varepsilon T))$.*

Proof. $|\mathbb{E}_p c_t - \mathbb{E}_{q_\theta} c_t| \leq \text{TV}$ between the laws of $y_{\leq t}$, which is at most $\sum_{k \leq t} e_k^{\text{off,TV}}$ by the hybrid argument (Proposition 1.14) applied to the first t tokens. Sum over t . In the example, the student is on track at step t with probability $(1 - \varepsilon)^{t-1}$. $\square$

Theorem 2.16 (On-policy: linear with recoverability). *[proved] [numerically checked]* The performance-difference identity (Kakade and Langford, 2002) holds:

$$J_c(q_\theta) - J_c(p) = \sum_{t=1}^T \mathbb{E}_{s \sim d_t^\theta} \left[\mathbb{E}_{a \sim q_t(\cdot|s)} Q_t^p(s, a) - V_t^p(s) \right], \quad V_t^p(s) := \mathbb{E}_{a \sim p_t(\cdot|s)} Q_t^p(s, a).$$

Hence $|J_c(q_\theta) - J_c(p)| \leq u T \varepsilon_{\text{on}}^{\text{TV}}$ and, by Pinsker per token and Jensen, $|J_c(q_\theta) - J_c(p)| \leq u T \sqrt{\varepsilon_{\text{on}}^{\text{KL}}/2}$.

Proof. Telescope $V_t^p(s) := \mathbb{E}_{a \sim p} Q_t^p(s, a)$ along the student’s trajectory. For any f with range at most u , $|\mathbb{E}_q f - \mathbb{E}_p f| \leq u \text{TV}(p, q)$. $\square$

Remark 2.17 (What on-policy training buys). Theorems 2.15 and 2.16 are the distillation versions of Ross and Bagnell (2010) and Ross et al. (2011). On-policy training replaces $T^2 \varepsilon_{\text{off}}$ by $u T \varepsilon_{\text{on}}$. It removes compounding only if (a) the teacher is *recoverable* ($u = O(1)$), and (b) training actually makes ε_{on} small, which DAgger-type no-regret arguments deliver relative to the best student *on its own state distribution*. In the non-recoverable cliff of Theorem 2.15, $u = T$, and no training distribution avoids $\Theta(T^2 \varepsilon)$. For language models, u is the worst-case cost of one bad token followed by the teacher taking over. It is small for fluency-type costs and can be $\Theta(T)$ for commitments such as a wrong early step in a proof. Related analyses of exposure bias include Bengio et al. (2015), Ranzato et al. (2016) and Arora et al. (2022). On-policy distillation is studied by Lin et al. (2020), Gu et al. (2024), Agarwal et al. (2024) and Ko et al. (2024).

Theorem 2.18 (KL chain rules and compounding). *[proved]*

- (i) Off-policy forward KL has $\mathcal{E}_{\text{shift}} = 0$ for $\mathcal{E} = \text{KL}(P \| Q_\theta)$, and $\mathbb{E} \text{TV}(P, Q_\theta) \leq \sqrt{T \varepsilon_{\text{off}}^{\text{KL}}/2}$.
- (ii) For cumulative costs, $|J_c(p) - J_c(q_\theta)| \leq \sum_{t=1}^T \sqrt{\frac{1}{2} \sum_{k \leq t} e_k^{\text{off,KL}}} \leq T^{3/2} \sqrt{\varepsilon_{\text{off}}^{\text{KL}}/2}$.
- (iii) On-policy reverse KL has $\mathcal{E}_{\text{shift}} = 0$ for $\mathcal{E} = \text{KL}(Q_\theta \| P)$. With recoverability, $|J_c(q_\theta) - J_c(p)| \leq u T \sqrt{\varepsilon_{\text{on}}^{\text{KL}}/2}$.

Proof. (i) and (iii) are Theorem 1.12. (ii) The law of $y_{\leq t}$ has KL $\sum_{k \leq t} e_k^{\text{off,KL}}$ by the truncated chain rule; apply Pinsker per t . The last claim is Theorem 2.16. $\square$

Bound (ii) improves on Theorem 2.15 only when errors are *diffuse* (Example 1.17(a)): there $e^{\text{off,TV}} \approx \sqrt{e^{\text{off,KL}}}$, so $T^{3/2}\sqrt{\varepsilon}$ beats $T^2\sqrt{\varepsilon}$. For concentrated errors (deterministic teacher, Example 1.17(b)) the TV route is tighter. In the cliff with $T = 100$ and $\varepsilon = 10^{-3}$, the true gap is 4.9, Theorem 2.15 gives 5.05, and (ii) gives 15.0.

Proposition 2.19 (Exposure-bias gap for mismatched pairings). *[proved] [numerically checked] Let $g_t : \mathcal{S}_t \rightarrow [0, G]$ be any bounded per-state loss, for example per-token TV ($G = 1$), JSD $_\beta$ ($G = h(\beta)$), or clipped KL. Then*

$$\left| \sum_{t=1}^T (\mathbb{E}_{d_t^\theta} - \mathbb{E}_{d_t^p}) g_t \right| \leq G \sum_{t=1}^T \text{TV}(d_t^\theta, d_t^p) \leq G \sum_{t=1}^T \sum_{k < t} e_k^{\text{off,TV}} \leq G \frac{T(T-1)}{2} \max_k e_k^{\text{off,TV}}.$$

Also $\leq G \sum_t \sqrt{\frac{1}{2} \sum_{k < t} e_k^{\text{off,KL}}}$. For a mixture prefix law $\nu_t = \lambda d_t^\theta + (1-\lambda)d_t^p$, the gap is multiplied by $1-\lambda$. For unbounded g_t , for example reverse KL on teacher prefixes, no bound in terms of the training loss exists. The training loss can be arbitrarily small while the on-policy loss is arbitrarily large, on states the teacher visits rarely but the student visits often. With $T = 2$, $p_1 = (1-\eta, \eta)$, $q_1 = (1-r\eta, r\eta)$, reverse KL $r^{3/2}$ at the rare state and $\eta = r^{-2}$, the training loss is $O(r^{-1/2})$ while the on-policy loss is $\Theta(r^{1/2})$. Exactly zero training loss does force $Q_\theta^x = P^x$ (Proposition 1.15).

Proof. Bounded-function duality for TV, then the hybrid argument for $\text{TV}(d_t^\theta, d_t^p)$ as in Proposition 1.14. $\square$

Proposition 2.20 (Covariate-shift term). *[proved] Let $\mathcal{E}(\theta) = \mathbb{E}_x e_\theta(x)$ with $e_\theta \geq 0$ pointwise, for example a per-input divergence. Under Assumption A2, $\mathcal{E}_{\text{cov}} \leq (\rho - 1)\mathcal{E}_\mathcal{D}(\tilde{\theta})$. If $0 \leq e_\theta \leq G$, then $\mathcal{E}_{\text{cov}} \leq G \text{TV}(\mu, \mathcal{D})$ (Proposition 1.18). Pointwise non-negativity is needed: a task regret can be negative at some inputs, and then the first bound fails.*

2.6 Assembling the bound, and which term dominates

Corollary 2.21 (End-to-end bound for off-policy soft-label forward-KL distillation). *[proved under Assumptions A1 and A2] Let $\mathcal{A} = \min_\theta \mathbb{E}_\mathcal{D} \text{KL}(P \| Q_\theta)$. With probability $\geq 1 - \delta$,*

$$\mathbb{E}_\mu \text{TV}(P^x, Q_\theta^x) \leq \sqrt{\frac{\rho}{2} \left[\mathcal{A} + 2 \left(2\mathfrak{R}_n + TB \sqrt{\frac{\log(2/\delta)}{2n}} \right) + \mathcal{E}_{\text{opt}} \right]}.$$

Under Assumptions A3 and A4, with hard labels and exact ERM, instead $\mathbb{E}_\mu \text{TV} \leq \min \{ \rho \eta_n, \eta_n + \text{TV}(\mu, \mathcal{D}) \}$ with $\eta_n = \sqrt{2(R \ln 2 + \log(1/\delta))/n}$.

Proof. Theorem 2.1 with $\mathcal{E}_{\text{shift}} = 0$ (Theorem 1.12), Theorem 2.6, Proposition 2.20, and Pinsker. For the second claim use Theorem 2.7 and Proposition 1.18. $\square$

Heuristic 2.22 (The dominant term in large-scale LLM distillation). *(1) $\mathcal{E}_{\text{est}}$ is small. Teacher data is effectively unlimited, soft labels remove label noise (Propositions 2.8 and 2.10), and realisable rates are horizon-free (Theorem 2.7). (2) $\mathcal{E}_{\text{opt}}$ is real but shrinks with compute (Theorem 2.12); it dominates only for undertrained students. (3) $\mathcal{A}$ is the floor. It dominates for small students (Theorem 2.3; Section 3) and, at sequence level, grows linearly in T (Corollary 2.5). (4) $\mathcal{E}_{\text{shift}}$ is zero for the exact pairings, but for mismatched pairings and cumulative criteria it amplifies $\mathcal{A}$ by up to $O(T)$ (Theorem 2.15 and Proposition 2.19). Hence in the capacity-limited regime the dominant term is approximation error as seen by the deployment criterion: the irreducible floor, amplified by prefix shift and misallocated by the choice of divergence (Proposition 4.13). Section 4 designs an objective against exactly this term.*

3 Lower bounds: is lossless even possible?

Framing. For a *fixed* teacher, whether lossless distillation is possible is purely a question of approximation theory: is p in the closure of $\{q_\theta\}$? Information theory enters once we ask what a class of 2^R students can do *uniformly over teachers*, either on average under a prior π over teachers or in the worst case over a class. We then view the student as a code of R bits for the teacher’s conditional law, with distortion $\mathbb{E}_x \text{KL}(p(\cdot | x) \| q_\theta(\cdot | x))$. The random teacher is denoted $\mathbf{T} \sim \pi$, and its conditionals $p_{\mathbf{T}}(\cdot | x)$.

3.1 An information-theoretic capacity floor

Theorem 3.1 (Capacity floor). *[proved under Assumption A4] [numerically checked] Let $\theta^*(\mathbf{T})$ be any measurable choice of student from a class of at most 2^R elements. Fix a design $x_{1:N} \in \mathcal{X}^N$ and a replication number $m \geq 1$. Let $Y = (Y_{ij})_{i \leq N, j \leq m}$ with $Y_{ij} \sim p_{\mathbf{T}}(\cdot | x_i)$ independent given $\mathbf{T}$. Then*

$$\mathbb{E}_\pi \frac{1}{N} \sum_{i=1}^N \text{KL}(p_{\mathbf{T}}(\cdot | x_i) \| q_{\theta^*(\mathbf{T})}(\cdot | x_i)) \geq \frac{I(\mathbf{T}; Y) - R \ln 2}{mN}.$$

In particular: (a) if $\mathcal{D}$ is uniform on $\{x_1, \dots, x_N\}$, the left side is $\mathbb{E}_\pi \mathbb{E}_{x \sim \mathcal{D}} \text{KL}$; (b) for general $\mathcal{D}$, $\mathbb{E}_\pi \mathbb{E}_{x \sim \mathcal{D}} \text{KL} \geq \sup_{N,m} \frac{I(\mathbf{T}; Y | X_{1:N}) - R \ln 2}{mN}$ with $X_i \stackrel{iid}{\sim} \mathcal{D}$.

Proof. Write $\Theta = \theta^*(\mathbf{T})$. KL tensorises, so $m \sum_i \text{KL}(p_{\mathbf{T}}(\cdot | x_i) \| q_\Theta(\cdot | x_i)) = \text{KL}(\otimes_{ij} p_{\mathbf{T}}(\cdot | x_i) \| \otimes_{ij} q_\Theta(\cdot | x_i)) = \mathbb{E}[-\log q_\Theta(Y) | \mathbf{T}] - H(Y | \mathbf{T})$. Average over $\mathbf{T}$. Since q_Θ is a valid conditional law of Y given Θ , the cross-entropy satisfies $\mathbb{E}[-\log q_\Theta(Y)] \geq H(Y | \Theta)$. Hence

$$m \mathbb{E} \sum_i \text{KL}_i \geq H(Y | \Theta) - H(Y | \mathbf{T}) = I(Y; \mathbf{T}) - I(Y; \Theta) \geq I(Y; \mathbf{T}) - H(\Theta) \geq I(Y; \mathbf{T}) - R \ln 2,$$

where the equality uses $H(Y)$ on both sides and the fact that Θ is a function of $\mathbf{T}$. For (b), apply the bound conditionally on $X_{1:N}$ and average, since $\mathbb{E}_{X_{1:N}} \frac{1}{N} \sum_i \text{KL}(\cdot; X_i) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}$. $\square$

Let $I_\infty := \sup_{N,m} I(\mathbf{T}; Y | X_{1:N})$ be the *functional information* of the teacher on $\mathcal{D}$: how much its input–output behaviour reveals about which teacher it is. Theorem 3.1 says that on average over π , lossless distillation is *impossible* whenever $R \ln 2 < I_\infty$. The student’s description length must be at least the teacher’s functional information. This is not the teacher’s parameter count: trained networks are highly redundant as functions, which is what makes pruning, quantisation and distillation work at all.

Corollary 3.2 (K independent facts; the floor is tight). *[proved] [numerically checked] Let the teacher be a uniformly random function $F : [K] \rightarrow \mathcal{V}$ with $p(\cdot | i) = \delta_{F(i)}$, and let $\mathcal{D}$ be uniform on $[K]$. Every R -bit student class satisfies $\mathbb{E} \mathbb{E}_{x \sim \mathcal{D}} \text{KL} \geq (\ln V - \frac{R \ln 2}{K})_+$. The student that memorises $\lfloor R / \log_2 V \rfloor$ facts and is uniform elsewhere attains this up to rounding.*

Proof. Take $N = K$, $x_i = i$, $m = 1$: $I(F; Y) = K \ln V$. The memorising student has $\text{KL} \ln V$ on each fact it does not store and 0 on the others. $\square$

Corollary 3.3 (Behavioural floor for K facts). *[proved] In the setting of Corollary 3.2 with $V \geq 2$, let $\bar{e} := \mathbb{E} \mathbb{E}_{x \sim \mathcal{D}} \text{TV}(p(\cdot | x), q_{\theta^*}(\cdot | x))$. Every R -bit student class satisfies*

$$h(\bar{e}) + \bar{e} \ln(V - 1) \geq \ln V - \frac{R \ln 2}{K}.$$

The left side increases from 0 to $\ln V$ as $\bar{e}$ goes from 0 to $1 - \frac{1}{V}$. So $\bar{e} > 0$ whenever $R < K \log_2 V$, and $\bar{e} \geq 1 - \frac{1}{V}$ at $R = 0$, which is tight. In particular $\bar{e} \geq (\ln V - \ln 2 - R \ln 2 / K) / \ln(V - 1)$ for $V \geq 3$. The KL floor of Corollary 3.2 is therefore not an artefact of KL: behavioural losslessness also fails once $R < K \log_2 V$.

Proof. Let $\hat{G}_i \sim q_{\theta^*}(\cdot | i)$ independently given θ^* . Then $e_i := \mathbb{P}(\hat{G}_i \neq F(i)) = \mathbb{E}TV_i$, because the teacher is a point mass. Since the $F(i)$ are independent and $\hat{G}$ depends on F only through θ^* , $\sum_i I(F(i); \hat{G}_i) \leq I(F; \hat{G}) \leq H(\theta^*) \leq R \ln 2$. Fano gives $I(F(i); \hat{G}_i) \geq \ln V - h(e_i) - e_i \ln(V-1)$. Sum over i and use concavity of h : $K[\ln V - h(\bar{e}) - \bar{e} \ln(V-1)] \leq R \ln 2$. Monotonicity follows from $\frac{d}{de}[h(e) + e \ln(V-1)] = \ln \frac{(1-e)(V-1)}{e} \geq 0$ for $e \leq 1 - \frac{1}{V}$. The explicit bound uses $h \leq \ln 2$. $\square$

Corollary 3.4 (Metric-entropy version). *[proved] Suppose Θ_S has an ε -cover Θ_ε in $\|\log q_\theta - \log q_{\theta'}\|_\infty$ (over the design and $\mathcal{V}^T$). Then $\text{KL}(p\|q_{\theta'}) \leq \text{KL}(p\|q_\theta) + \varepsilon$ whenever θ' is ε -close to θ . Hence*

$$\mathbb{E}_\pi \min_{\theta \in \Theta_S} \frac{1}{N} \sum_i \text{KL}_i \geq \frac{I(\mathbf{T}; Y) - \log \mathcal{N}(\varepsilon)}{mN} - \varepsilon.$$

For P parameters in $[-B_\theta, B_\theta]^P$ with $\theta \mapsto \log q_\theta$ being L -Lipschitz in ℓ_∞ , $\log \mathcal{N}(\varepsilon) \leq P \log[LB_\theta/\varepsilon]$, so the effective rate is $R_{\text{eff}}(\varepsilon) \approx P \log_2(LB_\theta/\varepsilon)$ bits. Pseudo-dimension and Rademacher bounds enter through the usual covering-number estimates.

Proof. $\text{KL}(p\|q_{\theta'}) - \text{KL}(p\|q_\theta) = \mathbb{E}_p[\log q_\theta - \log q_{\theta'}] \leq \varepsilon$. Apply Theorem 3.1 to the finite class Θ_ε . $\square$

Remark 3.5 (Bits per parameter). The rate that matters is *usable* bits. Allen-Zhu and Li (2024) report, in controlled synthetic settings, a knowledge capacity of about 2 bits per parameter for language models. Kumar et al. (2024) model how precision changes the effective parameter count. Combining such estimates with Theorem 3.1 gives a heuristic necessary condition. A student with N_S parameters cannot be lossless on content whose functional information exceeds roughly $2N_S$ bits. Treat this as an order-of-magnitude guide; the constant is an empirical estimate for a particular kind of knowledge.

3.2 Distillation as rate–distortion

Definition 3.6 (Distillation distortion–rate function). For a prior π on teachers, a reproduction alphabet of student functions, and distortion $d(\mathbf{T}, \hat{\mathbf{T}}) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(p_{\mathbf{T}}(\cdot | x) \| p_{\hat{\mathbf{T}}}(\cdot | x))$, define $D_\pi(R) := \inf\{\mathbb{E} d(\mathbf{T}, \hat{\mathbf{T}}) : I(\mathbf{T}; \hat{\mathbf{T}}) \leq R \ln 2\}$.

Theorem 3.7 (One-shot converse). *[proved] Every class of 2^R students satisfies $\mathbb{E}_\pi \min_\theta d(\mathbf{T}, q_\theta) \geq D_\pi(R)$.*

Proof. $\theta^*(\mathbf{T})$ is a deterministic function of $\mathbf{T}$ taking at most 2^R values, so $I(\mathbf{T}; \theta^*) \leq H(\theta^*) \leq R \ln 2$. It is therefore feasible in the infimum. $\square$

Unlike Shannon’s asymptotic theorem (Shannon, 1959), no block coding is needed. The converse holds for a single teacher draw because it uses only $I \leq H \leq \log(\#\text{codewords})$ (Cover and Thomas, 2006; Polyanskiy and Wu, 2025). For log-loss distortion the rate–distortion function has the simple form $R(D) = H - D$ (Courtade and Weissman, 2014). Corollary 3.2 is an instance of this.

Proposition 3.8 (Power-law spectrum gives a power-law floor). *[proved within the stated model] [numerically checked] Model. Near-lossless regime (Proposition 1.8), in Fisher-whitened centred log-probability coordinates, where $\text{KL} \approx \frac{1}{2} \|z - \hat{z}\|^2$. The teacher’s whitened log-probability function is a centred Gaussian vector with covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$. Reverse water-filling (Cover and Thomas, 2006, Thm. 10.3.3) gives*

$$D(R) = \frac{1}{2} \sum_i \min(w, \lambda_i), \quad \sum_i \frac{1}{2} \log_+ \frac{\lambda_i}{w} = R \ln 2.$$

If $\lambda_i = c i^{-\alpha}$ with $\alpha > 1$, then $D(R) = \frac{c}{2} \frac{\alpha}{\alpha-1} \left(\frac{2R \ln 2}{\alpha}\right)^{-(\alpha-1)} (1 + o(1))$, so $D(R) \asymp R^{-(\alpha-1)}$.

Proof sketch. With k active components, $w = \lambda_k$. The rate is $\frac{\alpha}{2} \sum_{i \leq k} \log \frac{k}{i} = \frac{\alpha}{2} (k - O(\log k))$ by Stirling, so $k \approx 2R \ln 2 / \alpha$. The distortion is $\frac{1}{2} [kw + \sum_{i > k} \lambda_i] = \frac{\epsilon}{2} k^{1-\alpha} (1 + \frac{1}{\alpha-1})(1 + o(1))$. The fitted exponents in `check_capacity.py` match $-(\alpha - 1)$ to within 0.04 for $\alpha \in \{1.5, 2, 3\}$. $\square$

Proposition 3.9 (Finite teacher dimension: from power law to exponential). *[proved within the stated model] [numerically checked] If $\lambda_i = 0$ for $i > k_T$, then as soon as $w < \lambda_{k_T}$, $D(R) = \frac{k_T}{2} (\prod_{i \leq k_T} \lambda_i)^{1/k_T} e^{-2R \ln 2 / k_T}$. The distortion decays exponentially in the student’s rate once the student can resolve every component the teacher has. For $k_T = 1000$ and $\alpha = 2$, D drops by a factor $e^4 \approx 55$ when $R \ln 2$ doubles from 2000 to 4000 nats. Below $R \ln 2 \approx k_T$ nats it follows the power law.*

Proposition 3.10 (No capacity gap under optimal rate-limited allocation). *[proved within the stated model] Let the ground truth have spectrum $(\lambda_i)_{i \geq 1}$ and let a teacher of size k_T know the first k_T components exactly. The best R -bit student’s distortion with respect to the ground truth is non-increasing in k_T .*

Proof. Let $E(k_T) = D_{k_T}(R) + \frac{1}{2} \sum_{i > k_T} \lambda_i$. For $k' > k_T$, one feasible code for the larger teacher uses the optimal code for the first k_T components and reproduces 0 elsewhere. Hence $D_{k'}(R) \leq D_{k_T}(R) + \frac{1}{2} \sum_{k_T < i \leq k'} \lambda_i$, so $E(k') \leq E(k_T)$. $\square$

3.3 What this implies for distillation scaling laws

Supervised pre-training follows power laws in parameters N and tokens D (Kaplan et al., 2020). A widely used joint form is the additive law $L(N, D) = E + A/N^a + B/D^b$ of Hoffmann et al. (2022). Busbridge et al. (2025) fit a distillation scaling law in which the student’s cross-entropy depends on the teacher’s cross-entropy and on the student’s supervised-scaling loss. That law captures a “capacity gap”: beyond some point, stronger teachers yield worse students (see also Cho and Hariharan, 2019; Mirzadeh et al., 2020; Zhang et al., 2023). We are confident of this qualitative content but not of the exact parametrisation in Busbridge et al. (2025). The claim of Zhang et al. (2023) that the optimal teacher is about $2.5\times$ the student is quoted from memory and should be checked. The results above imply the following.

Heuristic 3.11 (Student-size exponent). *With a constant number of usable bits per parameter, $R_S \propto N_S$. The approximation part of the gap should then scale as $N_S^{-(\alpha-1)}$, where α is the decay exponent of the teacher’s Fisher-weighted centred log-probability spectrum. This spectrum is measurable (Section 6.3). It should cross over to faster (exponential) decay once $R_S \ln 2$ approaches the teacher’s functional information (Proposition 3.9).*

Heuristic 3.12 (Data exponent: soft labels are steeper). *Consider a projection-estimator model. The target has coefficients of energy $\lambda_i = ci^{-\alpha}$ in an orthonormal basis under $\mathcal{D}$, and the student estimates the first k from n samples. The error is approximately $\sigma^2 \frac{k}{n} + \sum_{i > k} \lambda_i$, where σ^2 is the effective noise. With hard labels, σ^2 includes label noise $\sigma_0^2 > 0$. The optimum is $k \asymp n^{1/\alpha}$, giving error $\asymp n^{-(\alpha-1)/\alpha}$. With soft labels, the only noise is the aliased tail, $\sigma^2 \asymp k^{1-\alpha}$. Then $k \asymp n$ and the error is $\asymp n^{-(\alpha-1)}$. The rigorous version of this noiseless/noisy crossover in kernel regression is due to Cui et al. (2021). The prediction is that the distillation-token exponent should be steeper for full-logit distillation than for sequence-level (hard-label) distillation, with top- k logits in between.*

Heuristic 3.13 (Capacity gap as a finite-sample aliasing effect). *In the same model, let the teacher have k_T components and the student $k < k_T$. With soft labels, the student’s error with respect to the ground truth is approximately*

$$\frac{k}{n} \sum_{k < i \leq k_T} \lambda_i + \sum_{i > k} \lambda_i.$$

The first term increases in k_T (teacher content the student cannot represent aliases into the components it estimates). For $k_T \leq k$ the error is $\sum_{i>k_T} \lambda_i$, which decreases in k_T . Hence the optimal teacher has $k_T \approx k$, and the penalty for a stronger teacher is at most a factor $(1 + k/n)$ on the tail. It vanishes as $n \rightarrow \infty$. Together with Proposition 3.10, this predicts that capacity gaps are finite-data and optimisation effects, and should shrink as distillation tokens grow at fixed student size.

Conjecture 3.14 (Distillation gap law). *For a teacher whose Fisher-weighted log-probability spectrum decays as $i^{-\alpha}$, the per-token forward-KL gap of a well-optimised student with R_S usable bits and n distillation tokens satisfies*

$$\varepsilon_{\text{off}}^{\text{KL}} \asymp A R_S^{-(\alpha-1)} + B n^{-b} + \frac{k(R_S)}{n} \text{Tail}_T(k(R_S)),$$

with $b = \alpha - 1$ for full soft labels and $b = (\alpha - 1)/\alpha$ for hard labels. The first term switches to exponential decay once $R_S \ln 2 \gtrsim I_\infty$.

Remark 3.15 (Status). Section 7 proves Heuristics 3.12 and 3.13 exactly, and Conjecture 3.14 in a corrected form (Theorem 7.9), for deep linear students with a softmax read-out. The corrections are listed in Section 7.8. In particular, the switch to exponential decay does not survive spectra with infinitely many nonzero eigenvalues (Theorem 7.17), and its criterion should read $R_S \ln 2 \geq \rho_{k_T}$ rather than $R_S \ln 2 \gtrsim I_\infty$ (Corollary 7.18).

4 Objective design

4.1 Gradients with respect to student logits

Proposition 4.1 (Logit gradients of f -divergences). *[proved] [numerically checked] Let $q = \text{softmax}(z)$, $u = p/q$, and $\psi_f(u) := f(u) - u f'(u)$. Then $\nabla_z D_f(p||q) = q \odot (\psi_f(u) - \mathbb{E}_q \psi_f(u))$. In particular:*

$$\begin{aligned} \nabla_z \text{KL}(p||q) &= q - p, \\ \nabla_z \text{KL}(q||p) &= q \odot (\log \frac{q}{p} - \text{KL}(q||p) \mathbf{1}), \\ \nabla_z \text{JSD}_\beta(p||q) &= (1 - \beta) q \odot (\log \frac{q}{m} - \mathbb{E}_q \log \frac{q}{m} \mathbf{1}), \quad m = \beta p + (1 - \beta) q. \end{aligned}$$

Proof. $\partial_{q(y)} \sum_{y'} q(y') f(p(y')/q(y')) = f(u_y) - u_y f'(u_y)$. The softmax Jacobian is $\partial q_j / \partial z_i = q_j(\delta_{ij} - q_i)$. For KL, $\psi = -u$. For reverse KL, written as $D_{-\log}$, $\psi = 1 - \log u$. For JSD_β we have $f_\beta(u) - u f'_\beta(u) = -(1 - \beta) \log(\beta u + 1 - \beta) = (1 - \beta) \log(q/m)$. $\square$

4.2 Mode covering versus mode seeking

Proposition 4.2 (Support behaviour). *[proved] If $\text{KL}(p||q) < \infty$ then $\text{supp } p \subseteq \text{supp } q$ (zero-avoiding, mode covering). If $\text{KL}(q||p) < \infty$ then $\text{supp } q \subseteq \text{supp } p$ (zero-forcing, mode seeking). $\text{JSD}_\beta \leq h(\beta)$ is always finite and imposes neither constraint.*

Proposition 4.3 (Missing-mode gradients). *[proved] [numerically checked] Fix a token i with $p_i > 0$ and let $q_i \rightarrow 0$, the other coordinates staying bounded away from 0.*

- (i) *Forward KL: $\partial_{z_i} \text{KL}(p||q) = q_i - p_i \rightarrow -p_i$. The restoring force equals the missing mass.*
- (ii) *Reverse KL (assuming $\text{supp } p = \mathcal{V}$, otherwise $\text{KL}(q||p) = \infty$): $|\partial_{z_i} \text{KL}(q||p)| \leq q_i (|\log \frac{q_i}{p_i}| + \text{KL}(q||p)) = O(q_i \log \frac{1}{q_i}) \rightarrow 0$. A mode the student has dropped exerts no pull.*
- (iii) *JSD_β : exactly, the pull is $-\partial_{z_i} \text{JSD}_\beta = (1 - \beta) q_i [\log(1 - \beta + \beta p_i/q_i) + c]$ with $c = \text{KL}(q||m) \in [0, \log \frac{1}{1-\beta}]$. For a missing mode ($p_i > q_i$):*
 - *it is at most $\beta(1 - \beta)(p_i - q_i) + (1 - \beta) q_i \log \frac{1}{1-\beta}$, at most a quarter of the forward-KL pull when $q_i \ll p_i$;*

- while $q_i \gg \beta p_i$ it equals $\beta p_i - (1 - \beta) [\log \frac{1}{1 - \beta} - c] q_i + O(\beta^2 p_i^2 / q_i)$, which is forward-like with weight β ;
- as $q_i \rightarrow 0$ it equals $(1 - \beta) q_i \log \frac{\beta p_i}{q_i} (1 + o(1)) \rightarrow 0$, which is reverse-like.

A teacher mode is effectively recovered only while the student's mass on it is not $\ll \beta p_i$.

Proof. Read off Proposition 4.1. For (iii), $m_i/q_i = 1 - \beta + \beta p_i/q_i$, and $0 \leq c = \mathbb{E}_q \log \frac{q}{m} \leq \log \frac{1}{1 - \beta}$ because $m \geq (1 - \beta)q$. The upper bound uses $\log(1 - \beta + \beta r) \leq \beta(r - 1)$. The middle regime expands $\log(1 - \beta + \beta r) = \log(1 - \beta) + \frac{\beta r}{1 - \beta} + O(\beta^2 r^2)$. The last uses $\log(1 - \beta + \beta p_i/q_i) = \log \frac{\beta p_i}{q_i} + o(1)$. $\square$

Proposition 4.4 (M-projection versus I-projection). *[proved] (a), [known result; proof sketch] (b). (a) For an exponential-family student $q_\eta \propto e^{\eta^\top \phi}$, $\eta \mapsto \text{KL}(p \| q_\eta)$ is convex, and its stationarity condition is moment matching, $\mathbb{E}_{q_\eta} \phi = \mathbb{E}_p \phi$. Stationarity of $\text{KL}(q_\eta \| p)$ is $\text{Cov}_{q_\eta}(\phi, \log \frac{q_\eta}{p}) = 0$, which is generally non-convex. (b) Let $p = w \mathcal{N}(-a, 1) + (1 - w) \mathcal{N}(a, 1)$ with $w < \frac{1}{2}$, and let $q = \mathcal{N}(\mu, s^2)$. The forward minimiser is $\mu = (1 - 2w)a$, $s^2 = 1 + 4w(1 - w)a^2$. At it, $\text{KL}(q \| p) = \Theta(a^2)$ and $\text{TV}(p, q) \rightarrow 1$ as $a \rightarrow \infty$. The reverse KL has local minimisers near $\mu = \pm a$ with $s \rightarrow 1$, of value $\rightarrow \log \frac{1}{w}$ and $\log \frac{1}{1 - w}$ respectively; the global one is the heavier mode. At it, $\text{KL}(p \| q) = \Theta(a^2)$ but $\text{TV}(p, q) \rightarrow w$.*

Proof. (a) $\nabla_\eta \text{KL}(p \| q_\eta) = \mathbb{E}_{q_\eta} \phi - \mathbb{E}_p \phi$, and the Hessian is $\text{Cov}_{q_\eta} \phi \succeq 0$. For the reverse direction, differentiate $\mathbb{E}_{q_\eta} [\log q_\eta - \log p]$. (b) The forward case is moment matching. The mixture has mean $(1 - 2w)a$ and variance $1 + a^2 - (1 - 2w)^2 a^2$. The reverse values follow because $p \approx (1 - w) \mathcal{N}(a, 1)$ on the bulk of $\mathcal{N}(a, 1)$. Classical discussions are [Minka \(2005\)](#) and [Bishop \(2006\)](#). $\square$

Each direction's minimiser is catastrophic in the other direction. In this example, the mode-seeking solution is *behaviourally* better ($\text{TV} \approx w$ versus ≈ 1): it produces valid samples and drops a mode, whereas the covering solution produces mostly invalid samples. This is the formal content of “mode covering versus mode seeking”. Which is preferable depends on the deployment criterion (Section 1.4). Under misspecification, neither is TV-optimal (Proposition 4.13).

4.3 Temperature

Let $p^\tau = \text{softmax}(v/\tau)$, $q^\tau = \text{softmax}(z/\tau)$, and $\mathcal{L}_\tau(z) := \tau^2 \text{KL}(p^\tau \| q^\tau)$, the loss of [Hinton et al. \(2015\)](#).

Proposition 4.5 (Temperature and the τ^2 factor). *[proved] [numerically checked]*

- (i) $\nabla_z \mathcal{L}_\tau = \tau(q^\tau - p^\tau)$. Without the factor τ^2 the gradient is $\frac{1}{\tau}(q^\tau - p^\tau) = O(\tau^{-2})$ as $\tau \rightarrow \infty$. The factor τ^2 keeps gradient magnitudes $O(1)$, so the relative weighting against a hard-label term is stable across τ .
- (ii) (High temperature: logit matching.) $\tau(q_i^\tau - p_i^\tau) = \frac{1}{V}(\tilde{z}_i - \tilde{v}_i) + \frac{1}{2V\tau}[(\tilde{z}_i^2 - \tilde{v}_i^2) - \frac{1}{V}(\|\tilde{z}\|^2 - \|\tilde{v}\|^2)] + O(\tau^{-2})$, so $\mathcal{L}_\tau(z) = \frac{1}{2V}\|\tilde{z} - \tilde{v}\|^2 + O(1/\tau)$. This is centred-logit regression, as in [Ba and Caruana \(2014\)](#).
- (iii) (Low temperature: argmax matching.) If v has a unique maximiser i^* , then $\tau \text{KL}(p^\tau \| q^\tau) \rightarrow \max_j z_j - z_{i^*}$, a perceptron/hinge loss on the teacher's argmax, and $\mathcal{L}_\tau \rightarrow 0$.
- (iv) (Geometry, not solution.) The global minimisers $\{v + c\mathbf{1}\}$ do not depend on τ , and $\nabla^2 \mathcal{L}_\tau|_{z=v} = \text{diag}(p^\tau) - p^\tau p^{\tau\top} \rightarrow \frac{1}{V}C$ as $\tau \rightarrow \infty$. Temperature changes which errors are cheap, and hence the misspecified solution and the conditioning, but not the realisable solution.

Proof. (i) Chain rule: $\partial_z \text{KL}(p^\tau \| \text{softmax}(z/\tau)) = \frac{1}{\tau}(q^\tau - p^\tau)$. (ii) Expand $e^{\tilde{z}_i/\tau} = 1 + \tilde{z}_i/\tau + \tilde{z}_i^2/2\tau^2 + O(\tau^{-3})$, with $\sum_j e^{\tilde{z}_j/\tau} = V + \|\tilde{z}\|^2/2\tau^2 + O(\tau^{-3})$, and similarly for v . (iii) $p^\tau \rightarrow \delta_{i^*}$, and $-\tau \log q_{i^*}^\tau = \tau \text{lse}(z/\tau) - z_{i^*} \rightarrow \max_j z_j - z_{i^*}$; the entropy terms of p^τ vanish. (iv) The Hessian of $\tau^2 \text{lse}(z/\tau)$ is $\text{diag}(q^\tau) - q^\tau q^{\tau\top}$. $\square$

4.4 Dark knowledge as Fisher information

Proposition 4.6 (Low-probability tokens are invisible at $\tau = 1$). *[proved] [numerically checked]* For $\delta \in \mathbb{R}^V$ supported on $A \subseteq \mathcal{V}$, the curvature satisfies $\delta^\top (\text{diag } p - pp^\top) \delta \leq \max_{i \in A} p_i \|\delta\|^2 \leq p(A) \|\delta\|^2$. The exact cost of perturbing the matched logits v is

$$\text{KL}(p \parallel \text{softmax}(v + \delta)) = \log \left(1 + \sum_{i \in A} p_i (e^{\delta_i} - 1) \right) - \sum_{i \in A} p_i \delta_i \leq \frac{1}{2} \sum_{i \in A} p_i \delta_i^2 e^{(\delta_i)_+}.$$

So perturbing the logits of tokens of total teacher mass 10^{-6} by δ costs at most $\frac{1}{2} \cdot 10^{-6} e^{\max_i (\delta_i)_+} \|\delta\|^2$. Their relative logits, which is the “dark knowledge”, are essentially unconstrained at $\tau = 1$. At temperature τ , p is replaced by the flatter p^τ . (Proof: $\log(1+x) \leq x$ and $e^u - 1 - u \leq \frac{1}{2} u^2 e^{u_+}$.)

Proposition 4.7 (What soft labels add to hard labels). *[proved]*

- (a) At θ° , one hard label at s carries Fisher information $\mathcal{I}(s) = J^\top (\text{diag } p - pp^\top) J$, and m i.i.d. hard labels carry $m\mathcal{I}(s)$. The soft label is the $m \rightarrow \infty$ limit. It determines the centred teacher logits $\tilde{v}(s)$ exactly, because $p \mapsto C \log p$ is a bijection from the open simplex onto range C .
- (b) The per-state gradient-variance reduction from using the soft label is $\text{tr } \mathcal{I}(s)$ (Proposition 2.8). It is zero iff the teacher is deterministic at s (for J of full rank on range C). In the realisable linear case it upgrades a $k/2n$ error to exact recovery (Proposition 2.10).
- (c) For any $\tau > 0$, p^τ is a bijective function of p . Temperature adds no information; it reweights existing information in the loss (Proposition 4.5(iv), Proposition 4.6). It matters only when the channel is lossy, for example with top- k logits, finite precision, or finite optimisation budgets.
- (d) Under Assumption A5 for the teacher, $\tilde{v}(s) = CW_\top h_\top(s) + Cb_\top$. The soft label therefore reveals the teacher’s final hidden state up to $\ker(CW_\top)$ and a known offset: dark knowledge is an affine image of the teacher’s representation. This links to Section 5.

So soft labels carry information beyond hard labels *exactly when the teacher is non-degenerate*. The amount is unbounded in the noiseless realisable case and equals the Rao–Blackwell gap otherwise. Temperature is a preconditioner, not an information source. This sharpens the “dark knowledge” account of Hinton et al. (2015) and the gradient decompositions of Furlanello et al. (2018) and Tang et al. (2020). Relations to label smoothing are discussed by Müller et al. (2019).

4.5 Sequence-level objectives and unbiased gradient estimators

Proposition 4.8 (Sequence-level f -divergence gradients). *[proved] [numerically checked]* For sequence laws P and Q_θ , $\nabla_\theta D_f(P \| Q_\theta) = \mathbb{E}_{y \sim Q_\theta} [\nabla_\theta \log Q_\theta(y) \psi_f(P(y)/Q_\theta(y))]$. This gives $\nabla \text{KL}(Q_\theta \| P) = \mathbb{E}_{Q_\theta} [\nabla \log Q_\theta \cdot \log \frac{Q_\theta}{P}]$, $\nabla \text{JSD}_\beta(P \| Q_\theta) = (1 - \beta) \mathbb{E}_{Q_\theta} [\nabla \log Q_\theta \cdot \log \frac{Q_\theta}{M_\beta}]$ with M_β held fixed inside the logarithm, and $\nabla \text{KL}(P \| Q_\theta) = -\mathbb{E}_P [\nabla \log Q_\theta]$ (off-policy).

Proof. $\nabla D_f = \sum_y \nabla Q_\theta(y) \partial_{Q(y)} D_f$ with $\partial_{Q(y)} D_f = \psi_f(u_y)$ as in Proposition 4.1. Constants integrate to zero against ∇Q_θ . The divergence-minimisation view of imitation learning is developed by Ghasemipour et al. (2019) and Ke et al. (2020), and for sequence-level distillation by Wen et al. (2023). $\square$

Proposition 4.9 (Rao–Blackwellised unbiased estimator for sequence-level reverse KL). *[proved] [numerically checked]* Let $y \sim Q_\theta^x$, $g_t(s) := \text{KL}(q_\theta(\cdot | s) \| p(\cdot | s))$, and let b_t be any function of s_t (a baseline). Then

$$\hat{g} = \sum_{t=1}^T \left[\nabla_\theta g_t(s_t) \Big|_{s_t \text{ fixed}} + \nabla_\theta \log q_\theta(y_t | s_t) \left(\sum_{k>t} g_k(s_k) - b_t(s_t) \right) \right]$$

satisfies $\mathbb{E}\hat{g} = \nabla_{\theta}\text{KL}(Q_{\theta}^x\|P^x)$. The “stop-gradient” estimator that keeps only the first term is biased. Its bias is $-\sum_t \mathbb{E}[\nabla \log q_{\theta}(y_t | s_t) \sum_{k>t} g_k(s_k)]$, the effect of the student’s actions on its own future prefixes. Under Assumption A3 both estimators vanish in expectation at θ° . In general the stop-gradient fixed points are the θ that are stationary for $\theta' \mapsto \sum_t \mathbb{E}_{d_t^{\theta}} \text{KL}(q_{\theta'}\|p)$. This set includes the fixed points of the DAgger-type map $\theta \mapsto \arg \min_{\theta'} \sum_t \mathbb{E}_{d_t^{\theta}} \text{KL}(q_{\theta'}\|p)$, and is larger when that inner objective is non-convex (Proposition 2.11). These are not, in general, stationary points of the sequence-level KL.

Proof. $F(\theta) = \sum_t \mathbb{E}_{s \sim d_t^{\theta}} g_t(\theta; s)$ by Theorem 1.12. Differentiating gives a direct term and a score term through d_t^{θ} : $\sum_t \mathbb{E}[(\sum_{k<t} \nabla \log q(y_k | s_k)) g_t(s_t)] = \sum_k \mathbb{E}[\nabla \log q(y_k | s_k) \sum_{t>k} g_t(s_t)]$. Replacing the sampled per-token log-ratios by g_k is exact by the tower property, since $\mathbb{E}[\log \frac{q}{p}(y_k | s_k) | s_k] = g_k(s_k)$, and s_k contains y_t for $k > t$. Baselines are unbiased because $\mathbb{E}[\nabla \log q(y_t | s_t) | s_t] = 0$. The analytic per-token term is itself the Rao–Blackwellisation of $\nabla \log q(y_t | s_t) \log \frac{q}{p}(y_t | s_t)$. Gu et al. (2024) use a policy gradient for this objective (with further variance-reduction devices). Agarwal et al. (2024) do not differentiate through the student’s sampling; we believe this is their stated choice, but we have not re-verified the exact wording. $\square$

Proposition 4.10 (Sequence-level JSD_{β} decomposes with prefix-adaptive mixtures). *[proved]* The mixture $M_{\beta} = \beta P + (1 - \beta)Q_{\theta}$ is autoregressive, with conditionals

$$m_t(\cdot | s) = w_t(s) p_t(\cdot | s) + (1 - w_t(s)) q_t(\cdot | s), \quad w_t(s) = \sigma\left(\log \frac{\beta}{1-\beta} + \log P(s) - \log Q_{\theta}(s)\right),$$

where $P(s) = \prod_{k<t} p(y_k | s_k)$ and $Q_{\theta}(s)$ are prefix probabilities. Consequently

$$\text{JSD}_{\beta}(P^x\|Q_{\theta}^x) = \beta \sum_{t=1}^T \mathbb{E}_{s \sim P^x} \text{KL}(p_t(s) \| m_t(s)) + (1 - \beta) \sum_{t=1}^T \mathbb{E}_{s \sim Q_{\theta}^x} \text{KL}(q_t(s) \| m_t(s)).$$

The token-level JSD_{β} of Agarwal et al. (2024) differs in two ways. It fixes $w_t \equiv \beta$, and it evaluates both terms on one prefix law (student or mixed) rather than the β -term on teacher prefixes. Both differences vanish when $P(s) = Q_{\theta}(s)$ on all prefixes, since then the prefix laws also coincide. The token-level objective is therefore the near-lossless linearisation of the sequence-level one. Because $\log P(s_t) - \log Q_{\theta}(s_t)$ is a sum of $t - 1$ per-token log-ratios, the two diverge along long prefixes unless the student is already nearly lossless there.

Proof. $M_{\beta}(y_{\leq t}) = \beta P(y_{\leq t}) + (1 - \beta)Q_{\theta}(y_{\leq t})$. Dividing by $M_{\beta}(y_{<t})$ gives m_t . Now apply the chain rule (Theorem 1.12), which holds for any pair of autoregressive laws, to $\text{KL}(P\|M_{\beta})$ and to $\text{KL}(Q_{\theta}\|M_{\beta})$. $\square$

Proposition 4.11 (Unbiased per-token estimator for sequence-level JSD_{β}). *[proved]* Let $y \sim Q_{\theta}^x$, let $\rho_t(a; s) := \log \frac{q_t(a|s)}{m_t(a|s)}$ be evaluated with stop-gradient on m_t and w_t , and let b_t be any function of s_t . Then

$$\begin{aligned} \hat{g}_{\text{JSD}} = (1 - \beta) \sum_{t=1}^T \Big[\sum_{a \in \mathcal{V}} \nabla_{\theta} q_{\theta}(a | s_t) \rho_t(a; s_t) \\ + \nabla_{\theta} \log q_{\theta}(y_t | s_t) \left(\sum_{k>t} \text{KL}(q_k(s_k) \| m_k(s_k)) - b_t(s_t) \right) \Big] \end{aligned}$$

is unbiased for $\nabla_{\theta} \text{JSD}_{\beta}(P^x\|Q_{\theta}^x)$. It uses student samples only. The teacher enters through m_t .

Proof. By Proposition 4.8 and Proposition 4.10, $\nabla \text{JSD}_{\beta} = (1 - \beta) \mathbb{E}_Q[\sum_t \nabla \log q(y_t | s_t) \sum_k \rho_k(y_k; s_k)]$. Terms with $k < t$ vanish, because ρ_k is s_t -measurable. The $k = t$ term is replaced by its conditional expectation over y_t , and $k > t$ terms by $\mathbb{E}[\rho_k | s_k] = \text{KL}(q_k\|m_k)$. Baselines are unbiased as before. $\square$

Proposition 4.12 (A bounded, unbiased, horizon-free value estimator). *[proved] [numerically checked] Draw $y \sim M_\beta^x := \beta P^x + (1 - \beta)Q_\theta^x$, that is, a teacher sample with probability β and a student sample otherwise. Set $r = P^x(y)/M_\beta^x(y)$, $s = Q_\theta^x(y)/M_\beta^x(y)$ and*

$$\varphi(y) := \beta r \log r + (1 - \beta) s \log s.$$

Then $\mathbb{E}_{y \sim M_\beta^x} \varphi = \text{JSD}_\beta(P^x \| Q_\theta^x)$ and $0 \leq \varphi \leq \bar{h}(\beta) := \log \frac{1}{\min(\beta, 1-\beta)}$ for every y and every horizon T . The two coincide at $\beta = \frac{1}{2}$, where $\bar{h}(\frac{1}{2}) = h(\frac{1}{2}) = \log 2$. Only sequence log-probabilities under the two models are needed.

Proof. $\sum_y M(y) [\beta \frac{P}{M} \log \frac{P}{M} + (1 - \beta) \frac{Q}{M} \log \frac{Q}{M}] = \beta \text{KL}(P \| M) + (1 - \beta) \text{KL}(Q \| M)$. For the range, note $\beta r + (1 - \beta)s = 1$ with $r, s \geq 0$. Convexity of $u \log u$ gives $\varphi \geq 1 \cdot \log 1 = 0$. On the segment $\{\beta r + (1 - \beta)s = 1\}$, φ is convex, so it is maximised at an endpoint: $(1/\beta, 0)$ gives $\log \frac{1}{\beta}$ and $(0, 1/(1 - \beta))$ gives $\log \frac{1}{1 - \beta}$. $\square$

By contrast, summing the per-token KLs of Proposition 4.10 along sampled prefixes is unbiased but has a range growing linearly in T . For example, $p_t \equiv (\frac{1}{2}, \frac{1}{2}, 0)$ and $q_t \equiv (\frac{1}{2}, 0, \frac{1}{2})$ give $\frac{\beta T}{2} \log \frac{1}{\beta}$ on the all- a teacher path. That representation is used for gradients (Proposition 4.11), not for value estimates.

4.6 Bounded divergences are agnostically TV-competitive; KL is not

Proposition 4.13 (Agnostic TV-competitiveness). *[proved]*

(a) *For any student class, every $\theta_\beta \in \arg \min_\theta \mathbb{E}_x \text{JSD}_\beta(P^x \| Q_\theta^x)$ satisfies*

$$\mathbb{E}_x \text{TV}(P^x, Q_{\theta_\beta}^x) \leq \sqrt{\frac{h(\beta)}{2\beta(1 - \beta)}} \inf_\theta \mathbb{E}_x \text{TV}(P^x, Q_\theta^x).$$

The constant is minimised at $\beta = \frac{1}{2}$, giving $\sqrt{2 \ln 2 \cdot \text{TV}^} \approx 1.18 \sqrt{\text{TV}^*}$.*

(b) *No bound of the form $\text{TV}(\theta_{\text{KL}}) \leq \phi(\inf_\theta \text{TV})$ with $\phi(0^+) = 0$ holds for either KL direction. Let $P = (1 - \eta)\delta_a + \eta\delta_b$. Forward: for the class $\{\delta_a, \text{Unif}(\mathcal{V})\}$, the KL minimiser is the uniform law, with $\text{TV} \geq 1 - \eta - \frac{1}{V}$, while $\inf \text{TV} = \eta$. Reverse: for the class $\{(1 - \eta')\delta_a + \eta'\delta_c, \delta_b\}$ with $c \notin \text{supp } P$, the reverse-KL minimiser is δ_b (value $\log \frac{1}{\eta}$), with $\text{TV} = 1 - \eta$, while $\inf \text{TV} \leq \eta + \eta'$.*

Proof. (a) Let θ_{TV} attain (or approach) $\inf \mathbb{E} \text{TV}$. Using Proposition 1.7(i) twice and Jensen,

$$\mathbb{E} \text{TV}(\theta_\beta) \leq \sqrt{\frac{\mathbb{E} \text{JSD}_\beta(\theta_\beta)}{2\beta(1 - \beta)}} \leq \sqrt{\frac{\mathbb{E} \text{JSD}_\beta(\theta_{\text{TV}})}{2\beta(1 - \beta)}} \leq \sqrt{\frac{h(\beta) \mathbb{E} \text{TV}(\theta_{\text{TV}})}{2\beta(1 - \beta)}}.$$

The function $h(\beta)/\beta(1 - \beta)$ is symmetric about $\frac{1}{2}$ and is minimised there (direct check). (b) Direct computation. $\text{KL}(P \| \delta_a) = \infty$, and $\text{KL}(\cdot \| P) = \infty$ for any law charging c . $\square$

Proposition 4.13 is the formal reason to prefer a bounded divergence when the student is capacity-limited. KL-based objectives can place their unavoidable error where TV, and hence every bounded downstream task, is maximally hurt.

4.7 Proposed objective

Definition 4.14 (Prefix-adaptive sequence-level JSD distillation, PA-JSD).

$$\mathcal{J}_\beta(\theta) := \mathbb{E}_{x \sim \mathcal{D}} \text{JSD}_\beta(P^x \| Q_\theta^x), \quad \beta = \frac{1}{2} \text{ by default},$$

Values are estimated with the bounded mixture estimator of Proposition 4.12. Gradients use Proposition 4.11, which is already unbiased for the *full* gradient because its score form

accounts for θ inside M_β . An equivalent unbiased split, based on the exact decomposition of Proposition 4.10, is the pathwise gradient of the teacher-prefix term $\beta\text{KL}(P\|M_\theta)$ plus the score and pathwise gradients of the student-prefix term $(1 - \beta)\text{KL}(Q_\theta\|M_\theta)$. The two recipes are alternatives: adding the teacher-prefix pathwise term to Proposition 4.11 double-counts it and biases the estimator. Optionally add $\gamma\mathcal{R}_F$, the Fisher-aligned representation term of Definition 5.8.

Theorem 4.15 (Properties of PA-JSD). *[proved] (except where marked)*

- (i) Exactness (no exposure bias). $\mathcal{J}_\beta$ is a sequence-level divergence, and its per-token form is an identity (Proposition 4.10). With $\mathcal{E} = \mathbb{E}\text{JSD}_\beta$, Theorem 2.1 holds with $\mathcal{E}_{\text{shift}} \equiv 0$.
- (ii) Behavioural control. $\mathbb{E}_\mathcal{D}\text{TV}(P, Q_\theta) \leq \sqrt{\mathcal{J}_\beta(\theta)/(2\beta(1 - \beta))}$. Under input shift, $\mathbb{E}_\mu\text{JSD}_\beta \leq \mathcal{J}_\beta + h(\beta)\text{TV}(\mu, \mathcal{D})$; no density-ratio assumption is needed.
- (iii) Agnostic competitiveness. The population minimiser is TV-competitive (Proposition 4.13).
- (iv) Fisher consistency. $\mathcal{J}_\beta(\theta) = 0$ iff $Q_\theta^x = P^x$ for $\mathcal{D}$ -a.e. x . Under Assumption A3, the minimisers are exactly the lossless students.
- (v) No loss in the near-lossless regime. If $a \leq P^x/Q_\theta^x \leq b$, then $\mathcal{J}_\beta = \frac{\beta(1-\beta)}{2}\mathbb{E}\chi^2(1 + O(b-a)) = \beta(1 - \beta)\mathbb{E}\text{KL}(P\|Q_\theta)(1 + O(b-a))$.
- (vi) Unbiased, Rao-Blackwellised gradients (Proposition 4.11).
- (vii) Horizon-free estimation [proved under Assumption A4]. Let $\hat{\mathcal{J}}_n(\theta)$ average the estimator φ of Proposition 4.12 over n i.i.d. prompts, one mixture draw each, generated with common random numbers across θ (for example $y = G_\theta(x, \xi)$ with the noise ξ fixed). With probability $\geq 1 - \delta$, $\sup_\theta |\hat{\mathcal{J}}_n(\theta) - \mathcal{J}_\beta(\theta)| \leq \bar{h}(\beta)\sqrt{(R\ln 2 + \log(2/\delta))/(2n)}$, with no factor T . Compare the factor TB for the per-sequence KL loss of Theorem 2.6.

Proof. (i) Proposition 4.10. (ii) Proposition 1.7(i) with Jensen, and Proposition 1.18 with $G = h(\beta)$. (iii) Proposition 4.13. (iv) $\text{JSD}_\beta \geq 2\beta(1 - \beta)\text{TV}^2$. (v) Proposition 1.8(ii) applied to sequence laws, and $f''_\beta(1) = \beta(1 - \beta)$. (vi) Proposition 4.11. (vii) For each fixed θ the summands $\varphi_\theta(x_i, \xi_i)$ are i.i.d., unbiased, and $[0, \bar{h}(\beta)]$ -valued (Proposition 4.12). Apply Hoeffding, then a union bound over the 2^R students. $\square$

Remark 4.16 (Why this targets the dominant term). By Heuristic 2.22, the dominant term under capacity limits is approximation error as seen by the deployment criterion. PA-JSD attacks all three ways that term is inflated. Exactness removes its amplification by prefix shift. Agnostic competitiveness prevents its misallocation. Boundedness makes the estimation and covariate-shift terms horizon-free and additive. In the near-lossless regime it reduces to KL up to the constant $\beta(1 - \beta)$, so nothing is lost when losslessness is reachable. The price is on-policy sampling and the variance of a sequence-level score function. The saturation $w_t \in \{0, 1\}$ on prefixes that are already distinguishable limits the effective horizon of that variance, but does not remove it.

Proposition 4.17 (Likelihood-critical alternative: skew-Jeffreys). *[proved] [numerically checked]* Let $\mathcal{J}_\lambda^{\text{SJ}}(\theta) := (1 - \lambda)\sum_t \mathbb{E}_{d_t^p}\text{KL}(p_t\|q_t) + \lambda\sum_t \mathbb{E}_{d_t^q}\text{KL}(q_t\|p_t)$. It is estimated by off-policy soft-label forward KL plus on-policy reverse KL (Proposition 4.9). Then $\mathcal{J}_\lambda^{\text{SJ}} = \mathbb{E}[(1 - \lambda)\text{KL}(P\|Q_\theta) + \lambda\text{KL}(Q_\theta\|P)]$ exactly, and $\mathbb{E}\text{TV} \leq \sqrt{\mathcal{J}_\lambda^{\text{SJ}}/2}$. The log-loss regret is at most $\mathcal{J}_\lambda^{\text{SJ}}/(1 - \lambda)$. Both event bounds of Proposition 1.10 hold with KL replaced by $\mathcal{J}_\lambda^{\text{SJ}}/(1 - \lambda)$ (coverage) and $\mathcal{J}_\lambda^{\text{SJ}}/\lambda$ (precision), per input. For no $\lambda \in [0, 1]$ is it agnostically TV-competitive. Take $P = (1 - \eta, \eta)$ and the class $\{Q_1 = (1 - \varepsilon, \varepsilon), U = (\frac{1}{2}, \frac{1}{2})\}$. Then $\text{TV}(P, Q_1) \leq \eta$, but $\mathcal{J}_\lambda^{\text{SJ}}(Q_1) \geq (1 - \lambda)\eta(\log \frac{\eta}{\varepsilon} - 1)$, which exceeds the finite value $\mathcal{J}_\lambda^{\text{SJ}}(U)$ once ε is small enough. The minimiser is then U , with $\text{TV} = \frac{1}{2} - \eta$. For example, with $\eta = 0.01$, $\lambda = \frac{1}{2}$ and $\varepsilon = 10^{-120}$: $\mathcal{J}^{\text{SJ}}(Q_1) \approx 1.36 > \mathcal{J}^{\text{SJ}}(U) \approx 1.13$. For $\lambda = 1$ use Proposition 4.13(b).

Recommendation. If two-sided losslessness is within reach, the two objectives agree to second order (Proposition 1.8), and skew-Jeffreys additionally certifies likelihood-level guarantees. If

the student is capacity-limited, use PA-JSD to allocate its error where the deployment criterion is least hurt. A natural schedule is PA-JSD to convergence followed by a skew-Jeffreys fine-tune, monitoring both sequence-level quantities (Section 6.3).

5 Beyond output matching: intermediate representations

5.1 Formalisation

Let $h_T(s) \in \mathbb{R}^{d_T}$ be a teacher representation at some layer and $h_S(s) \in \mathbb{R}^{d_S}$ a student representation, both centred under a prefix law ν , with covariances $\Sigma_{TT}, \Sigma_{SS} \succ 0$ and cross-covariance Σ_{TS} . Representation matching adds a constraint to the output objective $\mathcal{J}$:

$$\min_{\theta} \mathcal{J}(\theta) \quad \text{s.t.} \quad \mathcal{R}(\theta) \leq \eta, \quad \text{or its Lagrangian} \quad \mathcal{J}(\theta) + \gamma \mathcal{R}(\theta).$$

Two standard choices are a learned linear map, as in FitNets (Romero et al., 2015),

$$\mathcal{R}_{LS}(\theta) := \min_{A \in \mathbb{R}^{d_T \times d_S}} \mathbb{E}_{\nu} \|Ah_S(s) - h_T(s)\|^2,$$

and linear CKA (Kornblith et al., 2019),

$$\mathcal{R}_{CKA}(\theta) := 1 - \frac{\|\Sigma_{ST}\|_F^2}{\|\Sigma_{SS}\|_F \|\Sigma_{TT}\|_F}.$$

Relation-based variants match attention maps or value relations (Zagoruyko and Komodakis, 2017; Jiao et al., 2020; Wang et al., 2020; Sun et al., 2020). Contrastive variants maximise a mutual-information bound (Tian et al., 2020).

Proposition 5.1 (The learned-map penalty is CCA; CKA constrains geometry). *[proved] [numerically checked] $\mathcal{R}_{LS} = \text{tr} \Sigma_{TT} - \text{tr}(\Sigma_{TS} \Sigma_{SS}^{-1} \Sigma_{ST})$. If the teacher features are whitened ($\Sigma_{TT} = I$), then $\mathcal{R}_{LS} = d_T - \sum_i \rho_i^2$, where the ρ_i are the canonical correlations. $\mathcal{R}_{LS}$ is invariant under $h_S \mapsto Gh_S$ for every invertible G , so it constrains only the linear information content of h_S . $\mathcal{R}_{CKA}$ is invariant only under orthogonal maps and isotropic scaling, so it also constrains the geometry (spectrum) of h_S . For example, with $d_S = d_T$ and $h_S = \Sigma_{TT}^{-1/2} h_T$: $\mathcal{R}_{LS} = 0$, but $1 - \mathcal{R}_{CKA} = \text{tr} \Sigma_{TT} / (\sqrt{d_T} \|\Sigma_{TT}\|_F) < 1$ unless $\Sigma_{TT} \propto I$.*

Proof. Normal equations give $A^* = \Sigma_{TS} \Sigma_{SS}^{-1}$. With $\Sigma_{TT} = I$, $\text{tr}(\Sigma_{TS} \Sigma_{SS}^{-1} \Sigma_{ST}) = \|\Sigma_{SS}^{-1/2} \Sigma_{ST}\|_F^2 = \sum_i \rho_i^2$. In the example, $\Sigma_{SS} = I$ and $\Sigma_{ST} = \Sigma_{TT}^{1/2}$. The CKA inequality is Cauchy–Schwarz on the eigenvalues. $\square$

5.2 When representation matching is redundant

Proposition 5.2 (Last-layer matching is implied by high-temperature logit matching). *[proved under Assumption A5] If CW_T has full column rank d_T (which requires $V - 1 \geq d_T$, true for language models), then*

$$\mathcal{R}_{LS} \leq \mathbb{E} \|(CW_T)^+(\tilde{z}_S - \tilde{z}_T)\|^2 \leq \frac{\mathbb{E} \|\tilde{z}_S - \tilde{z}_T\|^2}{\sigma_{\min}(CW_T)^2} = \frac{2V}{\sigma_{\min}(CW_T)^2} \lim_{\tau \rightarrow \infty} \mathbb{E} \mathcal{L}_{\tau}.$$

Final-layer feature matching therefore adds nothing to full-logit matching at high temperature. It is informative only when outputs pass through a lossy channel: hard labels, top-k logits, or $\tau = 1$ with a finite optimisation budget (Proposition 4.6).

Proof. Take $A = (CW_T)^+ CW_S$. With read-out biases, $\tilde{z} = C(Wh + b)$, and $(CW_T)^+ CW_T = I$ gives $Ah_S - h_T = Y - c$ with $Y := (CW_T)^+(\tilde{z}_S - \tilde{z}_T)$ and the constant $c := (CW_T)^+ C(b_S - b_T)$. The features are centred, so $\mathbb{E}[Ah_S - h_T] = 0$, hence $c = \mathbb{E}Y$ and $\mathbb{E}\|Ah_S - h_T\|^2 = \mathbb{E}\|Y - \mathbb{E}Y\|^2 \leq \mathbb{E}\|Y\|^2$. The last equality in the statement is Proposition 4.5(ii). $\square$

Proposition 5.3 (No identifiability gain in the realisable noiseless linear model). *[proved within the stated model] Let $u(s) \in \mathbb{R}^D$ be fixed input features, with teacher $h_T = B^\circ u \in \mathbb{R}^k$, $z_T = W^\circ h_T$, and student $h_S = Bu \in \mathbb{R}^k$, $z_S = WBu$ (realisable, $d_S = d_T = k$). With exact soft labels on states whose features span a subspace $\mathcal{U} \subseteq \mathbb{R}^D$, output matching identifies $CW^\circ B^\circ$ on $\mathcal{U}$. Adding exact representation targets identifies B° on the same $\mathcal{U}$. Neither yields any extrapolation beyond $\mathcal{U}$, and neither needs fewer states. Any gain must therefore come from noise, conditioning, or computation.*

5.3 When it provably helps

Proposition 5.4 (Noisy output supervision: fewer parameters learned from noise). *[known result; proof sketch] In the model of Proposition 5.3 (so $d_T = k$) with $\mathcal{U} = \mathbb{R}^D$, suppose outputs are observed only through hard labels $y \sim \text{softmax}(W^\circ B^\circ u)$ and $\text{rank}(W^\circ B^\circ) = k$. The MLE over rank- k factorisations satisfies $n \text{KL} \Rightarrow \frac{1}{2} \chi_k^2(V-1+D-k)$, with limiting mean $\frac{1}{2}k(V-1+D-k)$: the dimension of the manifold of rank- k $(V-1) \times D$ matrices, by the regular-model asymptotics of Proposition 2.10(b). If B° is instead supplied exactly by noiseless representation targets, only W is estimated, and the limiting mean drops to $\frac{1}{2}k(V-1)$. The relative reduction is $(D-k)/(V-1+D-k)$.*

In general, matching at a layer turns estimation of everything *below* that layer into noiseless regression. The statistical gain is the complexity of the sub-network below the matched layer relative to the whole. The gain disappears when the output channel is itself noiseless (Proposition 5.3).

Proposition 5.5 (Representation matching is a preconditioner). *[proved within the stated model] In the model of Proposition 5.3, near the optimum, with $W = W^\circ$ fixed and the near-lossless quadratic output loss, $\mathcal{L}_{\text{out}}(B) = \frac{1}{2} \text{tr}((B - B^\circ)\Sigma_u(B - B^\circ)^\top K)$ with $K = W^{\circ\top} F W^\circ$. Its Hessian is $\Sigma_u \otimes K$. Adding $\gamma \mathbb{E}\|Bu - B^\circ u\|^2$ gives $\Sigma_u \otimes (K + 2\gamma I)$, with the same minimiser. Gradient descent converges linearly at rate $1 - 1/\kappa$, where*

$$\kappa_\gamma = \kappa(\Sigma_u) \frac{\lambda_{\max}(K) + 2\gamma}{\lambda_{\min}(K) + 2\gamma} \searrow \kappa(\Sigma_u) \quad (\gamma \rightarrow \infty).$$

The ratio κ_0/κ_γ can be arbitrarily large, and $\kappa_0 = \infty$ when K is singular, without any change to the realisable solution. Heuristic: in language models K is badly conditioned, because $F = \text{diag } p - pp^\top$ is dominated by a few tokens (Proposition 4.6), so the speed-up can be large in practice.

Proposition 5.6 (Compositional hardness: a computational separation). *[known result; proof sketch] There are targets, for example k -sparse parities over D bits computed through intermediate partial parities, for which every statistical-query learner using only end-to-end labels needs $D^{\Omega(k)}$ queries. This follows from the statistical-query dimension of parities (Blum et al., 1994; Kearns, 1998). Supervising the intermediate partial parities makes each step learnable in polynomial time (Wies et al., 2023). The separation is computational. Information-theoretically, $\log \binom{D}{k} \approx k \log D$ samples suffice end-to-end (Theorem 2.7). Representation matching can therefore reduce the optimisation term $\mathcal{E}_{\text{opt}}$ exponentially, even when it cannot reduce $\mathcal{E}_{\text{est}}$.*

5.4 When it over-constrains, and a Fisher-aligned fix

Theorem 5.7 (Over-constraint in a linear-Gaussian model). *[proved within the stated model] [numerically checked] Let $h_T \sim \mathcal{N}(0, \Sigma)$ in $\mathbb{R}^{d_T}$, with teacher centred logits Wh_T and a constant Fisher matrix F . Let the student be $h_S = Bh_T$ with $B \in \mathbb{R}^{d_S \times d_T}$ and $d_S < \text{rank } \Sigma$, and let $\mathcal{L}_{\text{out}} = \frac{1}{2} \mathbb{E}\|F^{1/2}(W_S B - W)h_T\|^2$. Let $G := F^{1/2}W\Sigma^{1/2}$ have singular values $s_1 \geq s_2 \geq \dots$, and let (λ_i, u_i) be the eigenpairs of Σ .*

- (a) Output only ($\gamma = 0$): $\min \mathcal{L}_{\text{out}} = \frac{1}{2} \sum_{i>d_S} s_i^2$.
(b) Full feature matching ($\gamma \rightarrow \infty$ with $\mathcal{R}_{\text{LS}}$): the row space of B is forced onto $\text{span}(u_1, \dots, u_{d_S})$, provided $\lambda_{d_S} > \lambda_{d_S+1}$. Then $\min \mathcal{L}_{\text{out}} \rightarrow \frac{1}{2} \sum_{i>d_S} \lambda_i \|F^{1/2} W u_i\|^2$.
(c) The cost (b)–(a) is ≥ 0 . It is 0 iff the top- d_S principal subspace of Σ is output-optimal, and it can be arbitrarily large. For example, with $d_T = 2$, $d_S = 1$, $\Sigma = \text{diag}(\lambda_1, \lambda_2)$, $\lambda_1 > \lambda_2$, and $W = w e_2^\top$: (a) is 0 and (b) is $\frac{1}{2} \lambda_2 w^\top F w$. The student is forced to spend its only dimension on a high-variance nuisance direction.

Proof. (a) $\mathcal{L}_{\text{out}} = \frac{1}{2} \|F^{1/2} W_S B \Sigma^{1/2} - G\|_F^2$, and the first matrix has rank $\leq d_S$; apply Eckart–Young. (b) $\min_A \mathbb{E} \|A B h_T - h_T\|^2 \geq \sum_{i>d_S} \lambda_i$, with equality iff $B = M U_{d_S}^\top$ for invertible M (PCA, when $\lambda_{d_S} > \lambda_{d_S+1}$). Given this B , the best read-out reproduces W on the retained subspace, leaving $\frac{1}{2} \mathbb{E} \|F^{1/2} W (I - U_{d_S} U_{d_S}^\top) h_T\|^2$. (c) (a) is the unconstrained minimum. $\square$

Definition 5.8 (Fisher-aligned representation matching). Let $J(s) = \partial \tilde{z}_T(s) / \partial h_T(s)$ be the teacher’s centred-logit Jacobian at the matched layer; at the last layer $J = C W_T$. Let $K := \mathbb{E}_\nu [J^\top F J]$ be the output-relevance matrix and Σ the covariance of h_T . Let U_F collect the top- d_S eigenvectors of $\Sigma^{1/2} K \Sigma^{1/2}$. Match the *output-relevant projection* of the teacher:

$$\mathcal{R}_F(\theta) := \min_A \mathbb{E}_\nu \|A h_S(s) - U_F^\top \Sigma^{-1/2} h_T(s)\|^2.$$

Proposition 5.9 (Fisher alignment has no approximation cost). *[proved within the stated model] [numerically checked] In the model of Theorem 5.7, assume $s_{d_S} > s_{d_S+1}$. For every $\gamma \geq 0$, the minimisers of $\mathcal{L}_{\text{out}} + \gamma \mathcal{R}_F$ are exactly the output-optimal students of case (a). The constraint costs nothing in approximation. Heuristic: it should retain the preconditioning effect of Proposition 5.5 in the retained directions.*

Proof. Here $\Sigma^{1/2} K \Sigma^{1/2} = G^\top G$, so U_F are the top right-singular vectors of G . The Eckart–Young optimum is $F^{1/2} W_S B \Sigma^{1/2} = G U_F U_F^\top$. It is attained by $B = M U_F^\top \Sigma^{-1/2}$ and $W_S = W \Sigma^{1/2} U_F M^{-1}$, for which $h_S = M U_F^\top \Sigma^{-1/2} h_T$ and so $\mathcal{R}_F = 0$. Both terms are at their minima simultaneously, so every minimiser of the sum has $\mathcal{R}_F = 0$ and optimal $\mathcal{L}_{\text{out}}$. The spectral gap gives uniqueness up to M . $\square$

Heuristic 5.10 (Deep networks). *For intermediate layers $J(s)$ varies with s , and K averages over it. Alignment is then approximate, but K can be estimated with a few Jacobian–vector products per batch (Hutchinson-type probes). The theory predicts the following. Full-feature matching helps when the teacher’s high-variance directions are output-relevant. It hurts when they carry nuisance variance, for example the outlier or “massive” activation dimensions reported in large transformers (we are unsure of the canonical reference for this phenomenon). CKA matching over-constrains more than $\mathcal{R}_{\text{LS}}$ when $d_S < d_T$. Fisher-aligned matching should dominate both.*

regime	effect of representation matching
last layer, full logits, high τ	redundant (Proposition 5.2)
noiseless, realisable, linear	no identifiability gain (Proposition 5.3)
hard labels / top- k / sampled sequences	fewer parameters learned from noise (Proposition 5.4)
ill-conditioned output Fisher	preconditioning, faster optimisation (Proposition 5.5)
compositional targets	exponential computational gain (Proposition 5.6)
$d_S < \text{rank } \Sigma$, misaligned variance	over-constraint, unbounded cost (Theorem 5.7)
Fisher-aligned target	no approximation cost in the model (Proposition 5.9)

6 Synthesis

6.1 The full pipeline

Theorem 6.1 (End-to-end sandwich). (i) Necessity [proved under Assumption A4] . For any prior π over teachers, if $R \ln 2 < I_\infty(\pi, \mathcal{D})$ then

$$\mathbb{E}_\pi \min_\theta \mathbb{E}_{\mathcal{D}} \text{KL}(P \| Q_\theta) \geq \sup_{N,m} \frac{I(\mathbf{T}; Y \mid X_{1:N}) - R \ln 2}{mN} > 0.$$

In the K -facts model the behavioural error is bounded below too: $\bar{e} = \mathbb{E} \text{TV}$ satisfies $h(\bar{e}) + \bar{e} \ln(V - 1) \geq \ln V - R \ln 2/K$, so $\bar{e} > 0$ whenever $R < K \log_2 V$ (Theorem 3.1, Corollary 3.3).

- (ii) Sufficiency, realisable [proved under Assumptions A3 and A4] . Sequence-level hard-label KD with $n \geq 2(R \ln 2 + \log \frac{1}{\delta})/\varepsilon^2$ teacher sequences achieves $\mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_{\hat{\theta}}) \leq \varepsilon$ with probability $\geq 1 - \delta$, for every horizon T . Under input shift, $\mathbb{E}_\mu \text{TV} \leq \min\{\rho\varepsilon, \varepsilon + \text{TV}(\mu, \mathcal{D})\}$ (Theorem 2.7, Proposition 1.18). With exact soft labels and a linear-softmax student, $n \geq d$ states in general position give exact recovery (Proposition 2.10).
- (iii) Misspecified, behavioural [proved under Assumption A4] . Let $\hat{\theta}$ minimise, up to optimisation error $\mathcal{E}_{\text{opt}}$, the empirical PA-JSD objective built from the bounded mixture estimator (Proposition 4.12) with common random numbers, as in Theorem 4.15(vii). Let $\text{TV}^* = \min_\theta \mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_\theta)$ and $\Delta_n = \bar{h}(\beta) \sqrt{(R \ln 2 + \log \frac{2}{\delta})/(2n)}$. With probability $\geq 1 - \delta$,

$$\mathbb{E}_\mu \text{TV}(P, Q_{\hat{\theta}}) \leq \sqrt{\frac{h(\beta) \text{TV}^* + 2\Delta_n + \mathcal{E}_{\text{opt}}}{2\beta(1 - \beta)}} + \text{TV}(\mu, \mathcal{D}).$$

There is no dependence on T and no exposure-bias term. For $\beta = \frac{1}{2}$ the leading term is $\sqrt{2 \ln 2 \text{TV}^*}$.

- (iv) Misspecified, likelihood [proved under Assumptions A1, A2 and A5] . Off-policy soft-label forward KL gives Corollary 2.21. The approximation term obeys $\mathcal{A} \geq T\varepsilon_0$ (Corollary 2.5), and for students with a linear read-out ε_0 is at least the softmax-bottleneck floor (Theorem 2.3).

Proof. (i), (ii) and (iv) collect earlier results; Assumption A5 is needed only for the bottleneck clause of (iv). For (iii), write $\hat{\mathcal{J}}$ for the empirical objective. By Theorem 4.15(vii), $\mathcal{J}(\hat{\theta}) \leq \hat{\mathcal{J}}(\hat{\theta}) + \Delta_n \leq \hat{\mathcal{J}}(\theta_{\text{TV}}) + \mathcal{E}_{\text{opt}} + \Delta_n \leq \mathcal{J}(\theta_{\text{TV}}) + \mathcal{E}_{\text{opt}} + 2\Delta_n$ and $\mathcal{J}(\theta_{\text{TV}}) \leq h(\beta) \text{TV}^*$ (Proposition 1.7(i)). Then apply Theorem 4.15(ii) and Proposition 1.18 with $G = 1$. $\square$

Reading the sandwich. Near-lossless distillation is possible *only if* the student’s usable description length exceeds the teacher’s functional information on the deployment distribution. This condition is necessary, not sufficient: sufficiency needs realisability, which $R \ln 2 > I_\infty$ does not imply. When the teacher is realisable, the sample complexity of behavioural ε -losslessness is $\tilde{O}(R/\varepsilon^2)$ sequences independent of the horizon. Exposure bias is not a fundamental obstacle: it is an artefact of pairing a divergence with the wrong prefix law. When it is impossible, the best achievable behavioural error is TV^* , and PA-JSD is guaranteed to reach $O(\sqrt{\text{TV}^*})$, whereas KL objectives carry no such guarantee.

6.2 Status of the results

status	results
proved, unconditional	TV as worst-case regret (1.4); JSD _β sandwich (1.7); local equivalence (1.8); event bounds (1.10); margins (1.11); chain rules (1.12); hybrid bounds (1.14, 1.15); master decomposition (2.1); compounding theorems (2.14–2.19); curvature (2.11); gradient formulas and unbiased estimators (4.1, 4.8–4.11); temperature limits (4.5); bounded mixture estimator (4.12); agnostic competitiveness (4.13); facts floors and metric entropy (3.2–3.4); one-shot converse (3.7); CCA/CKA facts (5.1); quadratic sandwich (7.1); depth and bottleneck (7.4); truncation (7.11); fate of the exponential phase (7.17, 7.18)
proved under assumptions	capacity floor (A4); softmax-bottleneck floor (A5); uniform deviation (A1); horizon-free MLE (A3, A4); PA-JSD estimation (A4); end-to-end bounds; last-layer redundancy (A5)
proved within a stylised model	power-law and exponential floors (3.8, 3.9); no capacity gap under optimal allocation (3.10); no identifiability gain and preconditioning (5.3, 5.5); over-constraint and Fisher alignment (5.7, 5.9); in model $\mathcal{M}$: exact learning curve (7.5), Heuristics 3.12 and 3.13 as identities (7.7), corrected scaling law (7.9), softmax bottleneck (7.12), depth (7.14, 7.15)
known results cited	Pinsker/BH; SGD rates; $k/2n$ asymptotics; SQ lower bounds for parities; random-feature deterministic equivalents (7.19)
heuristics	dominant term (2.22); size exponent (3.11); deep-network representation matching (5.10). The data exponents (3.12) and the capacity gap (3.13) are proved within model $\mathcal{M}$
conjecture	distillation gap law (3.14): proved in corrected form within model $\mathcal{M}$ (7.9, Section 7.8)

6.3 Experiments that test the key assumptions

Several quantities recur below. Sequence-level TV has the unbiased, $[0, 1]$ -bounded estimator $\widehat{\text{TV}} = \frac{1}{n} \sum_i (1 - Q_\theta(y_i | x_i) / P(y_i | x_i))_+$ with $y_i \sim P^{x_i}$. It requires only exact sequence log-probabilities under both models (`check_sequences.py`), and admits Hoeffding intervals. Per-position curves $e_t^{\text{off,KL}}$, $e_t^{\text{on,rKL}}$ and e_t^{TV} are computed exactly over the vocabulary at sampled prefixes.

E1. Horizon, exposure and recoverability. *Hypotheses:* Corollary 1.16, Theorems 2.15, 2.16 and 2.18, and Proposition 4.10. *Protocol:* one teacher; two student sizes; five objectives: off-policy soft forward KL; on-policy reverse KL with the full estimator of Proposition 4.9; the same with stop-gradient; token-level GKD JSD_β; PA-JSD. *Measure:* (a) $\widehat{\text{TV}}(T')$ on prefixes truncated at $T' \in \{16, \dots, 2048\}$, against $\sqrt{\frac{1}{2} \sum_{t \leq T'} e_t^{\text{off,KL}}}$ and $\sum_{t \leq T'} e_t^{\text{off,TV}}$. The ratio of $\widehat{\text{TV}}$ to each bound locates the task in the diffuse or concentrated regime of Example 1.17. (b) Cumulative cost growth. With $c_t = \mathbf{1}\{\text{step } t \text{ violates a checkable constraint}\}$, for example a format, unit-test or arithmetic-step check, fit $J_c(q_\theta; T') - J_c(p; T') \propto T'^\kappa$. The predictions are $\kappa \approx 2$ for TV-type off-policy training on non-recoverable tasks, $\kappa \leq 1.5$ for off-policy forward KL, and $\kappa \approx 1$ for on-policy training when $u = O(1)$. (c) Recoverability. At student-visited prefixes, force each of the top- k tokens and estimate $\hat{u}_t(s) = \max_a \hat{Q}_t^p(s, a) - \min_a \hat{Q}_t^p(s, a)$ from teacher continuations. (d) The prefix-weight process $w_t(s)$ of Proposition 4.10: the fraction of positions where $|w_t - \beta| > 0.1$, which is where token-level GKD departs from sequence-level

JSD. *Falsifier*: on-policy gains that do not correlate with small $\hat{u}$, or κ unaffected by the prefix law, would refute the performance-difference account of exposure bias.

E2. Capacity floors and exponents. *Hypotheses*: Theorem 2.3, Propositions 3.8 and 3.9, Heuristics 3.11 and 3.13. *Protocol*: compute the teacher’s centred log-probability matrix on $N \approx 10^5$ held-out contexts, row-weighted by $\Pi^{1/2}$ and, separately, by per-context Fisher square roots. Estimate its spectrum and a decay exponent $\hat{\alpha}$. Train students across widths d_S , parameter counts N_S and quantisation levels b , at a token budget large enough that $\mathcal{E}_{\text{est}} + \mathcal{E}_{\text{opt}}$ is negligible. Check this by doubling the budget. *Measure*: (a) $\varepsilon_{\text{off}}^{\text{KL}}$ against the bottleneck floor for each d_S (the floor must never be violated, and its tightness is informative); (b) the slope of $\log \varepsilon_{\text{off}}^{\text{KL}}$ against $\log(N_S b_{\text{eff}})$, compared with $-(\hat{\alpha} - 1)$; (c) curvature of that curve as N_S approaches the teacher’s size (the predicted power-law-to-exponential transition); (d) for a fixed student, the ground-truth cross-entropy as a function of teacher size at several token budgets n . The optimal teacher size should grow with n , and the penalty for a too-strong teacher should shrink roughly as k/n . (Corollary 7.7 refines this: for a student of fixed capacity the optimal teacher equals that capacity at every n ; see Remark 7.8.) *Falsifier*: capacity gaps that persist undiminished as n grows would contradict Heuristic 3.13 and point to optimisation, not aliasing.

E3. Soft versus hard labels, and representation matching. *Hypotheses*: Proposition 2.8, Heuristic 3.12, Propositions 5.2, 5.5 and 5.9 and Theorem 5.7. *Protocol*: fixed student; supervision by full logits, top- k logits ($k \in \{1, 8, 64\}$), or sampled tokens; token budgets over three orders of magnitude. Representation terms at depths $\{L/4, L/2, 3L/4, L\}$: none, $\mathcal{R}_{\text{LS}}$, $\mathcal{R}_{\text{CKA}}$, or $\mathcal{R}_{\text{F}}$, with K estimated by Hutchinson probes. *Measure*: (a) fitted data exponents b in $\varepsilon_{\text{off}}^{\text{KL}} \approx A + Bn^{-b}$. The prediction is $b_{\text{soft}} > b_{\text{topk}} > b_{\text{hard}}$, with $b_{\text{soft}}/b_{\text{hard}} \approx \hat{\alpha}$. These are envelope exponents: fit them along the capacity-matched frontier or with early-stopped deep students, since at fixed size Theorem 7.9 predicts $b = 1$ (Remark 7.10). (b) Per-step gradient variance for soft versus hard labels, against the predicted gap $\mathbb{E}_\nu \text{tr} \mathcal{I}(s)$. (c) The alignment score $a = \|P_{\text{PCA}} U_F\|_F^2 / d_S$ at each layer, and the over-constraint cost $\Delta \mathcal{A}(\gamma) = \varepsilon_{\text{off}}^{\text{KL}}(\gamma) - \varepsilon_{\text{off}}^{\text{KL}}(0)$. Prediction: $\Delta \mathcal{A}$ is decreasing in a , and ≈ 0 for $\mathcal{R}_{\text{F}}$. (d) Steps to reach a target KL, against the condition-number ratio κ_0/κ_γ of Proposition 5.5. (e) Whether last-layer matching adds anything on top of full logits at high τ . It should not. *Falsifier*: benefits of $\mathcal{R}_{\text{LS}}$ under full-logit supervision that are uncorrelated with both conditioning and alignment would indicate a mechanism outside this theory.

7 A solvable distillation scaling law

This section settles open problem P4 (Section 8) within an explicit model: deep linear students with a softmax read-out, reading Gaussian context features, in the near-lossless regime where Proposition 3.8 already works. The model has both a softmax bottleneck (the narrowest layer) and depth. We prove:

- an *exact* finite-sample learning curve for every well-optimised deep student (Theorem 7.5). Heuristics 3.12 and 3.13 follow from it as identities (Corollary 7.7);
- a corrected form of Conjecture 3.14 with matching upper and lower bounds (Theorem 7.9);
- the effect of the softmax bottleneck (Theorem 7.12) and of depth (Proposition 7.14 and Theorem 7.15). Depth does not change the global minimisers. Its implicit bias under gradient flow, however, performs the capacity selection that the additive form of Conjecture 3.14 presupposes;
- that the exponential phase of Proposition 3.9 is a finite-rank phenomenon. It disappears for every spectrum with infinitely many nonzero eigenvalues and survives a spectral gap only as a window of explicit length (Theorem 7.17).

Section 7.8 lists what changes in Conjecture 3.14. Every claim marked *numerically checked* is verified by `verification/check_scaling_law.py`.

7.1 The model

Write $V' := V - 1$ and $\Gamma_p := \text{diag } p - pp^\top$, so that $\delta^\top \Gamma_p \delta = \text{Var}_p(\delta)$. The first lemma is the two-sided version of Lemma 2.2 that the model rests on.

Lemma 7.1 (Quadratic sandwich). *[proved] [numerically checked] Let p have full support, $\delta \in \mathbb{R}^V$, $q = \text{softmax}(\log p + \delta)$ and $o(\delta) := \max_y \delta_y - \min_y \delta_y$. Then*

$$e^{-o(\delta)} \frac{1}{2} \delta^\top \Gamma_p \delta \leq \text{KL}(p \| q) \leq e^{o(\delta)} \frac{1}{2} \delta^\top \Gamma_p \delta.$$

Proof. $\log p - \log q = -\delta + \log \mathbb{E}_p e^\delta$, so $\text{KL}(p \| q) = K(1) - K(0) - K'(0)$ for the cumulant generating function $K(t) = \log \mathbb{E}_p e^{t\delta}$. Hence $\text{KL} = \int_0^1 (1-t) K''(t) dt$ with $K''(t) = \text{Var}_{p_t}(\delta)$ and $p_t \propto p e^{t\delta}$. Since $p_t/p \in [e^{-to(\delta)}, e^{to(\delta)}]$ pointwise, and $\text{Var}_\nu(\delta) = \min_c \mathbb{E}_\nu(\delta - c)^2$, we get $e^{-to} \text{Var}_p \leq \text{Var}_{p_t} \leq e^{to} \text{Var}_p$. Integrating gives $2\varphi(-o) \leq \text{KL}/(\frac{1}{2} \text{Var}_p \delta) \leq 2\varphi(o)$ with $\varphi(o) = (e^o - 1 - o)/o^2$. Finally $2\varphi(o) \leq e^o$: compare the coefficients of o^j , which are $2/j! \leq 1/(j-2)!$. Also $2\varphi(-o) \geq e^{-o}$: the function $g(o) = 2(e^{-o} - 1 + o) - o^2 e^{-o}$ has $g(0) = 0$ and $g'(o) = 2 - 2e^{-o}(1+o) + o^2 e^{-o} \geq 0$. $\square$

Definition 7.2 (Model $\mathcal{M}$: linearised softmax with a deep linear student). (a) *Geometry.*

All teacher conditionals share one Fisher matrix, $\Gamma_{p_\top(\cdot|s)} = \bar{\Gamma}$ for every context s . This holds to leading order when teacher logits are uniformly small, where $\bar{\Gamma} \rightarrow C/V$. Otherwise it is the homogeneous-Fisher idealisation already used in Proposition 3.8. Whitened coordinates are $\zeta(s) := \bar{\Gamma}^{1/2} z(s) \in \text{range } \bar{\Gamma} \cong \mathbb{R}^{V'}$. They see only the centred logits. The per-context loss of a student is $\frac{1}{2} \|\zeta_\top(s) - \zeta_\text{S}(s)\|^2$, which by Lemma 7.1 equals $\text{KL}(p_\top(\cdot|s) \| q(\cdot|s))$ up to a factor $e^{\pm o}$. We write $\varepsilon_\top := \mathbb{E}_{s \sim \mathcal{D}} \frac{1}{2} \|\zeta_\top - \zeta_\text{S}\|^2$. For a ground truth ζ_G with the same Fisher matrix, ε_G is defined in the same way.

- (b) *Features.* The student reads $\phi(s) \in \mathbb{R}^k$. The triple $(\phi, \zeta_\top, \zeta_\text{G})$ is jointly centred Gaussian under $s \sim \mathcal{D}$, and $\Sigma := \mathbb{E} \phi \phi^\top \succ 0$. An intercept is one more feature.
- (c) *Student.* A depth- L linear network with softmax read-out: $z_\text{S}(s) = W_L \cdots W_1 \phi(s)$ with $W_l \in \mathbb{R}^{d_l \times d_{l-1}}$, $d_0 = k$, $d_L = V$. Its *bottleneck* is $d := \min(d_1, \dots, d_{L-1})$, with $d := V$ if $L = 1$. The student acts through the whitened end-to-end map $M := \bar{\Gamma}^{1/2} W_L \cdots W_1 \in \mathbb{R}^{V' \times k}$.
- (d) *Data.* Contexts $s_1, \dots, s_n$ are i.i.d. from $\mathcal{D}$, with targets $y_j = \zeta_\top(s_j) + \xi_j$. Here $\mathbb{E}[\xi_j | s_j] = 0$, $\mathbb{E}[\xi_j \xi_j^\top | s_j] = \sigma^2 I_{V'}$, and the ξ_j are independent across j . Soft labels have $\sigma^2 = 0$. For m hard labels per context, $\sigma^2 = 1/m$ (Remark 7.3). We write $\nu := \sigma^2 V'$.
- (e) *Well-optimised student.* Any global minimiser of the empirical loss $\hat{F}(M) = \frac{1}{2n} \sum_j \|y_j - M\phi(s_j)\|^2$ over the student class.

Remark 7.3 (Hard labels are noisy soft labels). *[proved within the stated model]* At context s with empirical label frequencies $\hat{p}$ from m teacher samples, the cross-entropy is $-\hat{p}^\top z_\text{S} + \text{lse}(z_\text{S})$. Put $\delta = z_\text{S} - z_\top$ and expand lse to second order at $z_\top$. Since $\hat{p} - p_\top \in \mathbf{1}^\perp = \text{range } \bar{\Gamma}$, the result is $\text{const} + \frac{1}{2} \|\bar{\Gamma}^{1/2} \delta - \xi\|^2 + O(\|\delta\|^3)$ with $\xi := \bar{\Gamma}^{+1/2}(\hat{p} - p_\top)$. This is least squares towards $\zeta_\top + \xi$, and $\text{Cov } \xi = \bar{\Gamma}^{+1/2}(\bar{\Gamma}/m)\bar{\Gamma}^{+1/2} = I_{V'}/m$. The noise ξ is bounded but heavy in low-probability directions. Only its conditional second moments enter Theorem 7.5, which is therefore exact for it (`check_scaling_law.py` uses the actual multinomial noise).

Lemma 7.4 (Depth enters only through the bottleneck). *[proved] [numerically checked] The set of whitened end-to-end maps of depth- L students with widths $d_1, \dots, d_{L-1}$ is exactly $\{M \in \mathbb{R}^{V' \times k} : \text{rank } M \leq d\}$. The population optimum and the set of well-optimised students therefore depend on the architecture only through d . The parameter count $\sum_l d_l d_{l-1}$ does not enter.*

Proof. The rank of a product is at most the smallest inner dimension. Conversely, let $\text{rank } M = r \leq d$ and $M = AB^\top$ with $A \in \mathbb{R}^{V' \times r}$, $B \in \mathbb{R}^{k \times r}$. Take $W_1 = \begin{pmatrix} B^\top \\ 0 \end{pmatrix}$, every middle layer $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$, and $W_L = (\bar{\Gamma}^{+1/2}A \quad 0)$, where $\bar{\Gamma}^{+1/2}$ maps $\mathbb{R}^{V'} \cong \text{range } \bar{\Gamma}$ back into $\mathbb{R}^V$. Then $\bar{\Gamma}^{1/2}W_L \cdots W_1 = AB^\top = M$. $\square$

So the softmax bottleneck is the rank constraint $\text{rank } M \leq d$, and depth is invisible to the well-optimised student. Depth does act on the *training path* (Section 7.5). The square loss of a deep linear network has no spurious local minima (Baldi and Hornik, 1989; Kawaguchi, 2016), so “well-optimised” is a reasonable idealisation. Proposition 7.14 computes *when* gradient flow gets there, and what it passes through on the way.

7.2 The exact learning curve

For a target $\zeta \in \{\zeta_\top, \zeta_\text{G}\}$ let $M_\zeta^\star := \mathbb{E}[\zeta \phi^\top] \Sigma^{-1}$ be the best linear read-out of the features and $A_\zeta := \mathbb{E} \|\zeta - M_\zeta^\star \phi\|^2$ its residual energy. Since $\zeta - M_\zeta^\star \phi$ is uncorrelated with ϕ , every M satisfies the Pythagorean identity

$$\varepsilon_\zeta(M) = \frac{1}{2} [A_\zeta + \|(M - M_\zeta^\star) \Sigma^{1/2}\|_F^2]. \quad (6)$$

Theorem 7.5 (Exact distillation learning curve). *[proved within the stated model] [numerically checked] Assume model $\mathcal{M}$ with no active bottleneck, $d \geq \min(k, V')$, and $n \geq k + 2$. The well-optimised student is almost surely unique: it is the least-squares map $\hat{M} = Y \Phi^\top (\Phi \Phi^\top)^{-1}$, where $\Phi = (\phi(s_1), \dots, \phi(s_n))$ and $Y = (y_1, \dots, y_n)$. Its expected loss with respect to the ground truth is*

$$\mathbb{E} \varepsilon_\text{G}(\hat{M}) = \frac{1}{2} \left[A_\text{G} + \|(M_\top^\star - M_\text{G}^\star) \Sigma^{1/2}\|_F^2 + (A_\top + \sigma^2 V') \frac{k}{n - k - 1} \right].$$

With respect to the teacher, $\mathbb{E} \varepsilon_\top(\hat{M}) = \frac{1}{2} [A_\top + (A_\top + \sigma^2 V') \frac{k}{n - k - 1}]$.

Proof. By Lemma 7.4 the class is all of $\mathbb{R}^{V' \times k}$. The loss $\hat{F}$ is strictly convex once $\Phi \Phi^\top \succ 0$, which holds almost surely for $n \geq k$, so its minimiser is $\hat{M}$. Write $Y = M_\top^\star \Phi + R + \Xi$, where R has columns $r_j = \zeta_\top(s_j) - M_\top^\star \phi(s_j)$ and Ξ has columns ξ_j . Each r_j is jointly Gaussian with $\phi(s_j)$ and uncorrelated with it, hence independent of it. So R is independent of Φ , with i.i.d. centred columns and $\mathbb{E} \|r_j\|^2 = A_\top$. Thus $\hat{M} = M_\top^\star + N$ with $N := (R + \Xi) \Phi^\top (\Phi \Phi^\top)^{-1}$, and $\mathbb{E}[N \mid \Phi] = 0$. Because r_j is a function of s_j and $\mathbb{E}[\xi_j \mid s_j] = 0$, the columns of $R + \Xi$ are uncorrelated given Φ , and $\mathbb{E}[(R + \Xi)^\top (R + \Xi) \mid \Phi] = (A_\top + \sigma^2 V') I_n$. Apply (6) with $\zeta = \zeta_\text{G}$ to $\hat{M} = M_\top^\star + N$. The cross term vanishes in conditional expectation, and

$$\mathbb{E} [\|N \Sigma^{1/2}\|_F^2 \mid \Phi] = (A_\top + \sigma^2 V') \text{tr}(\Sigma(\Phi \Phi^\top)^{-1}).$$

Write $\Phi = \Sigma^{1/2} X$ with $X \in \mathbb{R}^{k \times n}$ standard Gaussian. Then $\text{tr}(\Sigma(\Phi \Phi^\top)^{-1}) = \text{tr}((X X^\top)^{-1})$, and $\mathbb{E}(X X^\top)^{-1} = I_k / (n - k - 1)$ for $n \geq k + 2$. This is the mean of an inverse Wishart matrix (Muirhead, 1982, Thm. 3.2.12). $\square$

Remark 7.6 (Consistency with Section 2). (i) *Realisable, soft*: $A_\top = \sigma^2 = 0$ gives $\hat{M} = M_\top^\star$ exactly for $n \geq k$, which is Proposition 2.10(a). (ii) *Realisable, hard*: $\mathbb{E} \varepsilon_\top = \frac{(V-1)k}{2m(n-k-1)}$. This is the $k/2n$ law of Proposition 2.10(b) for $(V-1)k$ parameters, now as an exact finite- n expectation. The linearised model has no boundary, so it describes the non-degenerate part of the law of the true MLE, whose expectation is infinite. (iii) The theorem holds conditionally on any feature map for which $(\phi, \zeta_\top, \zeta_\text{G})$ is jointly Gaussian. In particular it covers random features (Proposition 7.19). (iv) The effective noise $A_\top + \sigma^2 V'$ has two sources: teacher content that the features cannot represent (*aliasing*), and label noise. Soft labels remove only the second. This is the mechanism behind Heuristics 3.12 and 3.13.

Corollary 7.7 (Nested model: Heuristics 3.12 and 3.13 are exact). *[proved within the stated model] [numerically checked] Let $x = (x_i)_{i \geq 1}$ be i.i.d. $N(0, 1)$ functions of the context, that is, an orthonormal Gaussian basis of $L^2(\mathcal{D})$. Let the ground truth be $\zeta_G = \sum_i g_i x_i$ with $g_i \in \mathbb{R}^{V'}$ and energies $\lambda_i := \|g_i\|^2$ non-increasing and summable. The teacher of size k_T is $\zeta_T = \sum_{i \leq k_T} g_i x_i$, the best teacher that uses k_T features. The student reads $\phi = (x_1, \dots, x_k)$. Put $\Lambda(k) := \sum_{i > k} \lambda_i$ and $\Lambda(k, k_T) := \sum_{k < i \leq k_T} \lambda_i$, which is 0 if $k_T \leq k$. For $n \geq k + 2$,*

$$\mathbb{E} \varepsilon_G = \frac{1}{2} \left[\Lambda(\min(k, k_T)) + (\Lambda(k, k_T) + \sigma^2 V') \frac{k}{n - k - 1} \right]. \quad (7)$$

- (a) Capacity gap (Heuristic 3.13). For fixed k, n and σ^2 , (7) is non-increasing in k_T on $[0, k]$ and non-decreasing on $[k, \infty)$. The optimal teacher has exactly the student's capacity, $k_T = k$, for every n and every kind of label. The strongest teacher ($k_T = \infty$) pays the additive penalty $\frac{1}{2} \Lambda(k) \frac{k}{n - k - 1}$. This is at most the factor $\frac{n-1}{n-k-1}$ over the optimum, and it vanishes as $n \rightarrow \infty$. With hard labels the penalty sits next to the larger term $\frac{1}{2} \sigma^2 V' \frac{k}{n - k - 1}$, so the capacity gap is diluted: it is a soft-label phenomenon.
- (b) Aliasing (Heuristic 3.12). The effective noise is $\Lambda(k, k_T) + \sigma^2 V'$. With soft labels only the aliased tail remains, which is the premise of Heuristic 3.12. The exponents follow in Theorem 7.9.

Proof. Here $\Sigma = I$, $M_G^* = (g_1, \dots, g_k)$ and $A_G = \Lambda(k)$. Also M_T^* keeps the columns g_i with $i \leq \min(k, k_T)$ and zeroes the rest, and $A_T = \Lambda(k, k_T)$. So $\|M_T^* - M_G^*\|_F^2 = \sum_{k_T < i \leq k} \lambda_i$, and $\Lambda(k) + \sum_{k_T < i \leq k} \lambda_i = \Lambda(\min(k, k_T))$. Substitute into Theorem 7.5. The monotonicity statements can be read off (7). $\square$

Remark 7.8 (Correction to experiment E2(d)). Section 6.3 predicts that the optimal teacher size grows with the token budget n . For a student of fixed capacity it does not: it equals the student's capacity at every n . Only the *penalty* for a too-strong teacher shrinks, like k/n . The optimal teacher grows with n only for students whose *effective* capacity grows with n , such as early-stopped deep students (Theorem 7.15).

7.3 The corrected distillation scaling law

Theorem 7.9 (Distillation scaling law: corrected Conjecture 3.14). *[proved within the stated model] [numerically checked] Take the nested model of Corollary 7.7 with $c_1 i^{-\alpha} \leq \lambda_i \leq c_2 i^{-\alpha}$, $\alpha > 1$, a teacher of size $k_T \geq k$ (for instance $k_T = \infty$), and label noise $\nu = \sigma^2 V' \geq 0$. Let $E(k, n)$ be the right side of (7), and let*

$$\varepsilon^*(k, n) := \min_{0 \leq k' \leq \min(k, n-2)} E(k', n)$$

be the loss of the best well-optimised student of capacity at most k .

- (a) Fixed capacity. $E(k, n) = \frac{1}{2} \Lambda(k) + \frac{1}{2} (\Lambda(k, k_T) + \nu) \frac{k}{n - k - 1}$ exactly. It is affine in $1/(n - k - 1)$, so at fixed capacity the data exponent is 1 for every kind of label.
- (b) Envelope. For $k \geq 1$ and $n \geq \max(4, \nu)$,

$$c \left[k^{-(\alpha-1)} + n^{-(\alpha-1)} + \left(\frac{\nu}{n} \right)^{\frac{\alpha-1}{\alpha}} \right] \leq \varepsilon^*(k, n) \leq C \left[k^{-(\alpha-1)} + n^{-(\alpha-1)} + \left(\frac{\nu}{n} \right)^{\frac{\alpha-1}{\alpha}} \right],$$

where $c, C > 0$ depend only on (α, c_1, c_2) . The data exponent is therefore $b = \alpha - 1$ for soft labels ($\nu = 0$) and $b = \frac{\alpha-1}{\alpha}$ for hard labels. The effective sample size of hard labels is $n/(\sigma^2(V - 1)) = mn/(V - 1)$. The two regimes cross over at $\sigma^2(V - 1) \asymp n^{-(\alpha-1)}$, which is the noiseless-to-noisy crossover of Cui et al. (2021).

(c) The aliasing term. The third term of Conjecture 3.14 is $\frac{1}{2}\Lambda(k, k_\top) \frac{k}{n-k-1} \leq \frac{k}{n-k-1} \cdot \frac{1}{2}\Lambda(k)$. For $k \leq n/2$ it multiplies E by at most 3, so it is invisible at the level of $\asymp$. It matters quantitatively, and it diverges only at the interpolation threshold $k \rightarrow n-1$.

Proof. (a) is (7). For (c), $\Lambda(k, k_\top) \leq \Lambda(k)$ and $\frac{n-1}{n-k-1} \leq 3$ when $k \leq n/2$ and $n \geq 4$. For (b), integral comparison gives $\kappa_1(k \vee 1)^{1-\alpha} \leq \Lambda(k) \leq \kappa_2(k \vee 1)^{1-\alpha}$ for all $k \geq 0$, with $\kappa_1 = c_1 \min(1, 2^{1-\alpha}/(\alpha-1))$ and $\kappa_2 = c_2 \alpha/(\alpha-1)$.

Upper bound. Take $k' = \min(k, \lfloor n/2 \rfloor, \lceil (n/\nu)^{1/\alpha} \rceil)$, with $(n/\nu)^{1/\alpha} := \infty$ if $\nu = 0$. Then $1 \leq k' \leq n/2$, so $n - k' - 1 \geq n/4$ and $\frac{k'}{n-k'-1} \leq 2$. Using $\Lambda(k', k_\top) \leq \Lambda(k')$,

$$E(k', n) \leq \frac{3}{2}\Lambda(k') + 2\nu k'/n.$$

Since $\nu \leq n$,

$$\frac{\nu k'}{n} \leq \frac{\nu}{n}((n/\nu)^{1/\alpha} + 1) \leq 2\left(\frac{\nu}{n}\right)^{\frac{\alpha-1}{\alpha}}, \quad \Lambda(k') \leq \kappa_2 \max\left(k^{1-\alpha}, \lfloor n/2 \rfloor^{1-\alpha}, \left(\frac{\nu}{n}\right)^{\frac{\alpha-1}{\alpha}}\right),$$

and $\lfloor n/2 \rfloor \geq n/4$. This gives the upper bound with $C = \frac{3}{2}\kappa_2 4^{\alpha-1} + 4$.

Lower bound. For $0 \leq k' \leq \min(k, n-2)$ we have $E(k', n) \geq \frac{1}{2}[\Lambda(k') + \nu k'/n]$. Three bounds follow: $\Lambda(k') \geq \Lambda(k) \geq \kappa_1 k^{1-\alpha}$; $\Lambda(k') \geq \Lambda(n-2) \geq \kappa_1 n^{1-\alpha}$; and $\Lambda(k') + \nu k'/n \geq c_3(\nu/n)^{(\alpha-1)/\alpha}$. For the third: if $k' = 0$ use $\Lambda(0) \geq \kappa_1 \geq \kappa_1(\nu/n)^{(\alpha-1)/\alpha}$. If $k' \geq 1$, minimise $x \mapsto \kappa_1 x^{1-\alpha} + \nu x/n$ over $x \geq 1$. If the unconstrained minimiser is at least 1, the minimum equals $\kappa_1^{1/\alpha}[(\alpha-1)^{(1-\alpha)/\alpha} + (\alpha-1)^{1/\alpha}](\nu/n)^{(\alpha-1)/\alpha}$. Otherwise it is attained at $x = 1$ and is at least $\kappa_1 \geq \kappa_1(\nu/n)^{(\alpha-1)/\alpha}$. So c_3 is the smaller of the two constants. Hence ε^* is at least $\frac{1}{2}$ of the largest of the three bounds, and at least $\frac{1}{6}$ of their sum. $\square$

Remark 7.10 (Reading Theorem 7.9 in the variables of Conjecture 3.14). (i) *Bits.* A capacity- k student has kV' free whitened coordinates (Lemma 7.4). At b bits each, $R_S \approx bV'k$, so the first term is $A R_S^{-(\alpha-1)}$ with $A \propto (bV')^{\alpha-1}$. Usable bits are bits on the rank- d manifold, whose dimension is $d(V' + k - d)$. Extra depth or width adds parameters but no usable bits. Put the Gaussian prior $g_i \sim N(0, \lambda_i I_{V'}/V')$ on the ground truth. Then Proposition 3.8 (with multiplicity V') needs $\approx \frac{\alpha}{2} V'k$ nats to reach distortion $\asymp k^{1-\alpha}$. So the architecture is within the factor $2b \ln 2/\alpha$ of the information-theoretic optimum, and both have the exponent $\alpha - 1$. (ii) *Bn^{-b} is an envelope term.* It holds for students whose capacity is matched to n . Take a fixed-capacity student trained to convergence with $k \gg k^*(n)$, where $k^* \asymp (n/\nu)^{1/\alpha}$ for hard labels. Its excess is $\frac{1}{2}\nu \frac{k}{n-k-1}$, which grows with k . Experiment E3(a) should therefore fit b along the capacity-matched frontier, or with early-stopped deep students (Theorem 7.15), and not at fixed size. At fixed size the model predicts $b = 1$. (iii) *Label types.* Soft labels give $\nu = 0$. With m sampled tokens per context, $\nu = (V-1)/m$. Soft labels computed from a noisy teacher interpolate between the two, with the crossover of (b).

7.4 The softmax bottleneck

Now let the bottleneck be active, $d < \min(k, V')$. Whitening the features, $\phi \mapsto \Sigma^{-1/2}\phi$, preserves ranks, so we may take $\Sigma = I_k$. Put $S := \Phi\Phi^\top$ and let $[A]_d$ denote a best rank- d approximation of A (Eckart and Young, 1936). Since $(Y - \hat{M}\Phi)\Phi^\top = 0$, we have $\|Y - \hat{M}\Phi\|_F^2 = \|Y - \hat{M}\Phi\|_F^2 + \|(M - \hat{M})S^{1/2}\|_F^2$. So by Lemma 7.4 the well-optimised width- d student is the reduced-rank regression estimator (Izenman, 1975)

$$\hat{M}_d = [\hat{M}S^{1/2}]_d S^{-1/2}.$$

Lemma 7.11 (Truncation). *[proved] [numerically checked] For $B, N \in \mathbb{R}^{p \times q}$ and $d \geq 1$, $\|[B + N]_d - B\|_F^2 \leq 2 \sum_{j>d} \sigma_j(B)^2 + 31 d \|N\|_{\text{op}}^2$.*

Proof. Let $A = B + N$, let P project onto a top- d left singular subspace of A , and let P_B do the same for B . Then $[A]_d = PA$ and $PA - B = PN - (I - P)B$, and these two terms have orthogonal column spaces. So $\|PA - B\|_F^2 = \|PN\|_F^2 + x^2$ with $x := \|(I - P)B\|_F$, and $\|PN\|_F^2 \leq d\|N\|_{\text{op}}^2 =: a^2$. Let $t := \|(I - P_B)B\|_F^2 = \sum_{j>d} \sigma_j(B)^2$. Optimality of P gives $\|PA\|_F^2 \geq \|P_B A\|_F^2$. Expanding both sides, $\|PB\|_F^2 \geq \|P_B B\|_F^2 - 2\langle PB - P_B B, N \rangle - \|PN\|_F^2$. The matrix $PB - P_B B = (I - P_B)B - (I - P)B$ has rank at most $2d$. So $|\langle PB - P_B B, N \rangle| \leq \sqrt{2d} \|PB - P_B B\|_F \|N\|_{\text{op}} \leq \sqrt{2} a(\sqrt{t} + x)$. Hence $x^2 = \|B\|_F^2 - \|PB\|_F^2 \leq t + 2\sqrt{2}a(\sqrt{t} + x) + a^2$. This gives $(x - \sqrt{2}a)^2 \leq (\sqrt{t} + \sqrt{2}a)^2 + a^2$, so $x \leq \sqrt{t} + (2\sqrt{2} + 1)a$. Therefore $\|PA - B\|_F^2 \leq a^2 + 2t + 2(2\sqrt{2} + 1)^2 a^2 \leq 2t + 31a^2$. $\square$

Theorem 7.12 (Softmax bottleneck). *[proved within the stated model] [numerically checked] Assume model $\mathcal{M}$ with $\Sigma = I_k$, Gaussian label noise, bottleneck $d < \min(k, V')$, and $W^* := M_\top^*$. Let $\Sigma_{\text{eff}} := \text{Err}^\top + \sigma^2 I_{V'}$ be the covariance of the effective noise, where $r = \zeta_\top - W^* \phi$.*

- (a) Approximation. *Every width- d student has $\varepsilon_\top(M) \geq \frac{1}{2}[A_\top + \sum_{j>d} \sigma_j(W^*)^2]$, with equality at $M = [W^*]_d$. This is the linearised, exact version of Theorem 2.3.*
- (b) Estimation. *Deterministically, $\|\hat{M}_d - W^*\|_F^2 \leq [2\lambda_{\max}(S) \sum_{j>d} \sigma_j(W^*)^2 + 31d\|(\hat{M} - W^*)S^{1/2}\|_{\text{op}}^2]/\lambda_{\min}(S)$. If $k \leq n/16$, the event $\mathcal{E} = \{\frac{n}{4}I \preceq S \preceq \frac{9}{4}nI\}$ has probability at least $1 - 2e^{-n/32}$, and*

$$\mathbb{E}[\varepsilon_\top(\hat{M}_d) | \mathcal{E}] \leq \frac{1}{2}A_\top + 9 \sum_{j>d} \sigma_j(W^*)^2 + \frac{62d}{n} [(\sqrt{\text{tr } \Sigma_{\text{eff}}} + \sqrt{k\|\Sigma_{\text{eff}}\|_{\text{op}}})^2 + \|\Sigma_{\text{eff}}\|_{\text{op}}].$$

With hard labels the last term is $\asymp \sigma^2 d(V' + k)/n$, which is σ^2/n times the dimension of the rank- d manifold, up to constants. So the bottleneck replaces $V'k$ by $d(V' + k)$ in the variance, as well as adding the approximation term.

Proof. (a) follows from (6) and Eckart–Young. (b) We have $\hat{M}_d S^{1/2} = [W^* S^{1/2} + (\hat{M} - W^*) S^{1/2}]_d$. Apply Lemma 7.11 with $B = W^* S^{1/2}$, and use $\sigma_j(W^* S^{1/2})^2 \leq \lambda_{\max}(S) \sigma_j(W^*)^2$ and $\|\Delta\|_F^2 \leq \|\Delta S^{1/2}\|_F^2 / \lambda_{\min}(S)$. For X standard Gaussian, $\sqrt{n} - \sqrt{k} - u \leq s_{\min}(X) \leq s_{\max}(X) \leq \sqrt{n} + \sqrt{k} + u$ with probability at least $1 - 2e^{-u^2/2}$ (Davidson and Szarek, 2001). With $u = \sqrt{n}/4$ and $k \leq n/16$ this gives $\mathcal{E}$. Next, $(\hat{M} - W^*) S^{1/2} = (R + \Xi)Q$ with $Q := \Phi^\top S^{-1/2}$, which has orthonormal columns. Given Φ , the columns of $R + \Xi$ are i.i.d. $N(0, \Sigma_{\text{eff}})$. By rotation invariance, $(R + \Xi)Q \stackrel{d}{=} \Sigma_{\text{eff}}^{1/2} Z$ with $Z \in \mathbb{R}^{V' \times k}$ standard Gaussian and independent of Φ . Chevet’s inequality (Chevet, 1978), (Vershynin, 2018, §8.7) gives $\mathbb{E}\|\Sigma_{\text{eff}}^{1/2} Z\|_{\text{op}} \leq \sqrt{\text{tr } \Sigma_{\text{eff}}} + \sqrt{k\|\Sigma_{\text{eff}}\|_{\text{op}}}$. Gaussian concentration adds a variance of at most $\|\Sigma_{\text{eff}}\|_{\text{op}}$. Finally, on $\mathcal{E}$ we have $\lambda_{\max}/\lambda_{\min} \leq 9$ and $1/\lambda_{\min} \leq 4/n$. $\square$

Remark 7.13 (Two capacity spectra). In the nested model the approximation error of a student with k features and bottleneck d is $\frac{1}{2}[\Lambda(k) + \sum_{j>d} \sigma_j(G_{\leq k})^2]$, where $G_{\leq k} = (g_1, \dots, g_k)$. There are two spectra: the *feature spectrum* λ_i , which decides how much of the teacher the features can see, and the *bottleneck spectrum* $\sigma_j(G_{\leq k})$, the singular values of the teacher’s head. For an aligned teacher (orthogonal g_i) they coincide, and the capacity is $\min(k, d)$: Theorem 7.9 holds with k replaced by $\min(k, d)$. A teacher with a linear read-out of width $d_\top$ has rank $G \leq d_\top$, and its bottleneck term vanishes *identically* once $d \geq d_\top$ (Remark 2.4). The feature term does not. This cutoff and exact recovery from soft labels when $A_\top = 0$ (Remark 7.6(i)) are the model’s only finite-rank phenomena. The cutoff is a transition in the width d , not in the number of bits.

7.5 Depth: implicit capacity selection

Theorem 7.9(b) is a statement about capacity-matched students. A student that is too large and trained to convergence pays $\frac{1}{2}\nu_{\frac{k}{n-k-1}}$ (Remark 7.10(ii)). We now show that depth, through the implicit bias of gradient flow, selects the capacity automatically. To make the dynamics

exactly solvable, we use the whitened sequence model: an orthogonal design $\Phi\Phi^\top = nI_k$, obtained for instance by empirically whitening the features, and Gaussian effective noise. Then $\hat{F}(M) = \frac{1}{2}\|M - \hat{S}\|_F^2 + \text{const}$ with $\hat{S} := Y\Phi^\top/n = W^* + N$ and $N \stackrel{d}{=} \Sigma_{\text{eff}}^{1/2}Z/\sqrt{n}$, with Z standard Gaussian. The fixed-design risk is $\varepsilon_\tau(M) = \frac{1}{2}[A_\tau + \|M - W^*\|_F^2]$. Write $\hat{S} = \sum_j \hat{s}_j \hat{u}_j \hat{v}_j^\top$ with $\hat{s}_1 \geq \hat{s}_2 \geq \dots$.

Proposition 7.14 (Gradient flow of deep students). *[proved] [numerically checked] Run gradient flow on $\frac{1}{2}\|W_L \cdots W_1 - \hat{S}\|_F^2$, with widths at least d , in the whitened parametrisation whose last layer is $\bar{\Gamma}^{1/2}W_L \in \mathbb{R}^{V' \times d_{L-1}}$. For $\bar{\Gamma} = C/V$ this is plain gradient flow with the last layer's learning rate multiplied by V . Start from the balanced spectral initialisation of scale $\varepsilon > 0$: $W_1(0) = \varepsilon \sum_{j \leq d} e_j \hat{v}_j^\top$, $W_l(0) = \varepsilon \sum_{j \leq d} e_j e_j^\top$ for $1 < l < L$, and $W_L(0) = \varepsilon \sum_{j \leq d} \hat{u}_j e_j^\top$ (Saxe et al., 2014). For $L = 1$ start from $W(0) = 0$.*

- (a) The end-to-end map is $M(t) = \sum_{j \leq d} s_j(t) \hat{u}_j \hat{v}_j^\top$, where $\dot{s}_j = L s_j^{2-2/L}(\hat{s}_j - s_j)$ and $s_j(0) = \varepsilon^L$. For $L = 1$, $M(t) = (1 - e^{-t})\hat{S}$: uniform shrinkage.
(b) Let $L \geq 2$, $c_2(\varepsilon) := \ln(1/\varepsilon)$ and $c_L(\varepsilon) := \varepsilon^{-(L-2)}/(L-2)$ for $L \geq 3$. For every $\tau > 0$ and every j with $\hat{s}_j \tau \neq 1$, $s_j(\tau c_L(\varepsilon)) \rightarrow \hat{s}_j \mathbf{1}\{\hat{s}_j \tau > 1\}$ as $\varepsilon \rightarrow 0$. Hence $M(\tau c_L(\varepsilon)) \rightarrow H_{1/\tau, d}(\hat{S})$, where

$$H_{\theta, d}(\hat{S}) := \sum_{j \leq d: \hat{s}_j > \theta} \hat{s}_j \hat{u}_j \hat{v}_j^\top$$

is singular-value hard thresholding, capped at rank d . Early-stopped deep students trace the hard-thresholding path, and depth-one students trace the shrinkage path.

Proof. (a) Try $W_1 = \sum_j a_j e_j \hat{v}_j^\top$, $W_l = \sum_j a_j e_j e_j^\top$, $W_L = \sum_j a_j \hat{u}_j e_j^\top$, with one scalar $a_j(t)$ per mode. The product is $\sum_j a_j^L \hat{u}_j \hat{v}_j^\top$, and the residual is $\sum_{j \leq d} (a_j^L - \hat{s}_j) \hat{u}_j \hat{v}_j^\top - \sum_{j > d} \hat{s}_j \hat{u}_j \hat{v}_j^\top$. The gradient with respect to W_l is $(W_L \cdots W_{l+1})^\top (\text{residual}) (W_{l-1} \cdots W_1)^\top$. The modes $j > d$ are annihilated by orthogonality, and each remaining mode contributes $a_j^{L-1}(a_j^L - \hat{s}_j)$ times its own rank-one block. So the ansatz is preserved with $\dot{a}_j = -a_j^{L-1}(a_j^L - \hat{s}_j)$. By uniqueness of ODE solutions it is the solution. With $s_j = a_j^L$ we get $\dot{s}_j = L a_j^{2L-2}(\hat{s}_j - s_j) = L s_j^{2-2/L}(\hat{s}_j - s_j)$. For $L = 1$ the flow is $\dot{M} = \hat{S} - M$. (b) For $L = 2$ the solution is logistic: $s(t) = \hat{s}/(1 + (\hat{s}\varepsilon^{-2} - 1)e^{-2\hat{s}t})$. At $t = \tau \ln(1/\varepsilon)$, $\hat{s}\varepsilon^{-2}e^{-2\hat{s}t} = \hat{s}\varepsilon^{2(\hat{s}\tau - 1)}$, which tends to 0 or ∞ . For $L \geq 3$, s increases in $(0, \hat{s})$. Below threshold: $\dot{s} \leq L \hat{s} s^{2-2/L}$ integrates to $s(t)^{-(L-2)/L} \geq \varepsilon^{-(L-2)} - (L-2)\hat{s}t$. So if $\hat{s}\tau < 1$, then $s(\tau c_L) \leq \varepsilon^L(1 - \hat{s}\tau)^{-L/(L-2)} \rightarrow 0$. Above threshold: using $\frac{1}{1-u} \leq 1 + 2u$ for $u \leq \frac{1}{2}$, the time to reach $\hat{s}/2$ is

$$\int_{\varepsilon^L}^{\hat{s}/2} \frac{ds}{L s^{2-2/L}(\hat{s} - s)} \leq \int_{\varepsilon^L}^{\infty} \frac{ds}{L \hat{s} s^{2-2/L}} + \frac{2}{L \hat{s}^2} \int_0^{\hat{s}/2} s^{2/L-1} ds = \frac{c_L(\varepsilon)}{\hat{s}} + O(1).$$

If $\hat{s}\tau > 1$, the time that remains after reaching $\hat{s}/2$ is at least $(\tau - 1/\hat{s})c_L(\varepsilon) - O(1) \rightarrow \infty$. During it, $\hat{s} - s$ decays at rate at least $L(\hat{s}/2)^{2-2/L}$. So $s \rightarrow \hat{s}$. $\square$

Theorem 7.15 (Depth selects capacity). *[proved within the stated model] [numerically checked] In the whitened sequence model, let $\sigma_1 \geq \sigma_2 \geq \dots$ be the singular values of W^* and $\gamma_n := (\sqrt{\text{tr } \Sigma_{\text{eff}}} + \sqrt{k\|\Sigma_{\text{eff}}\|_{\text{op}}})/\sqrt{n}$.*

- (a) Deep ($L \geq 2$, limit $\varepsilon \rightarrow 0$). Let the threshold θ be deterministic with $\theta \geq 2(\gamma_n + u\sqrt{\|\Sigma_{\text{eff}}\|_{\text{op}}/n})$. With probability at least $1 - e^{-u^2/2}$,

$$\|H_{\theta, d}(\hat{S}) - W^*\|_F^2 \leq 49 \min_{0 \leq r \leq d} \left[\sum_{j > r} \sigma_j^2 + r\theta^2 \right].$$

- (b) Shallow ($L = 1$). For every deterministic stopping time t ,

$$\mathbb{E}\|M(t) - W^*\|_F^2 \geq \frac{\|W^*\|_F^2 \cdot k \text{tr } \Sigma_{\text{eff}}/n}{\|W^*\|_F^2 + k \text{tr } \Sigma_{\text{eff}}/n} \geq \frac{1}{2} \min(\|W^*\|_F^2, k \text{tr } \Sigma_{\text{eff}}/n).$$

(c) Consequence. Take the aligned nested model with $\lambda_j \asymp j^{-\alpha}$, hard labels, and $\theta \asymp \gamma_n$, so that $\theta^2 \asymp \sigma^2(V' + k)/n$. The deep student's excess is $O(\Lambda(d) + (\sigma^2(V' + k)/n)^{(\alpha-1)/\alpha})$. This is the envelope law of Theorem 7.9, reached for every architectural size k between $k^*(n) \asymp (n/\nu)^{1/\alpha}$ and $O(V')$, without tuning the architecture. The shallow student is stuck at $\gtrsim \min(\sum_{j \leq k} \lambda_j, \nu k/n)$. When $\nu k/n \leq \sum_{j \leq k} \lambda_j$, this exceeds the envelope by a factor $\asymp k/k^*(n)$.

Proof. (a) $\|N\|_{\text{op}}$ has mean at most γ_n (Chevet's inequality, as in Theorem 7.12) and is $\sqrt{\|\Sigma_{\text{eff}}\|_{\text{op}}/n}$ -Lipschitz in Z . So $\|N\|_{\text{op}} \leq \theta/2$ with probability at least $1 - e^{-u^2/2}$. On this event, Weyl's inequality gives $|\hat{\sigma}_j - \sigma_j| \leq \theta/2$. Let $\hat{r} := \#\{j \leq d : \hat{\sigma}_j > \theta\}$, so $H_{\theta,d}(\hat{S}) = [\hat{S}]_{\hat{r}}$, and let $r_1 := \#\{j \leq d : \sigma_j > \theta/2\}$. Then $\hat{r} \leq r_1$, and every $j \in (\hat{r}, r_1]$ has $\sigma_j \leq 3\theta/2$. Lemma 7.11, which also holds trivially for $\hat{r} = 0$, gives

$$\|[\hat{S}]_{\hat{r}} - W^*\|_F^2 \leq 2 \sum_{j > \hat{r}} \sigma_j^2 + 31\hat{r}\|N\|_{\text{op}}^2 \leq 2 \sum_{j > r_1} \sigma_j^2 + \frac{9}{2}(r_1 - \hat{r})\theta^2 + \frac{31}{4}\hat{r}\theta^2 \leq 49 \left[\sum_{j > r_1} \sigma_j^2 + \frac{r_1\theta^2}{4} \right].$$

The function $h(r) = \sum_{j > r} \sigma_j^2 + r\theta^2/4$ has increments $\theta^2/4 - \sigma_r^2$. These are negative exactly when $\sigma_r > \theta/2$, so over $0 \leq r \leq d$ it is minimised at r_1 . (b) $M(t) = g\hat{S}$ with $g = 1 - e^{-t}$ deterministic. So $\mathbb{E}\|g\hat{S} - W^*\|_F^2 = (1 - g)^2\|W^*\|_F^2 + g^2\mathbb{E}\|N\|_F^2$, with $\mathbb{E}\|N\|_F^2 = k \text{tr} \Sigma_{\text{eff}}/n$. Minimise over $g \in \mathbb{R}$. (c) Insert $\sigma_j^2 = \lambda_j$ for $j \leq \min(k, k_{\text{T}})$ into (a) and minimise over r as in Theorem 7.9. The noise level is $\text{tr} \Sigma_{\text{eff}} + k\|\Sigma_{\text{eff}}\|_{\text{op}} \asymp \sigma^2(V' + k)$ once the aliased tail is small next to σ^2 . For (b), use $\text{tr} \Sigma_{\text{eff}} \geq \sigma^2 V'$. $\square$

Remark 7.16. (i) Depth 2 approaches the thresholding limit only logarithmically in $1/\varepsilon$. Depth $L \geq 3$ approaches it polynomially, and the transition sharpens with L (`check_scaling_law.py`). For generic small random initialisations the same incremental, saddle-to-saddle learning is known to hold approximately (Saxe et al., 2014; Gidel et al., 2019; Arora et al., 2019; Jacot et al., 2021). We prove only the spectral case. (ii) So the additive form of Conjecture 3.14, $AR^{-(\alpha-1)} + Bn^{-b}$ with the envelope exponent b , is the law of early-stopped *deep* students. Shallow students trained to convergence follow the fixed-capacity law of Theorem 7.9(a). In the whitened model, spectral bias has to come from depth. With anisotropic features, a depth-one student also has spectral bias, through the feature covariance (Bach, 2024; Lin et al., 2024; Paquette et al., 2024).

7.6 Does the exponential phase survive realistic spectra?

Return to the Gaussian source of Propositions 3.8 and 3.9. Its spectrum $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$ is summable and not identically zero. Measure rate in nats, $\rho := R \ln 2$, so that $D(\rho) = \frac{1}{2} \sum_i \min(w, \lambda_i)$ with $\frac{1}{2} \sum_i \ln_+(\lambda_i/w) = \rho$. Put

$$K(w) := \#\{i : \lambda_i > w\}, \quad T(w) := \sum_{i: \lambda_i \leq w} \lambda_i, \quad K_{\text{eff}}(w) := K(w) + T(w)/w.$$

Theorem 7.17 (Fate of the exponential phase). *[proved] [numerically checked]*

(i) Local rate. $\rho \mapsto D(\rho)$ is convex, C^1 and strictly decreasing on $(0, \infty)$, with $D'(\rho) = -w(\rho)$ and

$$-\frac{d \ln D}{d\rho} = \frac{2}{K_{\text{eff}}(w(\rho))}.$$

(ii) Finite rank. If exactly k_{T} eigenvalues are positive, then for $\rho \geq \rho_{k_{\text{T}}} := \frac{1}{2} \sum_{i \leq k_{\text{T}}} \ln(\lambda_i/\lambda_{k_{\text{T}}})$ we have $K_{\text{eff}} \equiv k_{\text{T}}$ and $D(\rho) = \frac{k_{\text{T}}}{2} (\prod_{i \leq k_{\text{T}}} \lambda_i)^{1/k_{\text{T}}} e^{-2\rho/k_{\text{T}}}$ (Proposition 3.9). For a power-law head, $\rho_{k_{\text{T}}} = \frac{\alpha}{2}(k_{\text{T}} - O(\log k_{\text{T}}))$.

- (iii) Infinite support. If infinitely many λ_i are positive, then $K_{\text{eff}}(w(\rho)) \rightarrow \infty$ and $-\ln D(\rho) = o(\rho)$: D decays more slowly than every exponential. For $\lambda_i \asymp i^{-\alpha}$, $K_{\text{eff}} \sim \frac{\alpha}{\alpha-1}K$ and $D \asymp \rho^{-(\alpha-1)}$ (Proposition 3.8).
- (iv) Analytic spectra. If $\lambda_i = ce^{-\beta i}$, then $-\ln D(\rho) = 2\sqrt{\beta\rho} + O(\ln \rho)$: a stretched exponential. The same computation gives $-\ln D \asymp \rho^{1/(1+\gamma)}$ for $\lambda_i = ce^{-\beta i^\gamma}$.
- (v) Spectral gap. Let $\lambda_{k_\top} > \lambda_{k_\top+1}$, $\eta > 0$ and $w_\eta := \max\{\lambda_{k_\top+1}, \Lambda(k_\top)/(\eta k_\top)\}$, where $\Lambda(k_\top) = \sum_{i>k_\top} \lambda_i$. The rates at which all $k_\top$ head components are active and the local rate is within the factor $1 + \eta$ of the finite-rank rate $2/k_\top$ form an interval of length

$$\Delta\rho = \frac{k_\top}{2} \ln_+(\lambda_{k_\top}/w_\eta) \text{ nats.}$$

Across it D falls by a factor between $(\lambda_{k_\top}/w_\eta)^{1/(1+\eta)}$ and $\lambda_{k_\top}/w_\eta$.

Proof. (i) The water level $w(\rho)$ is continuous and strictly decreasing. Between consecutive distinct eigenvalues, $\frac{dD}{dw} = \frac{K}{2}$ and $\frac{d\rho}{dw} = -\frac{K}{2w}$, so $D'(\rho) = -w(\rho)$. This is continuous and increasing in ρ , hence D is C^1 and convex. Then $-\frac{d \ln D}{d\rho} = \frac{w}{D} = \frac{2w}{Kw+T(w)} = \frac{2}{K_{\text{eff}}}$. (ii) For $w \leq \lambda_{k_\top}$ we have $K = k_\top$ and $T = 0$. So $D = k_\top w/2$ and $\ln w = (\sum_{i \leq k_\top} \ln \lambda_i - 2\rho)/k_\top$. For $\lambda_i = ci^{-\alpha}$, $\rho_{k_\top} = \frac{\alpha}{2}(k_\top \ln k_\top - \ln k_\top!)$, and Stirling's formula gives the claim. (iii) As $\rho \rightarrow \infty$, $w \downarrow 0$ and $K(w) \rightarrow \infty$. So $-\ln D(\rho) = -\ln D(\rho_0) + \int_{\rho_0}^\rho 2/K_{\text{eff}} \leq C + \int_{\rho_0}^\rho 2/K(w(r)) dr = o(\rho)$. For the power law, $T(w)/w \sim K/(\alpha-1)$ by integral comparison. (iv) For $w \in (\lambda_{K+1}, \lambda_K]$, $\ln(c/w) \in [\beta K, \beta(K+1))$, so $\rho = \frac{1}{2}[K \ln(c/w) - \beta K(K+1)/2] \in [\frac{\beta}{4}(K^2 - K), \frac{\beta}{4}(K^2 + K))$ and $K = 2\sqrt{\rho/\beta} + O(1)$. Also $T(w) = \lambda_{K+1}/(1-e^{-\beta}) \leq w/(1-e^{-\beta})$. So $\ln D = \ln w + \ln K + O(1) = -\beta K + O(\ln K)$. For $\lambda_i = ce^{-\beta i^\gamma}$ the same steps give $K \asymp \rho^{1/(1+\gamma)}$ and $-\ln D \asymp K^\gamma$. (v) For $\lambda_{k_\top+1} \leq w < \lambda_{k_\top}$ we have $K = k_\top$ and $T(w) = \Lambda(k_\top)$. So $K_{\text{eff}} \leq (1+\eta)k_\top$ if and only if $w \geq \Lambda(k_\top)/(\eta k_\top)$. For $w < \lambda_{k_\top+1}$, $K > k_\top$. On the resulting interval $[w_\eta, \lambda_{k_\top})$, $d\rho = -\frac{k_\top}{2} d \ln w$, which gives $\Delta\rho$. The drop of $\ln D$ is $\int 2/K_{\text{eff}} d\rho$ with $K_{\text{eff}} \in [k_\top, (1+\eta)k_\top]$. $\square$

Corollary 7.18 (Answer: the exponential phase is a finite-rank phenomenon). *[proved]*

- (a) $D(\rho)$ has an exponential phase if and only if the Fisher-weighted spectrum has finitely many nonzero eigenvalues. Realistic spectra, meaning power laws or smooth-function spectra $e^{-\beta i^\gamma}$, give polynomial or stretched-exponential decay at every rate. There is no transition.
- (b) With a gap, the phase survives as a transient. Take a power-law head whose tail continues at $1/g$ of its level. Then $\lambda_{k_\top}/w_\eta \approx \min\{g, \eta g(\alpha-1)\}$ and $\Delta\rho \approx \frac{k_\top}{2} \ln \min\{g, \eta g(\alpha-1)\}$, after which the power law resumes. Without a gap ($g=1$) the window is empty for every $\eta < 1/(\alpha-1)$. (`check_scalingLaw.py`: $k_\top = 200$, $\alpha = 2$, $\eta = \frac{1}{4}$ gives windows of 0, 92 and 553 nats for $g = 1, 10, 10^3$, as predicted.)
- (c) The switching criterion of Conjecture 3.14, $R_5 \ln 2 \gtrsim I_\infty$, is not the right one. For any continuous prior, such as the Gaussian prior of Proposition 3.9, $I_\infty = \infty$. As $m \rightarrow \infty$ the label frequencies converge to $p_\top(\cdot | x_i)$, which determines the teacher once the design identifies it (Proposition 4.7(a)). Data processing, and lower semicontinuity of mutual information under weak convergence (Pinsker, 1964), then force $I(\mathbf{T}; Y) \rightarrow \infty$. Yet the exponential phase starts at the finite rate $\rho_{k_\top} \approx \frac{\alpha}{2}k_\top$. The correct criterion is that the water level reaches the smallest teacher eigenvalue: $R_5 \ln 2 \geq \rho_{k_\top}$.
- (d) Statistical and architectural analogues. On the data side, the counterpart of the exponential phase is exact recovery from soft labels at $n \geq k$ (Remark 7.6(i)). It too needs finite rank. If $\Lambda(k) > 0$ for all k , then (7) gives $\mathbb{E}\varepsilon_G \geq \frac{1}{2}\Lambda(k) > 0$ for every k and n , and the envelope decays as the power law $n^{-(\alpha-1)}$ (Theorem 7.9). On the architecture side, the only sharp transition that survives is the softmax-bottleneck cutoff at $d \geq d_\top$ (Remark 7.13). It is a transition in width, not in bits, and leaves the power law of the feature spectrum intact.

7.7 Random-feature students

Proposition 7.19 (Random features cap the exponent). *[known result; proof sketch] [numerically checked] Let the context features be $x \sim N(0, \text{diag}(\mu))$ with $\mu_i \asymp i^{-a}$ and $a > 1$. Let the teacher be $\zeta_T = Bx$, with energies $\lambda_i := \mu_i \|Be_i\|^2 \asymp i^{-\alpha}$ and $\alpha > 1$. Let the student read $\phi = Fx$, where $F \in \mathbb{R}^{k \times \infty}$ has i.i.d. $N(0, 1)$ entries. Conditionally on F , Theorem 7.5 applies verbatim, with $A_T = A_T(F)$ the random-feature approximation error. The deterministic equivalent*

$$A_T(F) \approx \sum_i \lambda_i \frac{\kappa}{\mu_i + \kappa}, \quad \text{where } \kappa \text{ solves } \sum_i \frac{\mu_i}{\mu_i + \kappa} = k$$

(Bach, 2024; Atanasov et al., 2024; Defilippis et al., 2024) gives $A_T(F) \asymp k^{-\min(\alpha-1, a)}$, with an extra factor $\log k$ when $\alpha - 1 = a$. So for random-feature students the exponent $\alpha - 1$ in Conjecture 3.14 and Theorem 7.9 becomes $\min(\alpha - 1, a)$: the student’s own feature spectrum caps how much of the teacher’s smoothness it can use.

Derivation. $\sum_i \mu_i / (\mu_i + \kappa) \asymp \#\{i : \mu_i \geq \kappa\} \asymp \kappa^{-1/a}$, so $\kappa \asymp k^{-a}$. Split the sum at $i \asymp k$. The head gives $\kappa \sum_{i \leq k} \lambda_i / \mu_i \asymp k^{-a} \sum_{i \leq k} i^{a-\alpha}$. This is $\asymp k^{1-\alpha}$ if $\alpha < a + 1$, $\asymp k^{-a} \log k$ if $\alpha = a + 1$, and $\asymp k^{-a}$ if $\alpha > a + 1$. The tail gives $\sum_{i > k} \lambda_i \asymp k^{1-\alpha}$. The exact $A_T(F)$, averaged over draws of F , has fitted slopes -1.04 and -1.57 against the predicted -1 and -1.5 ($a = 1.5$; $\alpha = 2, 4$). $\square$

This is the saturation familiar from kernel regression and from solvable scaling-law models (Maloney et al., 2022; Bahri et al., 2024; Paquette et al., 2024; Lin et al., 2024). It is a genuine correction to Heuristic 3.11 and Conjecture 3.14. The exponent is set by the teacher’s Fisher-weighted spectrum *and* by the student’s feature spectrum, whichever is smaller.

7.8 Summary: the status of Conjecture 3.14

Within model $\mathcal{M}$ (Definition 7.2):

1. **Heuristics 3.12 and 3.13 are exact** (Corollary 7.7), once n is replaced by $n - k - 1$ and label noise by $\sigma^2(V - 1)$. The optimal teacher has exactly the student’s capacity. Its advantage over the strongest teacher is at most the factor $\frac{n-1}{n-k-1}$, and it is a soft-label effect.
2. **Conjecture 3.14 holds in corrected form** (Theorem 7.9). For the best student of capacity at most k ,

$$\varepsilon^*(k, n) \asymp k^{-(\alpha-1)} + n^{-(\alpha-1)} + \left(\frac{\sigma^2(V-1)}{n} \right)^{\frac{\alpha-1}{\alpha}},$$

with $R_S \approx bV'k$ usable bits. Six corrections:

- (a) Bn^{-b} is an *envelope* term. At fixed capacity the data exponent is 1. A shallow student that is too large and trained to convergence gets *worse* with size. Early-stopped deep students reach the envelope at every size (Theorem 7.15).
- (b) The aliasing term $\frac{k}{n} \text{Tail}_T(k)$ is at most a factor $\frac{n-1}{n-k-1}$. Up to constants it is redundant for $k \leq n/2$.
- (c) Hard labels use the effective sample size $mn/(V - 1)$. Soft and hard labels cross over at $\sigma^2(V - 1) \asymp n^{-(\alpha-1)}$.
- (d) R_S must count usable bits, meaning bits on the rank- d manifold. Depth adds none (Lemma 7.4).
- (e) The softmax bottleneck adds a second approximation spectrum (Remark 7.13). Random-feature students replace $\alpha - 1$ by $\min(\alpha - 1, a)$ (Proposition 7.19).
- (f) The switch to exponential decay has the wrong criterion (Corollary 7.18(c)). More importantly, it does not survive spectra with infinitely many nonzero eigenvalues (Theorem 7.17). Power laws stay power laws and analytic spectra give $e^{-2\sqrt{\beta\rho}}$. A spectral gap leaves only a window of $\frac{k_T}{2} \ln(\lambda_{k_T}/w_\eta)$ nats.

3. **Depth** is invisible to well-optimised students and decisive for early-stopped ones: it turns the fixed-capacity law into the envelope law (Proposition 7.14 and Theorem 7.15).
4. **Still open.** (i) Context-dependent Fisher matrices, where the whitened student is no longer linear. (ii) Non-Gaussian features, where residuals are uncorrelated with the features but not independent of them, so Theorem 7.5 becomes an approximation. (iii) Nonlinear students. (iv) Generic random initialisation in Proposition 7.14. (v) Measuring α , a and the bottleneck spectrum of real teachers (experiment E2).

8 Open mathematical problems

These are ordered by how much their solution would advance near-lossless distillation.

- P1. The functional information of trained teachers.** Characterise, or give computable bounds on, $I_\infty(\pi, \mathcal{D})$ from Theorem 3.1 for realistic priors, for example the distribution of networks produced by a training procedure. Is there a data-dependent estimator of the *functional* description length of a single trained teacher, such as the minimal R at which some R -bit student achieves ε -losslessness, with matching upper and lower bounds? This is the quantity that decides whether lossless distillation is possible at all.
- P2. Horizon-free agnostic guarantees with tractable estimators.** PA-JSD is horizon-free at the population level (Theorem 6.1(iii)), but its gradient estimator has sequence-level score-function variance. Is there an objective that is simultaneously (a) an exact sequence-level divergence, (b) agnostically TV-competitive with a constant (not square-root) factor, and (c) has a gradient estimator whose variance is bounded uniformly in T ? For deterministic experts, Foster et al. (2024) suggest that log-loss behaviour cloning is already horizon-free; the stochastic, misspecified case is open.
- P3. The optimal agnostic exponent.** Proposition 4.13 gives $\text{TV}(\hat{\theta}) = O(\sqrt{\text{TV}^*})$. Is the square root necessary for *any* smooth, tractable surrogate? Equivalently, what is the best ϕ with $\text{TV}(\arg \min \mathbf{D}) \leq \phi(\text{TV}^*)$ over divergences $\mathbf{D}$ that admit unbiased per-token gradients? A lower bound of the form $\phi(t) \gtrsim \sqrt{t}$ for all f -divergences with $f''(1) > 0$ would be a clean impossibility result.
- P4. A first-principles distillation scaling law.** Prove Conjecture 3.14, or a corrected form, in a model rich enough to exhibit both the softmax bottleneck and depth. Examples are deep linear networks with a softmax read-out, or random-feature models with a power-law spectrum. The model should include the soft-versus-hard data exponents (Heuristic 3.12) and the finite-sample capacity gap (Heuristic 3.13). Determine whether the transition to exponential decay (Proposition 3.9) survives realistic spectra. *Status: resolved in corrected form within model $\mathcal{M}$* (Section 7): exact learning curve (Theorem 7.5), corrected law (Theorem 7.9), bottleneck and depth (Theorems 7.12 and 7.15), and a negative answer on the exponential phase (Theorem 7.17). The remaining open cases are listed in Section 7.8: context-dependent Fisher metrics, non-Gaussian features, nonlinear students, and random initialisation.
- P5. Recoverability of language models.** Give conditions on the teacher under which the recoverability constant u of Theorem 2.16 is $O(1)$, or bounded in expectation rather than in the worst case. Relate u to measurable properties of the teacher, such as its value function under task rewards or the mixing of its induced Markov chain on prefixes. A distribution-dependent u would turn the $O(uT\varepsilon_{\text{on}})$ bound into a usable prediction.

- P6. Fixed points of stop-gradient on-policy distillation.** Characterise the fixed points of the DAgger-type map in Proposition 4.9. When do they coincide with minimisers of a sequence-level divergence? Is there a Lyapunov function? Under what conditions does the iteration converge for overparameterised students?
- P7. Representation matching in deep networks.** Extend Theorem 5.7 and Proposition 5.9 beyond the linear-Gaussian model. Is there a Fisher-aligned target, depending on s through the Jacobian, whose constraint provably has zero approximation cost for a nonlinear teacher–student pair? Can one quantify sample-complexity and computational gains of intermediate supervision for transformer students, beyond parity-type separations?
- P8. Optimal input design for distillation.** The teacher can be queried anywhere. Which input distribution $\mathcal{D}$, possibly adaptive, minimises the deployment risk under μ for a fixed query budget, given the covariate-shift terms of Proposition 1.18? This is an active-learning or optimal-design problem with a known labelling function. Minimax rates for it are unknown in the sequence setting.
- P9. Decoding-level losslessness.** Beyond the margin condition of Proposition 1.11, give sharp bounds on the probability that greedy or beam-search outputs of teacher and student differ, in terms of per-token divergences and the teacher’s margin distribution along its greedy paths. The analogous problem for speculative decoding, whose per-token acceptance probability is exactly $1 - \text{TV}$ (Leviathan et al., 2023), suggests that such bounds are within reach.

A Deferred proofs

A.1 Proof of Theorem 2.7 (horizon-free MLE bound)

Write $D_{\text{H}}^2(P, Q) = \sum_y (\sqrt{P(y)} - \sqrt{Q(y)})^2 = 2\text{H}^2(P, Q)$. For any P, Q on $\mathcal{Y}$,

$$\mathbb{E}_{y \sim P} \sqrt{Q(y)/P(y)} = \sum_y \sqrt{P(y)Q(y)} = 1 - \frac{1}{2}D_{\text{H}}^2(P, Q) \leq \exp\left(-\frac{1}{2}D_{\text{H}}^2(P, Q)\right). \quad (8)$$

Fix $\theta \in \Theta_{\text{S}}$ and let $Z_i(\theta) = \frac{1}{2} \log(Q_{\theta}^{x_i}(y^{(i)})/P^{x_i}(y^{(i)}))$. The pairs $(x_i, y^{(i)})$ are i.i.d., so by (8) and Jensen,

$$\mathbb{E} \exp\left(\sum_i Z_i(\theta)\right) = \left(\mathbb{E}_{x \sim \mathcal{D}} [1 - \frac{1}{2}D_{\text{H}}^2(P^x, Q_{\theta}^x)]\right)^n \leq \exp\left(-\frac{n}{2}\mathbb{E}_x D_{\text{H}}^2(P^x, Q_{\theta}^x)\right).$$

Markov’s inequality gives, for each θ , with probability $\geq 1 - \delta/|\Theta_{\text{S}}|$,

$$\sum_i Z_i(\theta) \leq -\frac{n}{2}\mathbb{E}_x D_{\text{H}}^2(P^x, Q_{\theta}^x) + \log(|\Theta_{\text{S}}|/\delta).$$

A union bound makes this hold for all θ simultaneously. By realisability and the definition of the MLE, $\sum_i Z_i(\hat{\theta}) \geq \sum_i Z_i(\theta^{\circ}) = 0$. Hence $\mathbb{E}_x D_{\text{H}}^2(P^x, Q_{\hat{\theta}}^x) \leq 2 \log(|\Theta_{\text{S}}|/\delta)/n$, that is, $\mathbb{E}_x \text{H}^2 \leq \log(|\Theta_{\text{S}}|/\delta)/n$. Finally, $\text{TV} \leq \text{H}\sqrt{2 - \text{H}^2} \leq \sqrt{2} \text{H}$ (Proposition 1.5), and Jensen gives $\mathbb{E}_x \text{TV} \leq \sqrt{2}\mathbb{E}_x \text{H}^2$. The horizon T never enters: y is treated as a single symbol from $\mathcal{Y}$, and only $\log|\Theta_{\text{S}}|$ appears (Zhang, 2006; Foster et al., 2024). $\square$

A.2 Proof of the performance-difference identity (Theorem 2.16)

Let $V_t^p(s) = \mathbb{E}_{a \sim p_t(\cdot|s)} Q_t^p(s, a)$, with $V_{T+1}^p \equiv 0$. Then $Q_t^p(s, a) = c_t(s, a) + \mathbb{E}[V_{t+1}^p(s_{t+1}) \mid s_t = s, y_t = a]$, and the transition $s_{t+1} = (s, a)$ is the same under both policies. Along a student

trajectory,

$$\sum_{t=1}^T c_t(s_t, y_t) - V_1^p(s_1) = \sum_{t=1}^T [c_t(s_t, y_t) + V_{t+1}^p(s_{t+1}) - V_t^p(s_t)].$$

Take expectations under q_θ . The t -th summand has conditional expectation $\mathbb{E}_{a \sim q_t(\cdot|s_t)} Q_t^p(s_t, a) - V_t^p(s_t)$ given s_t . Moreover $\mathbb{E} V_1^p(s_1) = J_c(p)$, since $s_1 = x \sim \mathcal{D}$ under both policies. This yields the identity. For the bound, note that for f with $\max f - \min f \leq u$, $|\mathbb{E}_q f - \mathbb{E}_p f| \leq u \text{TV}(p, q)$. $\square$

A.3 Reverse water-filling asymptotics (Propositions 3.8 and 3.9)

For a Gaussian vector with independent coordinates of variances λ_i and squared-error distortion, the distortion–rate function is $D(R) = \sum_i \min(w, \lambda_i)$, with w fixed by $\sum_i \frac{1}{2} \log_+(\lambda_i/w) = R$ nats (Cover and Thomas, 2006, Thm. 10.3.3). Take $\lambda_i = ci^{-\alpha}$, $\alpha > 1$, and let $k = \max\{i : \lambda_i > w\}$, so that $w \in [\lambda_{k+1}, \lambda_k]$. Then $R = \frac{\alpha}{2} \sum_{i \leq k} \log \frac{k}{i} + \frac{k}{2} \log \frac{\lambda_k}{w}$. The last term lies in $(0, \frac{k}{2} \alpha \log(1 + \frac{1}{k})) \subseteq (0, \frac{\alpha}{2}]$. By Stirling, $\sum_{i \leq k} \log \frac{k}{i} = k - \frac{1}{2} \log(2\pi k) + o(1)$. Hence $k = \frac{2R}{\alpha} (1 + o(1))$. The distortion is $k w + \sum_{i > k} ci^{-\alpha} = ck^{1-\alpha} (1 + \frac{1}{\alpha-1}) (1 + o(1))$. Substituting k , and halving for the $\frac{1}{2} \|\cdot\|^2$ form of the local KL, gives the constant in Proposition 3.8. For a finite spectrum of length $k_\top$, once $w < \lambda_{k_\top}$ all components are active. Then $R = \frac{1}{2} \sum_{i \leq k_\top} \log(\lambda_i/w)$ gives $w = (\prod_i \lambda_i)^{1/k_\top} e^{-2R/k_\top}$ and $D = k_\top w$, which is Proposition 3.9 after the change of units $R \mapsto R \ln 2$ and halving. $\square$

A.4 Numerical verification

The scripts in `verification/` check the results marked [numerically checked], by random search, exact enumeration or optimisation. For multi-part results, the parts covered are those listed below. `check_divergences.py` covers Propositions 1.5, 1.7, 1.8, 1.10, 2.11, 4.1, 4.3, 4.5 and 4.6, the counterexample of Proposition 4.17, and Lemma 2.2. This includes both comparison directions and the JSD $_\beta$ sandwich of Proposition 1.8, all three event bounds, and parts (ii)–(iv) of Proposition 2.11. `check_sequences.py` enumerates a $V = 3$, $T = 4$ autoregressive pair and verifies Theorems 1.12, 2.15 and 2.16 and Propositions 1.14, 2.19, 4.8, 4.9 and 4.12. It also checks the TV estimator, the bias of the stop-gradient estimator, the unbiasedness of the pathwise/score split in Definition 4.14, and the bias that results from adding the two recipes. `check_capacity.py` verifies Propositions 3.8 and 3.9, Theorem 2.3, and Corollary 3.2, the $R = 0$ tightness of Corollary 3.3, and Theorem 3.1 on 300 random instances by exact computation of $I(\mathbf{T}; Y)$. `check_representations.py` verifies Propositions 5.1 and 5.9 and Theorem 5.7 in a random instance of the linear-Gaussian model. All checks pass.

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